

Major Group Project Individual Report

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Group 1 BMM-A Electric Powertrain For A Formula Student Car

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Abstract

Brunel's MSc Major Group Project delivers a concept-stage Formula Student electric vehicle design for submission to the FS UK Concept Class at Silverstone in July 2026, to be inherited by Brunel Racing for the 2027 build. This report documents the motor subsystem: selection, performance envelope, controller architecture, and build-ready interface specifications.

The selected motor is the Emrax 228 MV Liquid Cooled, a 13.5 kg axial-flux PMSM, paired with the Cascadia PM150DZ inverter and a 108S5P accumulator (388.8 V nominal). It was chosen over four alternatives on compatibility, performance, and feasibility. Performance was analysed across the four FS dynamic events using a modular MATLAB framework with library functions, lap-time models, and a central parameter file. The framework set the motor-to-wheel ratio at 3.60 and predicted 4.256 s for 0–75 m, a 26.7 min endurance with 30.4% pack capacity remaining, and peak winding temperature of 78.5 °C against the 120 °C cutoff. It also quantified regen recovery and confirmed iMechE FS 2026 compliance.

Three findings emerged. First, the conventional reading of EV2.2.1 caps shaft power at 80 kW; at 92% efficiency, DC bus power reaches ~87 kW, exceeding the rule by 7 kW. The corrected DC-input cap controller derates the shaft cap to 73.6 kW so DC bus power clips at 80 kW. Second, the proposed SoC-aware adaptive controller resolves the crossover between high-SoC (rule-bound) and low-SoC (voltage-bound) regimes, saving 94 s of endurance against a conservative fixed map and 3.1 s against a calibrated one. Third, regen recovery reaches 7.1% of drive energy, bounded by the 2C pack acceptance (16.3 kW) rather than motor capability (~80 kW) — a 5:1 ratio. Recovery improvement is an accumulator problem, not a motor problem.

The deliverable supports the 2026 Concept Class submission and the 2027 build. At cohort level, BMM-A Electric Powertrain delivers seven co-ordinated subsystem submissions against a locked 400 V architecture, with build-sheet interfaces to BMM-B Chassis, BMM-C Suspension, and BMM-D Driver Controls. The submission shows component readiness with margin against every binding rule and event target, but defers closed-loop whole-vehicle verification to 2027 as a residual risk.

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Notation

Roman Symbols

Symbol	Unit	Meaning
$C_{DC,link}$	F	Inverter DC-link capacitance
$C_d A$	m ²	Aerodynamic drag coefficient × frontal area
$C_l A$	m ²	Aerodynamic lift coefficient × planform area
C_{rr}	–	Rolling resistance coefficient
C_{th}	J/K	Lumped winding thermal capacitance
E_{drive}	kWh	Total drive energy across endurance event
E_{regen}	kWh	Total regenerated energy across endurance event
E_{net}	kWh	Net pack energy consumption (drive – regen)
f_F	–	Front weight fraction
f_R	–	Rear weight fraction
f_{sw}	Hz	Inverter switching frequency
g	m/s ²	Gravitational acceleration
h_{CG}	m	Vehicle centre-of-gravity height
I_{cont}	A	Continuous-rated current
I_{peak}	A	Peak current
$I_{pk,RMS}$	A_{RMS}	Inverter peak phase current
i_g	–	Motor-to-wheel reduction ratio
K_E	V/RPM	Motor back-EMF constant
K_T	Nm/A	Motor torque constant
K_V	RPM/V	Motor voltage constant
L	m	Vehicle wheelbase
L_d	H	Direct-axis winding inductance
m_d	kg	Driver mass
m_m	kg	Motor mass
m_{total}	kg	Total vehicle + driver mass
m_v	kg	Empty vehicle mass
n_{max}	RPM	Motor maximum operating speed
N_{rear}	N	Rear-axle vertical load
$P_{cont,S1}$	W	Motor continuous (S1) shaft power rating
P_{DC}	W	DC bus power
P_{loss}	W	Motor heat dissipation
$P_{pk,S2}$	W	Motor peak (S2 2-min) shaft power rating
$P_{TS,max}$	W	FS regulatory tractive-system DC power ceiling
P_{shaft}	W	Motor shaft mechanical power

Symbol	Unit	Meaning
$P_{shaft,CAP}$	W	Shaft-power cap
R_{INT}	Ω	Pack internal resistance
R_{phase}	Ω	Motor phase resistance
r_{eff}	m	Effective rolling radius
SoC	– or %	Pack state of charge
T_{cap}	Nm	Adaptive controller commanded torque cap
T_{cont}	Nm	Motor continuous torque rating
T_{FS}	Nm	FS-rule-derived shaft torque cap
T_{inv}	Nm	Inverter-current-limited torque cap
$T_{max,ABS}$	$^{\circ}C$	Absolute thermal cutoff
$T_{max,OP}$	$^{\circ}C$	Sensor warning threshold
T_{pk}	Nm	Peak shaft torque
T_w	$^{\circ}C$	Winding temperature
UA	W/K	Lumped convective heat-transfer coefficient
V_{bus}	V	Instantaneous DC bus voltage
V_{DC}	V	DC bus voltage
V_{max}	V	Pack full-charge voltage
V_{min}	V	Pack end-of-discharge voltage
V_{nom}	V	Pack nominal voltage
V_{OCV}	V	Pack open-circuit voltage
$V_{threshold}$	V	Bus voltage at which FS rule and motor envelope cross

Greek Symbols

Symbol	Unit	Meaning
η_d	–	Drivetrain efficiency
μ	–	Tyre-road coefficient of friction
ρ	kg/m ³	Air density
ω	rad/s	Motor angular velocity

Acronyms and Abbreviations

Acronym	Meaning
AHP	Analytic Hierarchy Process
AMS	Accumulator Management System
APPS	Accelerator Pedal Position Sensor
BLDC	Brushless DC Motor
BMM	Brunel Masters Motorsport
BSPD	Brake System Plausibility Device
CAN	Controller Area Network

Acronym	Meaning
CG	Centre of Gravity
CFD	Computational Fluid Dynamics
DC	Direct Current
EMF	Electro Magnetic Field
EV	Electric Vehicle
FEA	Finite Element Analysis
FOC	Field Orientated Control
FS	Formula Student
FSUK	Formula Student United Kingdom
HV	High Voltage
IEC	International Electrotechnical Commission
IM	Induction Motor
IMD	Insulation Monitoring Device
IPM	Interior Permanent-Magnet
IPxx	Ingress Protection rating
ISO	International Organisation For Standardization
KTY	Silicon temperature sensor
LC	Liquid-Cooled
LPTN	Lumped-Parameter Thermal Network
LTS	Lap-Time Simulation
LV	Low Voltage
MTPA	Maximum Torque Per Ampere
MV	Medium Voltage
NdFeB	Neodymium-Iron-Boron (rare-earth magnet material)
NEDC	New European Driving Cycle
OEM	Original Equipment Manufacturer
PMSM	Permanent-Magnet Synchronous Motor
RPM	Revolutions Per Minute
RTD	Resistance Temperature Detector
RWD	Rear-Wheel Drive
S1	Continuous duty rating
S2	Short-time duty rating (2-minute)
SDC	Shutdown Circuit
SoC	State of Charge
SPM	Surface-mounted Permanent-Magnet (motor)
SRM	Switched-Reluctance Motor
TS	Tractive System
VCU	Vehicle Control Unit
WLTP	Worldwide Harmonized Light Vehicle Test Procedure
ZDAC	Zero d-axis Current Control

1. Introduction

1.1 Project Context

Brunel University London's MSc programmes undertake a year-long Major Group Project in which student cohorts develop concept-stage engineering deliverables suitable for university motorsport entry. The 2025-26 cohort runs four parallel projects sharing a single vehicle architecture: BMM-A Electric Powertrain, BMM-B Chassis, BMM-C Suspension and Wheels, and BMM-D Driver Controls. Together these four groups produce the design package for a single Formula Student (FS) electric vehicle, to be inherited by Brunel Racing for the 2027 build year, with the resulting vehicle entered into the Formula Student UK (FSUK) Concept Class at Silverstone in July 2026.

The FSUK competition, organised by the Institution of Mechanical Engineers, is the largest European university motorsport series. It is the principal route through which UK mechanical and automotive engineering students transition from academic study into industry. FSUK offers two entry routes. The Class 1 submission requires a fully constructed vehicle competing in dynamic events at the circuit. The Concept Class requires a complete documented concept-stage design package, judged on engineering design and on the feasibility of progression to a built vehicle. Concept Class is the route a team uses ahead of a build year, allowing the design to be evaluated, judged, and refined before commitment to manufacture. This work targets the 2026 Concept Class submission.

The commercial relevance of a strong Concept Class submission extends well beyond the event itself. The documented design package becomes the foundation on which the following year's build team plans component procurement, lead-time-critical orders, and integration testing. The depth and stability of the package determine whether the build year proceeds against a settled design or against a moving specification. For Brunel Racing, the 2026 submission also marks the team's transition from internal combustion to electric propulsion, making the cohort's design decisions formative for the team's competitive position over multiple subsequent seasons. The project therefore carries weight beyond a single competition placement. It sets the technical baseline for a multi-year programme and contributes to the UK EV engineering pipeline at a time of expanding industry demand for graduates with EV-specific experience.

The BMM-A Electric Powertrain project was framed at the November proposal stage with three group-level aims. The first is to develop a concept of an electric powertrain for the future Brunel Racing FS EV that improves reliability and performance over previous Brunel entries. The second is to design and justify the subsystem concepts to create a feasible 400 V single-motor configuration aligned with Formula Student rules. The third is to ensure efficient integration of the powertrain with the chassis,

suspension, and driver-controls cohort groups by establishing the required mechanical, thermal, electrical, and control interfaces.

Five group-level objectives follow from these aims. First, design subsystem concepts so that the architecture's structure can be understood and design decisions can be justified. Second, run simulations and analysis to validate each subsystem concept against the FS rule set. Third, define all interfaces so that the powertrain can be integrated with the chassis, suspension, and driver-controls cohort groups. Fourth, assess feasibility under available equipment and budget constraints. Fifth, produce a complete justified design package supporting Concept Class judging. The design must also remain financially feasible under a conceptual budget and must deliver controlled stable torque across all FS dynamic events, while maintaining predictable thermal and electrical behaviour, safe shutdown coordination, and rule-compliant insulation, monitoring, and discharge behaviour.

1.2 The Motor Subsystem within the Group

The BMM-A Electric Powertrain group divides the architecture into seven subsystem leads. Each lead is individually responsible for a complete concept-stage design within their assigned scope and for the interface specifications with adjacent subsystems. The task split is summarised in Table 1:

Table 1: BMM-A Electric Powertrain subsystem assignments

Subsystem	Lead
Accumulator Electrical	Roger Fuertes Piarpuezen
Accumulator Mechanical	Abdalla Abdalla
Inverter and High Voltage System	Powei Amughan
Low Voltage Systems	Harsha Vardhan Saka
Motor	Sharan Kalamohan
Drivetrain and Packaging	Viki Ganesan
Thermal Management	Ermal Shaqiri

The accumulator electrical subsystem covers cell-to-pack series-parallel architecture, segmentation strategy, accumulator management system (AMS) functional requirements, and integration with the shutdown circuit. The accumulator mechanical subsystem covers the cell module enclosure, mounting strategy, mechanical protection, and chassis-fit packaging of the assembled pack, in close coordination with the electrical and thermal leads. The inverter and high-voltage system covers inverter selection and operating-boundary specification, the high-voltage distribution path between accumulator and motor, pre-charge and discharge logic, and the protection devices required by the FS rule set. The low-voltage system covers LV power distribution, sensor integration, vehicle-controller logic for torque demand and subsystem coordination, the CAN communication architecture, and the safety state machines for plausibility checks, shutdown behaviour, and inter-subsystem signalling. The drivetrain

and packaging subsystem covers the mechanical reduction ratio, differential, half-shafts, motor mounting interfaces, and the spatial packaging of all powertrain subsystems within the chassis envelope. The thermal management subsystem covers the cooling-loop architecture, radiator and pump sizing, hose routing, and the integration of the cooling layout with the chassis packaging, sized against the heat-rejection requirements of the motor, inverter, and accumulator.

The motor subsystem occupies a central position within the group. It is the converter between the electrical domain owned by the accumulator and inverter leads and the mechanical domain owned by the drivetrain lead. Every motor decision therefore propagates outwards to multiple adjacent subsystems. The motor's peak torque sets the drivetrain reduction ratio and the mounting reaction loads. Its efficiency and operating envelope set the heat-rejection load that the cooling loop must remove and the bus-current profile that the high-voltage distribution must carry. Its regenerative behaviour interacts with pack acceptance limits to determine recoverable energy, and with the vehicle controller to determine when regen torque is commanded.

The Group 1 powertrain integrates with three other cohort groups across well-defined mechanical and electrical boundaries. The BMM-B Chassis group defines the envelope into which the powertrain must package, and the firewall thermal boundary against which the motor casing temperature must be evaluated. The BMM-C Suspension and Wheels group defines the rear-axle geometry, the tyre coefficient of friction that dominates the lap-time prediction sensitivity established later in this report, and the wheel hub interface that carries motor torque to the road. The BMM-D Driver Controls group defines the driver pedal inputs, the brake-by-wire interfaces feeding the regenerative-braking blending logic, the drive-mode paddle interface that selects between energy-management modes, and the wiring loom that physically routes the low-voltage signals between motor sensors and the VCU. Each of these cross-cohort interfaces imposes specific motor-side deliverables on the present work, and is documented through the committed parameter values that complete the integration.

1.3 Aims, Objectives and Report Structure

The motor subsystem aims, taken from the November project proposal, are threefold. The first is to identify a motor concept that aligns with the power, torque, thermal, and rule limits imposed by the FS Concept Class submission and the locked vehicle architecture. The second is to establish the motor's role in the 400 V architecture committed at the proposal stage, defining how the motor integrates with the accumulator voltage envelope and the inverter peak-current bound. The third is to provide performance data that informs the gearing selection delivered to the drivetrain lead, the thermal design delivered to the thermal-management lead, and the control-strategy specification delivered to the VCU lead.

Five objectives achieve these aims. The first is to evaluate feasible motor concepts against quantitative performance criteria, screening on compatibility, FS rule compliance, and performance before reaching a single committed selection. The second is to define the key mechanical and electrical interfaces of the selected motor, including rotation polarity, mounting flange geometry, phase cabling specification, position and temperature sensor specification, and HV connector interfaces, at a level of detail sufficient for cohort partners to commit their own design work. The third is to identify the thermal loads requiring cooling-system support across the FS dynamic event suite, delivering the heat-rejection specifications required by the thermal-management lead. The fourth is to confirm motor concept feasibility through both comparative ranking against alternative motor candidates and rule-based assessment against every relevant FS rule. The fifth is to provide the motor input parameters required by the drivetrain and VCU leads to complete their own designs.

The remainder of this report is structured as follows. Chapter 2 reviews the literature relevant to motor selection, performance modelling, control strategy, regenerative braking, and synthesises the gaps that motivate the contributions developed in Chapter 5. Chapter 3 translates the vehicle-level constraints, FS rules, and cohort interface inputs into specific motor-subsystem design targets. Chapter 4 describes the MATLAB simulation framework used to evaluate those targets. Chapter 5 presents the individual results, covering the motor selection outcome, the motor capability surface, baseline and final-design vehicle performance, the gear-ratio optimisation, the SoC-aware adaptive controller, the thermal envelope, regenerative behaviour and sensitivity analysis. Chapter 6 reviews the motor subsystem outputs in the wider Group 1 and cohort design context, and reviews the commercial, environmental, and social implications of the design. Chapter 7 draws conclusions. Chapter 8 sets out the recommended further work for the Brunel Racing 2027 build year. Appendix 1 contains the project management review.

2. Literature Review

2.1 PMSM Architectures for Formula Student Traction

Motor selection governs the boundary of an EV's performance envelope. It propagates constraints into every adjacent subsystem: drivetrain ratio, thermal load, controller architecture, and packaging volume. This subsection positions the Emrax 228 medium-voltage liquid-cooled axial-flux PMSM within the design space and justifies its selection over the alternative architectures available to a Formula Student (FS) Concept Class entrant.

The dominant traction-motor families in modern EV propulsion are induction motors (IMs), permanent-magnet synchronous motors in surface-mounted (SPM) and interior (IPM) configurations, switched-

reluctance motors (SRMs), and brushless DC variants of the PM family. Pellegrino et al. (2012) provide the canonical IEEE comparison of IM, SPM, and IPM drives at identical stack dimensions and inverter ratings, with FEA-validated simulation across the full torque-speed envelope. The IM is naturally de-excited under inverter fault, which is an inherent safety advantage, but is penalised by cage losses that require higher stator current than either PM type for the same continuous power. The SPM has the simplest construction and the shortest end connections, but suffers two limitations critical for racing: very limited transient overload power, capped by maximum-torque-per-voltage clamping at the motor's characteristic current, and a high uncontrolled generator voltage during inverter trip. The IPM offers the best overall performance compromise, with overload power comparable to the IM at low speed and lower losses than the SPM at high speed. Its multi-flux-barrier rotor is industrially complex, and its PM quantity can in fact exceed that of an SPM of equivalent rating (Pellegrino et al., 2012).

This last point is more consequential than it first appears. The intuition that IPM topologies save magnet mass through reluctance torque supplementation does not hold at FS scale. For an entrant where the supply chain for high-coercivity NdFeB is concentrated in a single region and rare-earth cost is a real procurement risk, the IPM's advantage weakens substantially. The SPM's higher uncontrolled generator voltage on inverter trip is a more relevant counterweight, but it can be managed at inverter-level interlock design rather than through motor reselection.

Within FS specifically, the architectural picture narrows. Kinoti et al. (2024) apply Saaty's Analytic Hierarchy Process to a five-criterion comparison of PMSM, IM, SRM, BLDC, and DC motors. They rank PMSMs first on efficiency and power density but conclude that induction motors are the top overall choice when cost and reliability are weighted equally with performance. The latter conclusion holds for general-purpose passenger-EV applications. It does not transfer to FS Concept Class, where mass minimisation dominates the scoring formulae and the cost penalty of a PMSM is absorbed by the project's component budget rather than by a production-volume cost case. The Kinoti et al. ranking is not wrong, but it is bounded by an objective function that does not match the FS one.

The empirical evidence from FS competition supports this distinction. Badawy et al. (2024) document that all electric racing cars participating in FSUK 2023 used AC machines, with synchronous PM motors dominating due to higher efficiency, more compact size, higher power density, and the absence of brush-wear maintenance. Electric cars formed more than two-thirds of FSUK 2023 entrants, and that share is expected to continue rising. The market evidence has already converged on the answer the AHP literature treats as contested.

Within the PMSM family, the axial-flux topology has emerged as a distinct sub-class with motorsport-relevant trade-offs against the conventional radial-flux configuration. Ivanov et al. (2022) survey high-

specific-power machines including the YASA 750, YASA P400, Emrax 208, and Emrax 228 in the context of electric aircraft and air-mobility applications. They report bench-test results from Bauman Moscow State Technical University in which the Emrax 208 was characterised under combined cooling, with the Emrax 228 used as the load machine. Their key finding is that the manufacturer-declared specific power values are achievable in practice, with measured shaft torque exceeding the rated continuous figure. This is one of very few independent experimental validations of Emrax-class motor performance in the open literature, and is the strongest available academic support for treating manufacturer datasheet figures as a defensible simulation input. Baker (2023), whose Honours dissertation documents the design of a custom inverter for an FSAE EV using the same Emrax 228 motor, additionally confirms successful field-oriented control commutation of the 228 using both a commercial Prohelion WaveSculptor200 inverter and a prototype STM32-based open-source controller. This provides peer-team precedent for the architectural lock-in adopted in this work.

The Emrax 228 sits within the design corner identified by Pellegrino et al. (2012) as the SPM family: high power density, simple construction, but limited transient overload margin and characteristic-current-bounded field-weakening behaviour. These limitations propagate directly into the methodology adopted later in this report. The inverter peak-current limit, rather than motor thermal capability, sets the effective peak torque. The bus-voltage limit, rather than thermal limit, sets the effective peak power at the operating bus voltage. The literature therefore identifies a priori the two binding constraints that define the torque-speed envelope used in the present work.

2.2 Motor Selection Methodology

The literature on motor selection divides into two camps. The first is multi-criteria decision-making (MCDM) based on subjective weighting. The second is simulation-based comparative ranking based on validated performance models. The former is the dominant published methodology; the latter is the methodology adopted in this work.

The Analytic Hierarchy Process is the most widely cited MCDM framework in the EV motor literature. Kinoti et al. (2024) provide the canonical application, comparing PMSM, IM, SRM, BLDC, and DC motors across criteria including efficiency, power density, torque ripple, cost, and reliability. They use Saaty's pairwise-comparison scale and an eigenvector-based weighting method with a consistency ratio threshold of $CR \leq 0.1$. AHP's strengths are well-established: a structured hierarchy, an explicit weighting mechanism, and an internal-consistency check. The Saaty method has decades of demonstrated application across procurement, supplier selection, and design selection problems.

Three limitations expose AHP's unsuitability for the present application. First, AHP's pairwise comparisons are subjective. Relative-importance values come from the decision-maker's judgement, not measurement, and the resulting ranking holds only within the assumptions encoded in those weights. Kinoti et al. (2024) explicitly acknowledge that "the ranking becomes less clear when considering the role of power sources, electronic components, intelligent control, and diagnostic processes". Second, AHP rankings are static. Each criterion is scored independently, but a motor's effective performance depends on interactions. A 220 Nm peak-torque, 124 kW peak-power motor performs differently when paired with a 300 A inverter than with a 400 A inverter, and AHP has no mechanism to capture this. Third, AHP rankings are sensitive to weight perturbation. A small change in the decision-maker's pairwise judgement can flip the top-ranked candidate, which is an unacceptable source of decision risk for a final selection between similarly rated PMSMs.

The Pugh matrix offers a simpler alternative MCDM approach widely used in FS team reports. It relies on relative scoring against a single datum candidate rather than the eigenvector mathematics of AHP. The Pugh matrix carries the same three weaknesses: subjective scoring, static criterion evaluation, and weight-perturbation sensitivity. Both AHP and Pugh are appropriate for early-stage shortlisting, where the goal is to eliminate clearly unsuitable candidates, but neither is appropriate as the final selection mechanism when the remaining candidates are similarly rated.

The alternative methodology adopted in this work runs each candidate motor through a validated MATLAB simulation of the full FS dynamic event suite and ranks the candidates by simulated lap-time and energy-efficiency outcomes. This addresses each limitation of AHP directly. The criteria emerge naturally from the FS scoring formulae (iMechE, 2025) rather than being independently weighted. Interactions with inverter, bus voltage, gear ratio, and thermal envelope are captured automatically. The ranking is robust to small parameter perturbations because the FS scoring formulae are continuous around the predicted operating points.

This methodology is not novel. Doyle et al. (2019) explicitly frame their LTS as enabling "comparative analysis between different vehicle concepts, without the inconsistencies of a driver" and use it for gear ratio and battery capacity selection. Pellegrino et al. (2012) perform conceptually the same exercise at component level via FEA-validated simulation across three motor designs. The approach is the appropriate generalisation of established simulation-based comparison to the motor-selection problem in an FS context.

The methodology carries one residual limitation worth surfacing. A simulation-based comparison is only as defensible as the underlying simulation models and the parameter inputs to those models. Where candidate motors have parameters of differing fidelity, the simulation can bias the ranking toward the

motor whose envelope is best characterised in published data. The present work mitigates this risk in two ways. The K_v interpolation method described in §2.4 is applied uniformly across all candidates, removing systematic differences in how the field-weakening envelope is computed. Candidate motors whose datasheets omit parameters required by the simulation framework are eliminated at screening stage, ensuring the comparison is between motors with comparable-fidelity inputs. Both preconditions are documented in §5.1.

2.3 Torque-Speed Envelope Modelling

A defensible motor selection rests on a defensible model of the torque-speed envelope. The textbook idealisation, a constant-torque region below the corner speed and a constant-power region above it, captures the dominant envelope behaviour but conceals three constraints central to FS EV design. First, the constant-power region droops asymptotically toward an upper bound set by the motor's characteristic current and available terminal voltage, with the droop rate depending on whether the machine is SPM or IPM (Pellegrino et al., 2012). Second, the envelope's upper boundary is set by the minimum of three constraints: inverter current, bus voltage, and any externally imposed power cap such as the FS Energy Meter rule (iMechE, 2025). Which constraint binds varies with speed and bus voltage. Third, for a pack-voltage-limited EV, the envelope itself is voltage-dependent, with the field-weakening corner speed and post-corner droop scaling with bus voltage.

Lu and Kar (2010) provide the canonical voltage-limit-ellipse and current-limit-circle framework in the (i_d, i_g) reference frame. They formalise the two flux-weakening operating modes that any PMSM envelope model must respect: the current-and-voltage-limited Mode I, and the voltage-limited Mode II beyond the maximum-torque-per-voltage point. The Mode I to Mode II boundary depends on the motor's characteristic current $i_{chl} = \lambda_m / L_d$, which determines whether the machine has a bounded or unbounded speed range under field weakening. For an SPM machine of the Emrax class, the characteristic current is comparable to the rated current, meaning the constant-power region is finite and the envelope droops above the corner speed. Alwaqfi et al. (2025) update the field-weakening literature with a comprehensive 2025 review, classifying modern methods into analytical direct calculation, direct open-loop with experimental LUTs, single current regulator, unified direct flux vector control, torque and flux control with LUTs, and vector current control. Their critical synthesis confirms that no single approach dominates all regimes, and that selection depends on motor type, performance requirement, computational budget, and acceptable complexity.

Three approaches to envelope construction appear in the literature. Full d-q simulation (Pellegrino et al., 2012; Lu and Kar, 2010) requires the motor's d-axis and q-axis inductances, magnet flux linkage, stator resistance, characteristic current, and saturation curves. This is the methodological gold standard,

but it is unsuitable for FS Concept Class work because these parameters are typically not published in commercial datasheets. The flat-datasheet approach treats published peak-torque and peak-power as constants across all bus voltages, as in Doyle et al. (2019). This is defensible at nominal voltage but carries the SoC-dependent bias documented in §2.2. The K_v interpolation approach occupies the middle ground. The motor's voltage constant K_v (rpm/V), available in the Emrax datasheet, defines no-load speed as a linear function of bus voltage. Combined with the torque constant K_T (Nm/A) and the inverter peak-current limit, this allows construction of a three-region envelope capturing the bus-voltage dependence that Doyle et al. (2019) omit. The accuracy penalty against full d-q simulation is estimated at 5% to 10% in the field-weakening region, and is acknowledged in §4.3.

The torque-speed envelope is bounded thermally as well as electrically (Cadence, 2022). Sequeira et al. (2022) validate a lumped-parameter thermal network (LPTN) of an interior PM motor against both FEA and experimental data. They report maximum LPTN-versus-experiment error of 4% at the slot-winding and end-winding hot spots, maximum LPTN-versus-FEA error of 2%, and a sensitivity analysis identifying the liner-to-lamination contact resistance as the dominant uncertainty. This error band justifies treating the simulation's thermal predictions as defensible within $\pm 5\%$ of true winding temperature for concept-stage work, where multiple candidate motors must be evaluated and full FEA is computationally prohibitive.

The duty cycle against which the thermal envelope is evaluated is also methodologically significant. Standard road-EV cycles such as NEDC (Pellegrino et al., 2012), WLTP, or IEC 62660-1 are calibrated for road-driving thermal loads. Kellner et al. (2018) develop a high-performance duty cycle representative of race-circuit driving and validate it experimentally against a 53 Ah pouch cell. They show that high-performance scenarios produce higher cell temperatures, gradients, and degradation than road cycles, concluding that "the nature of high-performance electric vehicle use cases when compared to driving on public roads is fundamentally different". This is the academic justification for using the FS endurance event, rather than NEDC or WLTP, as the thermal validation input in this work.

The envelope model adopted in §3 follows the K_v interpolation method with timestep-by-timestep recomputation as bus voltage varies through the SoC range. It is overlaid with an LPTN-derived continuous-power thermal soft limit calibrated against the Emrax datasheet's S1 ratings.

2.4 EV Powertrain Control Strategies

This subsection identifies the specific limitation of the published control consensus when applied to a pack-voltage-limited FS EV. It frames the SoC-aware adaptive torque map proposed in this work as the methodological extension that fills the resulting gap.

In the dominant production-EV control architecture, motor torque demand is computed from a static look-up table indexed on driver pedal position and motor speed, calibrated for a single nominal bus voltage. Lu and Kar (2010) frame the limitation directly: the voltage-limit ellipse "shrinks inversely with speed", meaning the achievable operating envelope contracts as bus voltage falls. A fixed map calibrated at nominal bus voltage commands torque the motor cannot deliver at low SoC, and under-utilises the motor at high SoC (MathWorks, 2025). For a road EV operating across a narrow SoC band, this error is tolerable. For an FS EV operating across the full SoC range from a high pre-grid voltage to a deeply depleted endurance-end voltage, it becomes performance-limiting.

Lu and Kar (2010) classify PMSM flux-weakening control into feedforward, feedback, hybrid, and non-linear categories, each with documented strengths and limitations. Feedforward methods have good stability but are strongly dependent on motor parameters (Pan et al., 2022). Feedback methods are robust to parameter variation but have poor transient response near voltage saturation. Hybrid methods combine pre-computed MTPA tables with feedback adjustment, at the cost of large LUT characterisation effort. Non-linear methods provide on-line parameter adaptation but with substantially higher computational complexity. Alwaqfi et al. (2025) update this taxonomy with the 2025 review summarised in §2.4. Crucially, both reviews frame the bus voltage V_{max} as a constant input parameter to the controller. Neither addresses the case where V_{max} varies dynamically with battery SoC over the duration of a single endurance event.

The most directly relevant published work on FS EV motor control is Ibrahim et al. (2025), who optimise the d-q axis current ratio for the Formula Student Tallinn FEV's 35 kW IPMSM using Covariance Matrix Adaptation Evolution Strategy. They report 22% average torque improvement over the conventional ZDAC strategy across seven dyno test points. Three limitations of their study are directly relevant to the gap this work addresses. First, all experimental tests were conducted with a Cinergia battery emulator at fixed bus voltage; the controller is optimised in the speed domain only, not in the bus-voltage domain. Second, the paper acknowledges that the optimal MTPA strategy increases motor current and reduces battery range, but does not address how the optimal current ratio should adapt as available bus voltage falls. Third, the controller is optimised against a steady-state operating point rather than against an FS event scoring formula. The Ibrahim et al. contribution is therefore complementary to this work: it advances the motor-speed-domain control problem at fixed bus voltage, while the present work addresses the bus-voltage-domain control problem at fixed event-scoring objective.

Reading Lu and Kar (2010), Alwaqfi et al. (2025), and Ibrahim et al. (2025) together exposes a gap. Under FS Concept Class rules (iMechE, 2025), the motor's effective envelope is bounded

simultaneously by the inverter peak-current limit at low speed, the FS 80 kW Energy Meter rule in the mid-speed region, and the bus-voltage-limited motor envelope at high speed. Which constraint binds varies with both motor speed and instantaneous bus voltage. At high SoC the FS rule binds across most of the speed range. At low SoC the voltage-limited envelope falls below the rule across the upper speed range and the rule becomes non-binding. A fixed map calibrated at any single bus voltage cannot represent both regimes correctly. The EV control literature is dominated either by road-EV applications, where the FS-style fixed power cap does not exist, or by FS applications where bus voltage is treated as constant. The gap is structural, sitting at the intersection of two well-developed but separate research strands, rather than a failure of any individual paper to address its own scope.

A further observation strengthens the case for the proposed approach. The FS 80 kW rule (EV2.2.1) is defined at the DC bus, measured by the FS Energy Meter, not at the motor shaft. The motor's published peak-power figure is a shaft-power figure. A controller that simply caps shaft power at 80 kW allows DC input to climb to approximately shaft-power divided by motor efficiency, which at typical 92% efficiency exceeds the rule by approximately 7 kW. The literature reviewed above does not engage with this subtlety, because road-EV controllers do not face a regulatory DC power cap and the limited FS EV control literature does not explicitly distinguish DC-side from shaft-side cap. The present work treats the distinction as a first-order design consideration, developed in §5.5.

This work fills the gap by proposing an SoC-aware adaptive torque map that resolves the regime crossover by adapting the torque cap in real time as a function of instantaneous DC-link voltage and pack state of charge. Three implementation framings (bus-voltage-aware, SoC-aware via coulomb counting, and combined) are detailed and quantified in §3 and §4. The proposed approach is positioned as a methodological extension of the established flux-weakening control taxonomy (Lu and Kar, 2010; Alwaqfi et al., 2025) rather than as an alternative to it. The inner-loop flux-weakening controller continues to operate as published, but its outer-loop torque reference is no longer drawn from a fixed map. The contribution is methodological rather than implementational; firmware-level validation is identified in §8 as further work.

2.5 Regenerative Braking and Energy Recovery

In a battery EV the motor operates bi-directionally. Under deceleration it functions as a generator, recovering kinetic energy back into the pack. Szumska (2025) provides the most comprehensive 2025 review of regenerative braking, synthesising 89 peer-reviewed articles from 2005 to 2024, and identifies three universal constraints on regen acceptance that any defensible simulation must respect. The low-speed cutoff arises because the electric machine cannot generate sufficient back-EMF below approximately 10 km/h, where only mechanical brakes are employed. The SoC ceiling reduces battery

acceptance once SoC exceeds approximately 80% (Cummins, 2025). The high-deceleration cutoff occurs above approximately 3 m/s^2 , where the regen system cannot develop sufficient torque and mechanical brakes assume the dominant role. These constraints are physical limits imposed by motor back-EMF, battery electrochemistry, and tyre-grip-limited brake force, respectively. Two of the three (the SoC ceiling and the high-deceleration cutoff) interact directly with the bus-voltage-aware controller framework established in §2.5.

Szumska (2025) reports recovery efficiencies of 79% under gentle braking (1 to 2 m/s^2) and 29% under medium-intensity braking (3 to 4 m/s^2) for typical road EVs. Under urban cycles, recovered braking energy can reach 20% to 40% of energy consumed for propulsion, while highway recovery falls below 5%. FS endurance events fall between these regimes: braking is frequent but at deceleration rates well above the 3 m/s^2 regen-favourable cutoff. No published FS-EV study has quantified the recovery efficiency band specific to FS endurance under bus-voltage-limited motor regen. This is the specific gap §5.8 of the present work fills via simulation.

A second methodological consideration is the architectural distribution of regen torque. Szumska (2025) reports that all-wheel-drive EVs with in-wheel motors achieve 23% and 31% greater energy recovery than front-wheel-drive and rear-wheel-drive single-motor configurations, respectively. The mechanism is straightforward: the front axle carries the dominant load transfer during deceleration and therefore the dominant tyre-limited brake force. This finding is methodologically significant for the architectural choice locked in §2.1. The single-motor RWD layout cedes approximately 23% to 31% of theoretical regen recovery relative to the in-wheel optimum. The trade-off was selected on grounds of mass, complexity, and cost, but it is acknowledged here as a constraint on achievable regen contribution.

Szumska (2025) further classifies regen control strategies into a hierarchy that parallels the flux-weakening taxonomy of §2.5: velocity-threshold, torque-based, brake-pressure, ideal-force-distribution, PID, fuzzy-logic with genetic algorithms, dynamic programming, neural networks, and Model Predictive Control, with reported gains of 10% to 53% across these methods. The critical evaluation in Szumska (2025) notes that "the computational load of these methods limits their deployment in mass-market vehicles". For a Concept Class FS EV operating without on-board GPS or route preview, the realistic regen control strategy sits closer to the ideal-braking-force-distribution end of the spectrum than the MPC end.

The most directly relevant academic precedent for endurance EV simulation is van Kampen et al. (2022), whose bi-level mixed-integer convex optimisation framework for a fully electric LMP3-class endurance vehicle reports 30% greater driven distance under the optimal pit-and-charge strategy versus a flat-out baseline over a 6-hour Zandvoort race. Their controller treats the regen-mechanical brake split

implicitly through a bi-directional EM model, and they explicitly note that "the recuperated energy during braking before pit entry... can be neglected, since it does not affect the powertrain operation and stint time". This assumption is defensible for a fixed-time pit-stoppable race with charging, but does not hold for an FS endurance event with no mid-event charging. Every joule of recovered energy directly extends the achievable distance because the pack must complete the full event on its initial energy budget.

A further observation deserves attention because the published literature does not engage with it. The regen literature treats the binding constraint on recovery efficiency as a function of the controller, the brake-force distribution strategy, or the architectural choice of motor placement. None of the reviewed sources explicitly considers the case where the binding constraint is not in the motor or in the controller but in the accumulator's electrochemical acceptance limit. For a high-power, low-energy FS pack with a moderate C-rate, the pack's regenerative current acceptance may be lower than the motor's regenerative power capability by a substantial margin. In that case, control-loop refinement returns nothing, and recovery improvement becomes a cell-chemistry decision rather than a motor or controller decision. This observation is developed quantitatively in §5.8 and §6.3 as the central cohort-level finding of this work.

The regen treatment in §4 of this work therefore commands regen torque as the minimum of brake-pedal demand, the negative-torque envelope at instantaneous bus voltage, an SoC-dependent battery acceptance limit derived from coulomb counting, and the rear-axle tyre-grip limit accounting for load transfer. Mechanical brakes provide the balance to total deceleration demand. This methodology is consistent with the implicit bi-directional EM model adopted by van Kampen et al. (2022), extended with explicit SoC-acceptance and tyre-grip constraints. The no-charging FS endurance context renders the van Kampen et al. regen-as-implicit assumption inappropriate.

2.6 Synthesis and Identified Gaps

The literature surveyed in §2.1 to §2.6 establishes a clear methodological precedent for FS EV motor selection, performance prediction, control, and regenerative behaviour. It also exposes three structural gaps that the present work addresses, each tied to a specific finding developed in §5 and consolidated in §6.

The first gap concerns the interpretation of the FS 80 kW power rule (EV2.2.1, iMechE, 2025) in controller architecture. Within the FS-EV control literature reviewed for this project, no source examined makes the DC-vs-shaft distinction explicit; this should not be read as a claim of universal novelty against the full electrical-engineering literature. §5.5 demonstrates that the distinction is non-

trivial: the naive shaft-power interpretation exceeds the rule by approximately 7 kW at typical motor efficiency. The corrected DC-input cap controller derived here satisfies the rule by construction.

The second gap concerns the interaction between the FS power rule and the motor's voltage-limited envelope across the full pack SoC range. The flux-weakening literature (Lu and Kar, 2010; Alwaqfi et al., 2025) treats bus voltage as constant. The FS EV control literature (Ibrahim et al., 2025) optimises in the motor-speed domain at fixed bus voltage. Neither addresses the regime crossover at the bus voltage where the FS rule and the motor envelope intersect. The present work proposes an SoC-aware adaptive torque map that resolves this crossover. The benefit is quantified at 94 seconds of endurance event time against a conservative fixed-map alternative.

The third gap concerns the location of the binding constraint on regenerative recovery in an FS EV. The regen literature treats recovery efficiency as a function of controller architecture, brake-force distribution, or motor placement. None of the reviewed sources explicitly considers the case where the pack's regenerative acceptance limits recovery below the motor's regenerative capability. §5.8 and §6.3 establish that for the locked architecture, the motor can deliver approximately 80 kW of regen at typical braking speeds against a pack that accepts 16.3 kW, a 5:1 ratio. Recovery improvement is therefore an accumulator design problem, not a motor or controller design problem.

These three findings are developed in §5 and integrated into the cohort design discussion in §6.

3. Design Requirements and Constraints

This chapter translates the whole-vehicle constraints established by the cohort and the Formula Student rule set into quantitative motor-subsystem design targets. It sits between the literature review, which positions this work in the published landscape, and the methodology in Chapter 4, which describes how those targets are tested. Five subsections proceed in order. §3.1 sets the vehicle-level performance targets against which the design is evaluated. §3.2 identifies the FS 2026 rules that bound the motor design space. §3.3 documents the cohort interface parameters, separated by whether they are formally locked or carried as conservative concept-stage assumptions pending cohort refinement. §3.4 describes the locked architecture within which the motor must integrate. §3.5 derives the seven motor-specific requirements that emerge from the intersection of the preceding four subsections.

3.1 Vehicle-Level Performance Targets

The FSUK Concept Class 2026 dynamic event scoring formula assigns 525 points across the four timed events. Endurance carries 275 points, autocross 100, and acceleration and skidpad 75 each. The

dominant weighting on endurance reflects the competition's emphasis on energy management over peak performance and drives the order in which the motor subsystem requirements are prioritised in §3.5.

Group 1 set its November proposal performance targets to be competitive against the upper quartile of 2025 FSUK entrants while remaining defensible against the cohort's locked vehicle mass, accumulator capacity, and inverter sizing. The targets are binding design objectives; the 2025 winner times in Table 2 appear as competitive benchmarks for context, not as design commitments.

The 0–75 m acceleration target of 4.3 s places the vehicle within 16 ms of the 2025 acceleration winner. It requires the motor subsystem to deliver peak shaft torque that fully exploits the rear-axle traction limit at launch and continues delivering near-peak power until the field-weakening corner. The skidpad target of 5.3 s is a lateral-grip-bounded outcome dependent on tyre coefficient of friction, satisfied automatically provided the gear ratio places the steady-state cornering speed within the constant-torque region. The autocross event is left to a competitive-pace target rather than a fixed time because the FS rules permit only a representative track rather than a fixed circuit, making absolute time comparison between teams unsound. The endurance target of ≤ 27 minutes total event duration with $\geq 20\%$ pack energy margin is the binding integrated requirement: it demands energy-weighted motor efficiency above approximately 90%, controller management of state-of-charge depletion across 22 laps within a single energy budget, and a thermal envelope that sustains the duty cycle without derate. The pack margin requirement is a conservative engineering allowance against the residual uncertainty in cohort-pending parameters discussed in §3.3.

Table 2: Vehicle-level performance targets and 2025 FSUK winning times for comparison.

Event	Group 1 Target	2025 Winner Time	Winning Team
Acceleration	≤ 4.3 s	4.284 s	Oxford Brookes
Skidpad	≤ 5.3 s	5.826 s	UPV/EHU
Autocross	Competitive pace	n/a	n/a
Endurance	≤ 27 min, $\geq 20\%$ pack margin	32.2 min	Birmingham

The targets in Table 2 are deliberately calibrated to be ambitious but not aspirational. Two are within 50 ms of the 2025 winning time and one (skidpad) is 600 ms below it. The skidpad target reflects the conservative $\mu = 1.30$ baseline assumption used in §3.3, which the §5.9 sensitivity analysis identifies as the dominant uncertainty in the lap-time prediction. The motor subsystem is not the binding constraint on this event under any plausible refinement.

3.2 FS 2026 Rules Bounding the Motor Subsystem

The FS 2026 rule set imposes an extensive specification environment on the motor subsystem. Table 3 consolidates the rules that materially bound the design space, grouped functionally rather than by rule number. Each row states the rule reference, its quantitative content where applicable, and the design implication for the motor subsystem.

Table 3: FS 2026 rules materially bounding the motor subsystem design space, grouped by functional category.

Rule(s)	Quantitative Content	Design Implication for Motor Subsystem
Power and voltage ceilings		
EV2.2.1, D9.4.1	TS DC bus power ≤ 80 kW or 500 A (whichever lower), 500 ms moving-average enforcement	Binding regulatory ceiling at high SoC. Forces controller to clip DC-side, not shaft-side; see §5.5.2.
EV2.2.2	Regen power unrestricted in magnitude, 500 A ceiling applies	Pack acceptance binds far below 500 A in practice; see §5.8.
EV4.1.1	≤ 600 VDC general TS, ≤ 630 VDC for motor controllers and AMS internal signals	Pack $V_{max} = 453.6$ V leaves comfortable headroom; not binding.
EV4.1.3	Components rated for max possible operating temperature	Motor winding insulation class must match worst-case hot-spot, not nameplate.
Motor hardware and cooling		
T7.3.4 (EV)	Axial-flux motors with rotor rotating around stator require ≥ 1 mm 6061-T6 Al or steel scatter shield	Triggered by Emrax 228 topology. Scatter shield mass and packaging penalty; see §5.10.1.
T7.2.2	TS coolant: plain water, air, or oil only	Eliminates all water-glycol candidates at motor screening; see §5.1.

3.3 Cohort Interface Constraints

The motor subsystem operates within a vehicle architecture in which some parameters are fixed by other cohort members. Designing only against fully locked numbers would have prevented progress; designing against unverified numbers would have produced unsupported results. The methodology adopted distinguishes locked parameters from cohort-conservative assumptions, treats each appropriately, and bounds the effect of the latter through the sensitivity analysis presented in §5.9. Every parameter used in the downstream MATLAB analysis is listed in Table 4 with its status and provenance, so the basis of every numerical result is fully traceable. A locked parameter is one for which the responsible cohort member has committed a formal numerical value with documentation in the powertrain or vehicle specification sheets. A cohort-conservative assumption is a value used in the

analysis where the responsible cohort member has not yet finalised the parameter at the time of analysis which is presented in Table 8.

Table 4: Cohort interface parameters used in the MATLAB analysis.

Parameter	Value	Status
Empty vehicle mass	230 kg	Locked
Driver mass allowance	68 kg	Locked
Total mass	298 kg	Locked
Wheelbase	1.56 m	Locked
Static ride height	38 mm	Locked
Tyre outer diameter	0.4115 m	Locked
Effective rolling radius	0.1955 m	Locked
Pack series count	108	Locked
Pack parallel count	5	Locked
Pack nominal voltage	388.8 V	Locked
Pack maximum voltage	453.6 V	Locked
Pack minimum voltage	270 V	Locked
Pack capacity	8.16 kWh	Locked
Pack peak DC current	325 A	Locked
Pack regen acceptance	2C (42 A)	Locked
Peak phase current	300 A_{RMS}	Locked
Continuous phase current	225 A_{RMS}	Locked
DC bus operating range	100–800 V	Locked

3.4 Architecture Context

The locked architecture into which the motor must integrate is established before motor selection itself, because the inverter, accumulator, and drivetrain configuration each impose constraints that narrow the candidate motor space. All three were selected by the responsible cohort member during the November proposal stage and are treated as fixed inputs.

The Cascadia PM150DZ inverter sets the upper bound on motor phase current at 300 A peak, regardless of the motor's intrinsic thermal capability. For a typical torque constant near 0.6 Nm/A, this corresponds to a peak shaft torque ceiling of approximately 180 Nm; any motor with a peak-current rating exceeding 300 A under-utilises its capability through this inverter (quantified in §5.1). The inverter's 100–800 V DC bus range comfortably spans the 270–453.6 V accumulator envelope, leaving voltage capability over-specified. Internal active discharge of the DC link capacitor below 50 V in under 1 s satisfies EV6.1.5 without additional circuitry, and ISO 26262 certification permits overcurrent protection to rely on programmable logic per the COTS exemption in EV3.2.5.

The 108S5P accumulator sets the bus voltage envelope across which the motor must operate. The 40% voltage swing from 270 V at end of discharge to 453.6 V at full charge is large in absolute terms and dynamically traversed during a single endurance event, distinguishing the application from typical road-EV problems where bus voltage may be treated as quasi-static. The 8.16 kWh capacity sets the energy envelope for the 22-lap endurance event. The 2C regen acceptance corresponds to 42 A maximum charge current at the pack terminals and is the binding regenerative-recovery constraint; the architectural consequence is developed in §5.8 and reframed as a cohort-level finding in §6.3.

The single-motor rear-wheel-drive configuration with open differential places all motor torque through a single mechanical reduction stage at the rear axle, eliminating the regen advantage of AWD in-wheel architectures (§2.6) in exchange for reduced mass, complexity, and cost. The motor itself, identified in §5.1 as the Emrax 228 MV LC, is introduced here as an architectural input rather than an outcome; its selection rationale appears in §5.1 only after Chapter 4's methodology, so the selection can be defended against the full simulation framework rather than as an a priori choice. The architecture is therefore: Cascadia PM150DZ inverter, 108S5P accumulator, single-motor RWD with open differential, 400 V class — with motor selection performed against this fixed reference frame.

3.5 Translation to Motor-Subsystem Targets

The motor-subsystem requirements emerge from the intersection of the §3.1 performance targets, §3.2 rule constraints, and §3.3 cohort interface parameters evaluated against the §3.4 locked architecture. Seven requirements are derived below and compared against the Emrax 228 MV LC datasheet in Table 5. Two derivations use values committed downstream: gear ratio $i_g = 3.60$ from the §5.4 sweep, and endurance-average shaft power ≈ 30 kW from the §5.6 simulation. Both are stable across the §5.9 sensitivity sweep. Peak shaft torque is set by the rear-axle traction limit at launch. At the 0.9 g target, rear-axle vertical load combines the static 60% rear contribution of 1754 N with longitudinal transfer of $m \times a \times h_{CG} / L = 298 \times 8.83 \times 0.375 / 1.56 = 632$ N, giving 2386 N peak rear-axle load where downforce is approximately zero. At $\mu = 1.30$, maximum tractable rear longitudinal force is 3102 N, requiring $3102 \times 0.1955 = 606$ Nm wheel torque and $606 / (3.60 \times 0.95) \approx 180$ Nm peak shaft torque. The inverter's 183 Nm cap ($300 A_{RMS} \times K_T = 0.61$ Nm/A) closely matches, consistent with the §5.1 selection rationale. Peak shaft power is set by acceleration target subject to the FS rule cap. At the 4.3 s target and ≈ 27 m/s finish-line speed, peak instantaneous shaft power demand reaches ≈ 75 kW including drag and rolling resistance. The 80 kW DC regulatory ceiling, derated at 92% motor efficiency, gives a 73.6 kW shaft cap — binding for high-SoC acceleration and governing peak shaft power. The motor must deliver this through the constant-power region, not just a single corner point.

Continuous shaft power is set by the endurance load. The §5.6 simulation gives event-average DC input ≈ 32 kW, equivalent to ≈ 30 kW shaft after losses. The motor's S1 rating must exceed this with margin for corner-exit excursions, against the $1.5\text{--}2\times$ peak-to-continuous ratio characteristic of PM motors at FS duty. Requirement: ≥ 30 kW shaft, used as a thermal-soft-limit boundary condition in §5.7. Maximum operating speed is set by top speed and gear ratio. At ≈ 30 m/s (108 km/h) on the longest endurance straight, $i_g = 3.60$ and $r = 0.1955$ m give ≈ 5300 RPM. A 20% margin against autocross corner-exit transients gives ≥ 6400 RPM, setting the constant-power-region upper bound.

Motor mass is set by cohort allocation against the 230 kg empty-vehicle target. The combined motor-plus-inverter allocation is ≈ 25 kg (inverter 10 kg, mounts ≈ 1.5 kg), leaving ≈ 13.5 kg for the motor — tight rather than generous, ruling out heavier oil-cooled axial-flux candidates. Cooling fluid and rating are set by T7.2.2 and T7.2.7: liquid-cooled with plain water, air, or oil (no water-glycol), rated for at least 120°C or maximum operating fluid temperature, whichever is higher. Voltage compatibility is set by the locked accumulator envelope. Back-EMF constant K_E , d-axis inductance L_d , and rated terminal voltage must support operation across $V_{\min} = 270$ V to $V_{\max} = 453.6$ V. The constraint binds in both directions: insufficient headroom at $V_{\min}$ prevents the inverter from driving rated phase current through winding inductance, while operation at $V_{\max}$ must not exceed continuous insulation rating..

Table 5: Motor-subsystem requirements derived in §3.5 against the Emrax 228 MV LC datasheet capability (Emrax, 2026).

Requirement	Target	Emrax 228 MV LC Capability	Margin
Peak shaft torque	≥ 180 Nm	220 Nm (S2 2-min)	+22%
Peak shaft power	≥ 75 kW	124 kW (S2)	+69%
Continuous shaft power	≥ 30 kW	64 kW (S1 continuous)	+113%
Maximum motor speed	≥ 6400 RPM	6500 RPM	+1.6%
Motor dry mass	≤ 13.5 kg	13.5 kg	0%
Cooling fluid	Plain water	Plain water (LC variant)	-
Voltage compatibility	270–453.6 V	100–630 V rated, 833 V tested	+39%

The motor capability margins in Table 5 quoted against the Emrax 228 MV LC's own rating are healthy across all seven requirements. Two of these margins are misleading at the system level and must be read with care. The peak-torque margin of +22% on the motor's own rating is collapsed to system-level parity by the inverter's 183 Nm cap; the binding peak-torque constraint is therefore the inverter selection in §3.4, not the motor itself. The peak-power margin of +69% on the motor's S2 rating is collapsed to a regulatory-bound 73.6 kW by EV2.2.1; the binding peak-power constraint is the FS rule, not the motor.

The maximum-speed margin of +1.6% is the tightest in the table and forms the basis for the corner-speed analysis in §5.2.

The combined picture in Table 5 is that the Emrax 228 MV LC meets every motor-subsystem requirement, that the binding system-level constraints on peak performance sit outside the motor at the inverter, FS rule, and accumulator interfaces, and that the motor's own thermal and speed envelopes are exercised but not exceeded by the FS duty cycle. The methodology in Chapter 4 develops the simulation framework that demonstrates each of these constraints quantitatively across the four FS dynamic events.

4. Methodology

This chapter describes the methodology by which the motor-subsystem design targets derived in Chapter 3 were tested. §4.1 frames the constraints that drove methodology selection. §4.2 documents the four-layer simulation architecture. §4.3 to 4.6 describe the physical models used for motor capability, efficiency, vehicle dynamics, and thermal behaviour. §4.7 describes the gear-ratio optimisation procedure. §4.8 to 4.10 describe the validation approach, the numerical assumptions, and the acknowledged limitations of the resulting framework. §4.11 describes how the methodology was scheduled to integrate with the cohort-wide design timeline.

4.1 Overview and Rationale

The motor-subsystem analysis required a methodology capable of predicting performance, energy consumption, thermal behaviour, and rule compliance across the four FSUK dynamic events from a small set of manufacturer-published parameters and cohort-supplied vehicle data. Three constraints drove methodology selection.

First, the work is undertaken at concept stage with no physical hardware available for measurement. This precludes empirical validation and forces reliance on first-principles modelling backed by manufacturer datasheet values, with sensitivity analysis substituting for direct measurement of the parameters most likely to differ between datasheet and as-built. Second, the breadth of analysis required demanded a modular simulation architecture in which each analysis builds upon validated lower-level functions rather than being implemented in isolation. Third, the deliverable depth required by the Engineering Design Score Sheet S6.1.5 for justified design decisions called for every numerical result to be traceable to its underlying assumptions, allowing the limitations of the methodology to be quantified rather than acknowledged.

The methodology adopted is therefore a quasi-static, physics-based simulation framework implemented in MATLAB, structured in four layers. This structure supports two properties critical to the assessment criteria. It allows any single result to be regenerated by re-running the relevant script, satisfying the reproducibility requirement of the Guide. It also forces every analysis to share a single consistent set of vehicle and motor parameters, eliminating the risk of intermediate inconsistencies between subsections of the work.

4.2 Simulation Architecture

The MATLAB project comprises fifteen library functions, twenty-one figure-generating scripts, and a single parameter file. The four architectural layers are summarised in Table 6.

Table 6: Four-layer simulation architecture used throughout this work.

Layer	Function
Parameters	Returns a nested structure containing every motor, inverter, pack, vehicle, aerodynamic, drivetrain, rule, thermal, and physical-constant value used in the analysis. Each value is annotated with its source.
Library functions	Implement the underlying physics: motor capability surface, regenerative envelope, lap-time integration, efficiency-map interpolation. Reusable across multiple analyses.
Event wrappers	Parameterise the library functions for the specific FS dynamic events. Return the metrics required for scoring.
Post-processing	Aggregate event-wrapper outputs, generate report figures, run gear-ratio sweeps and sensitivity analyses by overriding parameter-file fields at runtime.

The parameter file is the single point at which any numerical input may be modified. Derived quantities are computed within the parameter file itself rather than being recalculated in each script, preventing inconsistent assumptions across analyses. Both the gear-ratio sweep and the sensitivity analysis operate by overriding fields of this structure at runtime rather than by maintaining parallel parameter sets. Efficiency-map interpolation uses persistent caching of a precomputed scatteredInterpolant, which avoids repeated initialisation costs across the 144-simulation gear-ratio sweep.

4.3 Motor Capability Modelling

The motor capability surface was constructed in four constraint regions, with the binding constraint at any operating point being the minimum of all four. The inverter current limit defines a horizontal torque ceiling at all speeds below the corner point, computed as $T_{inv} = K_T \times I_{pk,inv} = 0.61 \times 300 = 183$ Nm. This sits below the motor's intrinsic 220 Nm S2 rating because the Cascadia PM150DZ's 300 A_{RMS} peak phase-current capability binds before the motor's own 360 A_{RMS} thermal capability is reached. The voltage envelope was modelled by linear interpolation of the three published K_v values at $I = 0$,

I_{cont} , and I_{peak} , computing the achievable phase current at the required RPM/V ratio and thence the torque via $T = K_T \times I$. This interpolation is a simplification of the full d-q axis flux-weakening behaviour and is estimated accurate to within $\pm 5\text{--}10\%$ in the field-weakening region. Full d-q modelling would require parameters not published by EMRAX and was rejected as offering precision unsupported by the data available.

The Formula Student 80 kW power ceiling required particular care. The rule applies to DC bus power measured by the Energy Meter under D9.4.1, not to motor shaft power. The relationship $P_{DC} = P_{shaft} / \eta$ means that capping shaft power at the rule limit allows DC input to rise to approximately 87 kW at typical 92% motor efficiency, exceeding EV2.2.1 by 7 kW under steady-state operation at the binding point. The capability function was therefore implemented with a default efficiency factor $\eta_{FS} = 0.92$ such that the shaft cap becomes $T_{FS} = (P_{TS,max} \times \eta_{FS}) / \omega$, ensuring DC input remains below 80 kW under all operating conditions. This correction is demonstrated quantitatively in §5.5.2 using a worst-case rolling endurance lap at full state of charge; the present subsection records only that the capability surface was constructed with the corrected shaft cap built in by default.

The capability surface was finally extended into a state-of-charge-aware adaptive controller. At full charge the FS rule binds before the motor's own voltage envelope; at discharged conditions the voltage envelope binds well below 80 kW and the rule becomes irrelevant. Rather than calibrating a fixed map at one operating voltage, the adaptive controller recomputes the envelope continuously from real-time bus voltage measurement. Three logical strategies were defined and simulated for comparison: a conservative controller using the V_{min} envelope as a fixed reference, a nominal controller using the V_{nom} envelope, and an adaptive controller using the instantaneous V_{OCV} . The methodological motivation for this comparison is the regime crossover problem described in §3.4 and quantified in §5.5.

4.4 Efficiency Map Construction

Two methods were considered for efficiency modelling: analytical decomposition into copper, iron, and mechanical losses with Steinmetz coefficients back-fitted to the manufacturer's published peak-efficiency point, and direct digitisation of the manufacturer's published efficiency contour map at typical FS bus voltage. The analytical method was rejected. The published peak-efficiency value is a single number, requiring two additional assumptions to fit the three Steinmetz parameters; the resulting map would carry the uncertainty of all three assumptions while delivering no information not already implicit in the published map. Direct digitisation was selected as the more honest approach. Six iso-efficiency contour lines from the Emrax 228 MV LC datasheet at 400 V_{DC} were sampled, shown in Figure 1:

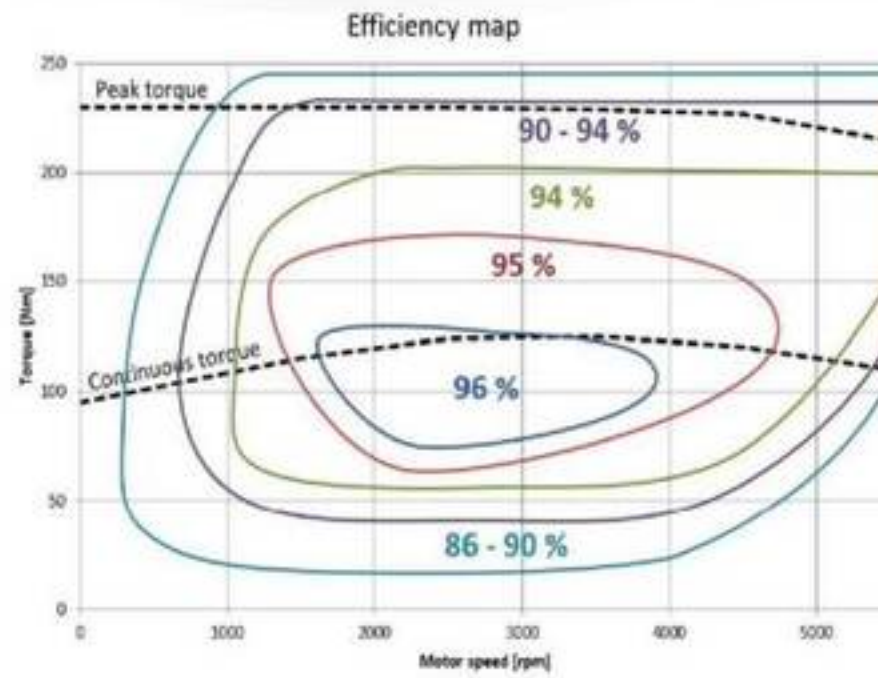


Figure 1: Emrax 228 MV LC Efficiency Contour Map

Twenty to thirty points each were sampled, producing a 6271-point dataset assembled into a MATLAB scatteredInterpolant. This interpolant returns efficiency for any (RPM, T) operating point within the motor's envelope and is queried by both the lap-energy calculations and the thermal model. The acknowledged limitation is the absence of decomposition into loss mechanisms; the model cannot be queried for the proportion of total loss attributable to copper losses at a given operating point. This was judged an acceptable trade-off because no analysis in this project required loss decomposition, while all analyses required accurate total-efficiency values which the digitised map provides directly from manufacturer data.

4.5 Vehicle Dynamics Modelling

The four FS dynamic events require fundamentally different modelling approaches because their binding constraints differ. The acceleration event is one-dimensional and dominated by traction and motor torque; skidpad is two-dimensional and dominated by lateral grip; autocross and endurance are spatial events along a defined track and require quasi-static forward-backward lap simulation. The event-specific models are summarised in Table 7.

Table 7: Vehicle-dynamics model used for each FS dynamic event

Event	Integration domain	Step size	Binding constraint	Rationale
Acceleration	Temporal	1 ms	Rear-axle traction at launch; motor torque thereafter	Newton's second law along a straight line; short duration justifies fine temporal step.

Event	Integration domain	Step size	Binding constraint	Rationale
Skidpad	Steady-state	n/a	Lateral tyre grip on R = 9.125 m inscribed circle	Single algebraic solve for V from μRg with downforce feedback; no integration required.
Autocross	Spatial	1 m segments	Lateral grip in corners, motor torque on straights	Quasi-static forward-backward pass on track centreline; resolves corner-entry braking by backward propagation.
Endurance	Spatial	1 m segments	Same as autocross plus bus-voltage tracking across SoC	Autocross simulator extended with per-lap $V_{OCV}(\text{SoC})$ update and regenerative-braking energy return.

The acceleration model integrates a one-dimensional Newton's-second-law equation of motion at 1 ms timesteps. At each step the available wheel force is the lesser of the rear-axle traction limit (from rear-axle vertical load including longitudinal weight transfer and aerodynamic downforce) and the motor torque transmitted through the gear ratio and drivetrain efficiency. Effective vehicle mass includes wheel and rotor inertia projected through gear ratio squared, giving 352.1 kg against vehicle mass 298 kg — an 18.2% rotational penalty significant for short-duration acceleration prediction.

The skidpad model assumes steady-state cornering at the inscribed circle of the FS-defined corridor ($R = 9.125$ m). Maximum corner speed follows from the lateral tyre-friction limit: $V^2 = \mu Rg(1 + 0.5\rho C_{LA} \cdot V^2/(mg))/m$ solved iteratively with downforce contribution to vertical load included. Motor torque demand of 6.5 Nm is predicted against 183 Nm available, confirming skidpad is lateral-grip-limited rather than power-limited; gear ratio is therefore a non-factor, as verified by the §5.4 sweep.

Autocross and endurance lap times are computed by quasi-static forward-backward simulation on a 1 m spatial grid along a track centreline. Tracks are synthesised from the FS rule envelope: autocross is 1300.5 m ($\approx 30\%$ straights, 40% constant-radius turns, 5% hairpins, 25% slalom at 10 m cone spacing); endurance uses 12 m slalom spacing and 1075 m lap length, producing $\approx 23,650$ segment samples across 22 laps. Quasi-static was chosen because dominant cornering timescales are steady-state lateral acceleration, not suspension transients; the 3–5% accuracy penalty against full transient simulation enabled the 144-simulation gear-ratio sweep to complete in seven seconds. For each 1 m segment, maximum corner speed is: $V^2 = \mu Rg / (1 - \mu \times 0.5 \times \rho \times C_{LA} / m)$ where the denominator captures the downforce-feedback effect. The forward pass propagates speed subject to available longitudinal force; the backward pass propagates braking deceleration from each corner-entry constraint subject to the friction-plus-downforce braking limit. The minimum of the two passes defines the achievable speed profile, integrated to give lap time and energy. Slalom segments reduce to an equivalent constant-radius turn through: $R_{\text{eff}} = (s^2 + 4 \times 1.5^2) / (8 \times 1.5)$ where s is cone spacing and 1.5 m is assumed lateral offset. The endurance model extends autocross to 22 consecutive laps with a four-minute driver change

after lap 11 (entry speed reset to zero). Bus voltage updates at the start of each lap from a linear approximation: $V = V_{\min} + (V_{\max} - V_{\min}) \times \text{SoC}$ with SoC depleted by integrated input energy of the previous lap. Regenerative recovery during decelerating phases is the minimum of the available regen envelope (set by motor current limit, bus voltage, and pack acceptance C-rate at instantaneous SoC) and the demanded braking force; the remainder is absorbed by friction brakes. Regen charge returned to the pack is reduced by drivetrain and motor efficiency in the regen direction, giving round-trip pack-to-pack efficiency of approximately 72%.

4.6 Thermal Modelling

Motor winding temperature was modelled as a single lumped thermal mass governed by the first-order ODE $C_{th} \times dT/dt = P_{loss} - UA \times (T_w - T_{coolant})$. The thermal capacitance C_{th} was estimated as motor mass times a copper/steel/aluminium blended specific heat capacity, giving $13.5 \times 480 = 6480 \text{ J/K}$. The convective coefficient UA was back-fitted from the manufacturer's S1 continuous rating: at S1 the motor delivers 64 kW at approximately 95% efficiency, dissipating 3.2 kW, and the steady-state winding temperature approaches the 100 °C sensor warning level against a 50 °C coolant inlet. $UA = P_{loss} / \Delta T$ therefore gives $3200 / 50 = 64 \text{ W/K}$. The model is integrated by forward Euler at 0.1 s timesteps using the per-instant heat-generation profile from the endurance simulation. The four-minute driver change is inserted with zero heat input, allowing the model to capture the thermal recovery period.

The single-node lumped approach carries three acknowledged limitations: it cannot resolve the temperature gradient between winding hot-spot, stator iron, and motor housing; the constant UA assumption ignores the temperature- and flow-dependence of real cooling jackets; and coolant inlet temperature is held constant at 50 °C rather than modelled as rising during operation. These simplifications would matter for accurate transient prediction at thermal limits, but are inconsequential here because the predicted peak winding temperature of 78.5 °C lies 39.3 °C below the absolute cutoff, and the simplifications affect this margin by no more than 5 to 10 °C per the LPTN-vs-experiment validation reported by Sequeira et al. (2022).

4.7 Design Optimisation

The motor-to-wheel gear ratio was selected by exhaustive sweep across 36 values from $i_g = 2.5$ to 6.0 in 0.1 increments. At each ratio, all four event simulators were executed and their predicted times converted into FSUK Concept Class scoring per the formulas in S5/D4–D7, using each event's own minimum simulated time as the reference T_{min} . The total LTS-relevant score was computed as the sum of acceleration (75 max), skidpad (75 max), and sprint/autocross (100 max) scores. Endurance was

excluded from the optimum-selection criterion because the endurance scoring formula is dominated by total-time differences across the long event duration, which over-weights the optimisation toward lower ratios at the cost of the shorter events. The endurance result is reported separately for reference in §5.4.

4.8 Validation Approach

Direct empirical validation against measured data is not possible at concept stage. Two indirect validation methods were therefore applied. First, simulated event times were compared against the published 2025 FSUK winning times. The simulated acceleration time of 4.256 s agrees with the Oxford Brookes winning time of 4.284 s to within 0.7%, well within the methodology's stated accuracy bound. The simulated skidpad time of 5.217 s is faster than the UPV/EHU winning time of 5.826 s by approximately 10%, indicating that the conservative $\mu = 1.30$ assumption may be slightly optimistic relative to the friction coefficient achievable on the Silverstone surface; the §5.9 sensitivity analysis confirms that μ is the dominant uncertainty source. The simulated autocross time was compared qualitatively because the FS rules permit each entrant a different representative track layout, making absolute comparison between teams unsound. Second, a one-at-a-time sensitivity analysis was conducted in which each major assumed parameter (tyre friction, vehicle mass, drag area, downforce area, drivetrain efficiency, rolling resistance, wheel rotational inertia) was swept between conservative and aggressive values. The resulting tornado plot, presented in §5.9, identifies tyre friction as the dominant uncertainty, with all other parameters contributing below two seconds of endurance lap-time variation each. The methodology is therefore robust to most cohort-pending values and the principal residual uncertainty is correctly localised.

4.9 Assumptions

The numerical assumptions on which the analysis depends are summarised in Table 8, with their source and the variation explored in the sensitivity analysis. Manufacturer-published values for the Emrax 228 MV LC and Cascadia PM150DZ are taken as ground truth and are not listed as assumptions; they appear in the parameter file with full citation to the source datasheet.

Table 8: Numerical assumptions used in the simulation framework

Parameter	Value	Source	Sensitivity range explored
Tyre coefficient of friction μ	1.30	Concept-stage assumption pending suspension tyre-data refinement	1.20 to 1.50
Drag area C_{dA}	1.00 m ²	Concept-stage assumption pending full-vehicle CFD	0.80 to 1.20

Parameter	Value	Source	Sensitivity range explored
Downforce area C_{LA}	1.50 m ²	Concept-stage assumption pending full-vehicle CFD	1.00 to 2.50
Vehicle mass	298 kg	Group-level cohort estimate	280 to 320 kg
Drivetrain efficiency	0.95	Single-stage spur reduction estimate	0.92 to 0.96
Rolling resistance C_{rr}	0.020	Standard motorsport-tyre estimate	0.015 to 0.025
Wheel rotational inertia	0.40 kg·m ² each	Standard 13" lightweight wheel estimate	0.25 to 0.50
Pack regen acceptance	2C (42 A)	Conservative Sony VTC6-class baseline pending accumulator cell selection	1C to 5C explored separately
$V_{OCV}(\text{SoC})$ relation	Linear V_{min} to V_{max}	Approximation pending pack characterisation	Not varied
Motor thermal capacitance	6480 J/K	Mass × blended specific heat capacity	Not varied
Motor convective coefficient UA	64 W/K	Back-fit from manufacturer S1 point	Not varied

4.10 Methodology Limitations

The principal limitations of the methodology and their implications for result interpretation are summarised in Table 9. Each was judged acceptable in aggregate because none alters the qualitative conclusions of the analysis, and the principal quantitative uncertainty has been explicitly bounded through the sensitivity analysis described in §4.9 and presented in §5.9.

Table 9: Acknowledged methodology limitations and their quantitative implications for result interpretation.

Limitation	Implication
Voltage-limited region uses linear K_v interpolation rather than full d-q model	±5–10% torque accuracy in the field-weakening region
Quasi-static lap simulator with no transient suspension dynamics	±3–5% lap-time accuracy against full transient simulation
Synthesised representative tracks rather than actual Silverstone layouts	Comparison to real winner times is qualitative, not quantitative
Single-node lumped thermal model	Cannot resolve winding hot-spot vs housing gradient (~5–10 °C effect)
Linear $V_{OCV}(\text{SoC})$ approximation	Real Li-ion has flatter mid-SoC plateau and sharper extremes
No internal-resistance voltage sag in real-time controller	~25 V sag at 75 A typical, not modelled in capability surface
Pack regen acceptance held constant across event	Real cells may derate regen current at extreme SoC or temperature
Round-trip motor efficiency uses drive-side η -map for both directions	Real regen efficiency typically 1–3% lower than drive at same operating point

4.11 Cohort Integration and Methodology Schedule

The methodology was structured to permit early delivery of motor-derived numbers to dependent cohort members while accepting their inputs incrementally as they became available. Cohort dependencies were established at the November proposal stage and managed through the locked-vs-conservative parameter discipline described in §3.3.

Tyre friction, vehicle mass, and aerodynamic coefficients were required at the start of methodology development to permit the vehicle-dynamics models to function. In their absence, the conservative concept-stage assumptions documented in Table 8 were used and the resulting uncertainty was bounded by the sensitivity analysis. Pack open-circuit voltage curve, internal resistance, and regen acceptance current were required for the SoC-aware controller and the regen analysis. Pending the accumulator electrical lead's final cell selection, the conservative 2C acceptance baseline was used and the recovery sensitivity to higher C-rates was explicitly characterised, generating the cohort-level finding developed in §5.8 and §6.3.

Numbers delivered to other cohort members were sequenced to match their consumption. Motor mass and casing dimensions were finalised within the parameter file by week two of project execution to permit the chassis and packaging leads to commence layout work. The motor torque-speed envelope was completed before the gear-ratio decision was required by the drivetrain lead. Thermal heat-load profiles were delivered to the thermal-management lead once the endurance simulator was validated.

5. Results and Discussion (Individual)

This chapter presents the analytical findings of the motor-subsystem investigation. The structure follows the design narrative from initial selection through to final integrated performance. §5.1 establishes the selected motor and the rejection rationale for alternatives. §5.2 characterises the motor capability surface that bounds every subsequent analysis. §5.3 reports baseline vehicle performance at a starting gear ratio of 4.20, against which the gear-ratio optimisation in §5.4 is evaluated. §5.5 develops the SoC-aware adaptive controller, which is the principal innovation contribution of this work. §5.6 reports the mid-design performance achieved with the optimised gear ratio and adaptive controller in place. §5.7 layers in the thermal envelope and §5.8 layers in the regenerative behaviour. §5.9 then consolidates the final integrated headline numbers covering top speed, peak power, peak torque, and event timings. §5.10 reports the input-parameter sensitivity analysis that bounds the cohort-pending uncertainties. §5.11 consolidates the hardware specification decisions that complete the build-ready package.

5.1 Motor Selection Outcome

The motor selection process evaluated five commercially available motorsport-grade machines against a four-stage screening procedure derived from the architectural and rule constraints established in Chapter 3. Table 10 summarises the longlist specifications. Each candidate was screened in sequence against compatibility with the locked Cascadia PM150DZ inverter (300 A_{RMS} peak phase current, 100 to 800 V DC bus operating range, resolver position feedback), compatibility with the locked 108S5P accumulator voltage envelope (270 to 453.6 V), compliance with FS rule T7.2.2 governing coolant type, and availability of the manufacturer-published electrical parameters required for envelope construction under the K_V interpolation method described in §4.3. Two candidates failed at the screening stage and were eliminated before performance ranking.

Table 10: Candidate motor specifications evaluated for the BMM-A motor subsystem. All values from manufacturer datasheets.

Parameter	Emrax 228 MV LC	Emrax 208 MV LC	YASA P400 R	Phi PHI27	ASRUK AIM- Lbe
Topology	Axial flux PMSM	Axial flux PMSM	Axial flux PMSM	Axial flux PMSM	Axial flux inrunner
Mass [kg]	13.5	10.3	28.2	20.0	12.4
Casing OD [mm]	228	208	305	270	260
Axial length [mm]	85	~80	106.7	106	80
Max RPM	6500	7000	8000	12,000	5000
Peak torque (motor) [Nm]	220 (S2)	150	370 (@450 A)	240	390
Continuous torque [Nm]	112	84	200	140	N/A
Peak power (motor) [kW]	124 (S2)	86	160 (@700 V)	150	170
Continuous power [kW]	64	52	60	90	120

The Phi Motors PHI27 and ASRUK AIM-Lbe were both eliminated on T7.2.2 coolant grounds: the PHI27 specifies water-glycol as standard coolant, and the AIM-Lbe a 50/50 water-glycol mix, neither permitted under the rule's plain water, air, or oil restriction. The AIM-Lbe also failed the parameter-availability screen — peak phase current, phase resistance, phase inductance, back-EMF constant, and position sensor type were all absent from the datasheet. Plain-water variants may exist as configurable options, but neither was documented in publicly available data.

The YASA P400 R-Series passed the coolant screen as oil-cooled but introduced cohort-level concerns. An oil-cooled motor requires a second cooling loop independent of the water-cooled inverter loop, with mass, packaging, pump-count, and catch-can penalties per T7.2.6. Its 28.2 kg dry mass exceeds the cohort budget of 13.5 kg (§3.5), and its 450 A_{RMS} rated peak phase current cannot be fully exploited through the Cascadia's 300 A_{RMS} limit, capping peak torque at approximately 247 Nm against the published 370 Nm. The YASA was eliminated as a poor system-level fit despite competitive standalone performance.

The Emrax 208 MV LC passed every screening criterion and was retained for direct comparison against the 228 MV LC. The two share architectural family, liquid cooling, KTY 81/210 thermal sensor, TAMAGAWA TS2620N21E11 resolver, and similar mounting interfaces. The 208 is 3.2 kg lighter, but its lower torque constant (0.38 vs 0.61 Nm/A) caps peak torque through the Cascadia at 114 Nm against the 228's 183 Nm. At the LTS-optimum gear ratio of 3.60 and drivetrain efficiency 0.95, this delivers wheel torque of 411 Nm against the 228's 626 Nm. The rear-axle traction limit of 606 Nm at launch sits between the two: the 208 leaves 32% of rear-axle grip unexploited, while the 228's wheel torque is fully exploitable through controller clipping. Combined with the 208's 1167 Hz peak electrical frequency sitting at the upper edge of Cascadia control bandwidth, the smaller motor was rejected.

The Emrax 228 MV LC therefore emerges as the selection: an axial-flux PMSM of 13.5 kg dry mass with 220 Nm peak torque and 124 kW peak power (S2 2-min rating), liquid-cooled per T7.2.2, factory-fitted with TAMAGAWA resolver and KTY winding temperature sensor, and rated across the full 270–454 V envelope of the locked accumulator. Every alternative was eliminated on cohort-compatibility (coolant, mass, second-loop) or inverter-system fit (peak phase current under-utilisation, electrical bandwidth). The methodology addresses the AHP critique developed in §2.3: candidate ranking emerges from quantitative simulation against the locked architecture rather than subjective pairwise weighting, with binding-criterion identification preceding performance scoring. The capability surface of the selected motor is presented in §5.2.

5.2 Motor Capability Surface

The motor capability surface bounds every subsequent analysis in this chapter. §5.2.1 establishes the ideal envelope as published by the manufacturer; §5.2.2 reduces this ideal envelope to the usable region available at nominal bus voltage after the four physical and regulatory constraints identified in §3.4 are applied; §5.2.3 reports the efficiency map within the usable region and quantifies its alignment with the duty cycle.

5.2.1 Ideal Envelope

Figure 2 presents the ideal motor envelope: peak torque 219.6 Nm flat from standstill to a corner speed of 5392 RPM, beyond which constant peak power of 124.0 kW is held to the manufacturer-set speed limit of 6500 RPM, at which torque has fallen to 182.2 Nm. The envelope describes the Emrax 228 MV LC in isolation against its own electromagnetic and mechanical constraints, with no inverter, accumulator, or regulatory boundary applied.

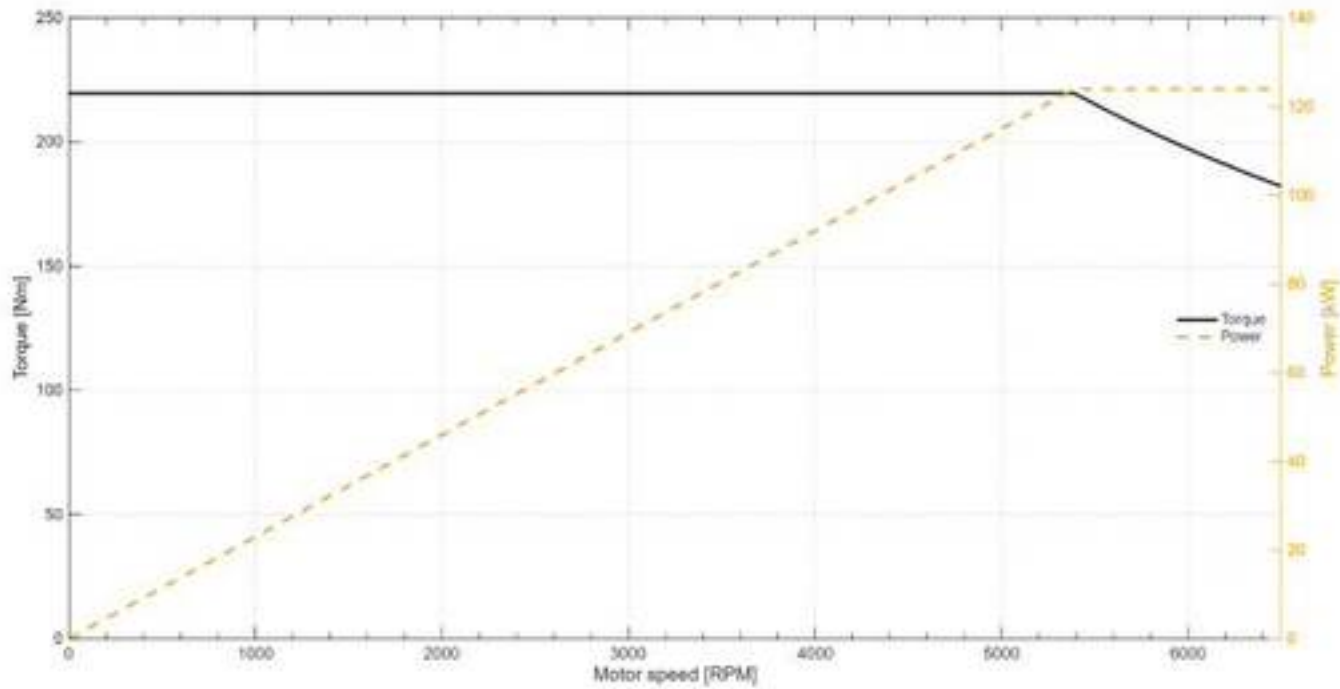


Figure 2: Ideal motor capability envelope of the Emrax 228 MV LC.

Two features of the ideal envelope determine the analytical framework adopted for the rest of this chapter. First, the corner speed of 5392 RPM sits well above the 4000 RPM region in which the vehicle spends most of its time at the LTS-optimum gear ratio established in §5.4. The motor is therefore intrinsically a constant-torque machine across the dominant operating region of the duty cycle, a property the gear-ratio sweep result in §5.4 depends upon and which would invert if the corner shifted into the typical operating range. Second, the peak power of 124 kW exceeds the FS regulatory ceiling of 80 kW DC by 55%. The motor's own thermal and electromagnetic envelope is therefore not the binding constraint on system-level peak performance. This is consistent with the SPM characterisation in Pellegrino et al. (2012) reviewed in §2.1. SPM-class machines deliver a high but finite peak-power region, and how much of that region is actually usable in a regulated FS application comes down to external constraints rather than the motor itself. These constraints are the inverter sizing, the bus voltage, and the regulatory cap. The same pattern shows up throughout §5 and is the physical reason the SoC-aware adaptive controller in §5.5 has a problem worth solving.

5.2.2 Real Usable Envelope

Figure 3 presents the usable envelope at nominal accumulator voltage $V_{nom} = 388.8$ V, with the ideal envelope from §5.2.1 overlaid for comparison. The four constraints identified in §4.3 reduce the operating region from ideal capability to physically realisable command space. Each binds in a specific speed range, and the overall envelope is the minimum of all four at every speed.

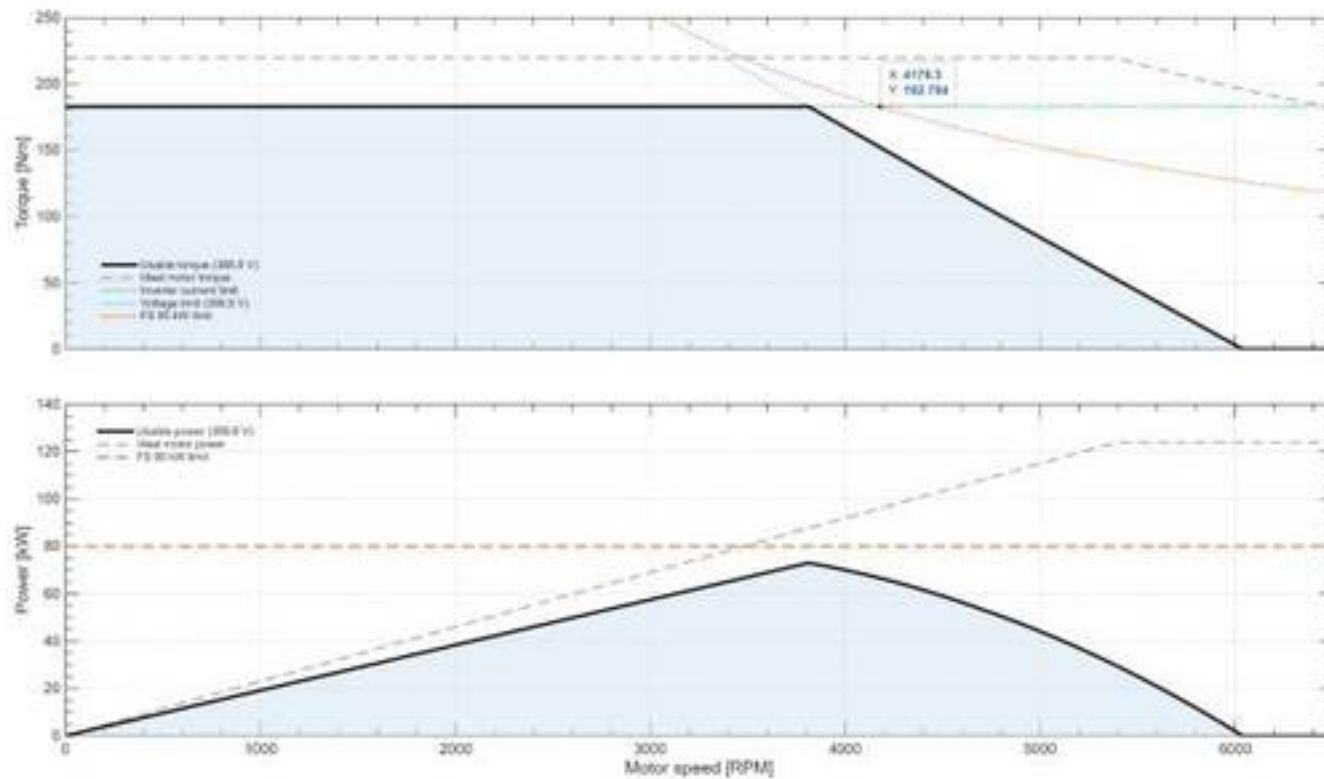


Figure 3: Real usable envelope of the Emrax 228 MV LC at nominal accumulator voltage $V_{nom} = 388.8$ V (shaded). Upper panel: shaft torque versus motor speed. Lower panel: shaft power versus motor speed with the FS 80 kW DC ceiling shown for reference.

The inverter peak phase current of $300 A_{RMS}$ caps shaft torque at 183 Nm across the entire constant-torque region, a 17% reduction from the motor's intrinsic 219.6 Nm capability. This cap is an architectural consequence of the inverter sizing decision documented in §3.4, not a motor limitation. The motor itself can sustain $360 A_{RMS}$ phase current at the peak rating; the Cascadia PM150DZ was selected for thermal envelope, packaging, and cost compatibility with the FS application rather than to match the motor's full peak capability. The consequence is that the motor operates well within its own thermal stress limits at all times, demonstrated in §5.7 as a 39.3°C margin to absolute thermal cutoff during the worst-case endurance event.

The voltage-limited region begins at 3812 RPM at V_{nom} , substantially below the ideal envelope's 5392 RPM corner. This shift reflects the physical reality that at 388.8 V there is insufficient bus voltage to overcome motor back-EMF while driving the required phase current through the winding inductance. Beyond this corner the controller transitions into the field-weakening region described by Lu and Kar

(2010) in §2.4, and the available torque decreases monotonically toward the speed limit. The K_V interpolation method used to compute the field-weakening region carries the $\pm 5\text{--}10\%$ accuracy bound documented in §4.3; this corresponds to approximately ± 150 RPM uncertainty on the corner location at V_{nom} , smaller than the speed-range resolution required for any downstream conclusion in this chapter, and identified as the priority dynamometer-validation activity in §8.

Peak shaft power at V_{nom} is 73.0 kW at the corner point. Translated to the DC bus through the motor's 92% efficiency at this operating point, the corresponding DC bus power is approximately 79.3 kW, sitting 0.7 kW below the FS 80 kW DC ceiling. The FS rule therefore does not bind at V_{nom} : the voltage-limited motor envelope itself is the operative ceiling, and the rule sits at the very edge of the usable region rather than across it. The FS limit appears in Figure 3 only as a reference curve.

This near-binding-but-non-binding behaviour at V_{nom} is the engineering finding that motivates the entire SoC-aware adaptive controller analysis in §5.5. As bus voltage rises toward $V_{max} = 453.6$ V at full charge, the corner shifts upward and the FS rule begins to bind across part of the operating range. As voltage falls toward $V_{min} = 270$ V at low SoC, the voltage envelope falls further below the rule and peak available power drops below 51 kW. A controller calibrated to a single bus voltage cannot represent both regimes correctly. The four-region structure visible in Figure 3 (inverter-bound below the corner, voltage-bound above the corner, FS-bound only at high SoC, mechanical-speed-limit at the upper bound) is the physical foundation of the regime-crossover problem resolved by the adaptive controller in §5.5.

The shaded usable region in Figure 3 represents the operating envelope within which any commanded (torque, RPM) pair is physically achievable at V_{nom} . All subsequent analyses in this chapter, including acceleration prediction (§5.3), gear-ratio optimisation (§5.4), thermal transient (§5.7), and regenerative behaviour (§5.8), operate on commands lying strictly within this envelope, recomputed for instantaneous bus voltage at every simulation step rather than evaluated against a single fixed reference.

5.2.3 Efficiency Map

Figure 4 presents the motor efficiency map across the operating region, digitised from the manufacturer's published contour map at 400 V_{DC} by the methodology of §4.4. The peak-efficiency plateau of 96% sits between approximately 1500 and 4500 RPM at 90 to 150 Nm, a broad central region characteristic of a well-designed permanent-magnet synchronous machine. Efficiency degrades in three directions: at low torque it falls through the 88% and 82% bands as fixed iron and mechanical losses dominate the small useful output; at low speed it falls slightly because the torque-to-current ratio is

unfavourable below approximately 500 RPM; and along the inverter current limit at the top of the operating region it falls through 94% and 92% as copper losses scale with the square of phase current.

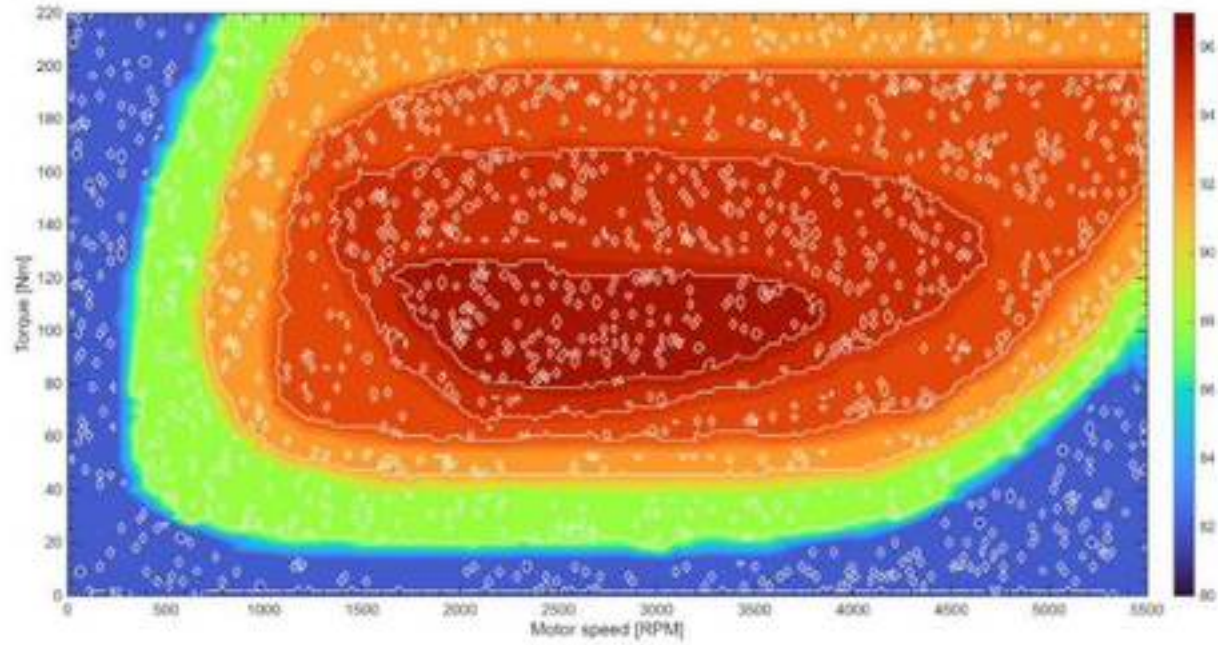


Figure 4: Digitised motor efficiency map across the (RPM, torque) operating plane at 400 V_{DC}

The position of the peak-efficiency plateau relative to the duty cycle is the most consequential feature of this map for downstream analysis. Two aspects of the duty cycle align directly with the plateau. The gear ratio of 3.60 places the typical endurance cruise speed of approximately 50 km/h at 2300 RPM, and §5.6.2 demonstrates quantitatively that this operating point falls within the 95–96% efficiency region. Corner-exit acceleration peaks during autocross and endurance laps occur at moderate speed and high torque, also within the 94–95% region. The motor is therefore well-matched to the application not only in capability but in efficiency: the energy-weighted analysis in §5.6.2 confirms that approximately 73.98% of all energy passing through the motor during the endurance event traverses operating points with $\eta \geq 92\%$, with an energy-weighted overall efficiency of 91.4%.

The contrary outcome, operation predominantly in the lower-left of the map below the 88% band, would indicate a motor either too large for the application (operating points clustered at low load) or too small (operating points clustered at the inverter cap). Neither applies to the Emrax 228 MV LC at the selected gear ratio. The efficiency map provides evidence that the motor selection is well-matched to the application. It backs up the capability-based argument in §5.2.2 with an energy-efficiency argument rooted in manufacturer measurement data rather than in datasheet ratings alone. The independent Emrax-class dyno validation reported by Ivanov et al. (2022) reviewed in §2.1, in which measured shaft torque exceeded the published continuous rating during bench testing, supports treating the digitised efficiency map as a defensible simulation input for the lap-energy and thermal-load calculations that depend on it.

5.3 Vehicle Performance at Baseline

This section establishes the predicted vehicle performance across the three timed dynamic events at the baseline gear ratio of $i_g = 4.20$ and bus voltage $V_{nom} = 388.8$ V. These results form the reference against which the gear-ratio optimisation in §5.4 is evaluated. The operating-point profiles surfaced here also motivate three findings developed in later sections: the traction limit on acceleration that prevents larger motors from being beneficial (§5.4), the lateral-grip limit on skidpad that makes gear ratio a non-factor for that event (§5.4), and the regime-crossover behaviour that the adaptive controller resolves (§5.5).

Figure 5 presents the four-channel acceleration trace for the FS 75 m standing-start event, plotting velocity (km/h), distance (m), motor shaft torque (Nm), and motor shaft power (kW) against time. The vehicle covers the distance in 4.421 s, reaching 97.2 km/h at the finish line. Peak longitudinal

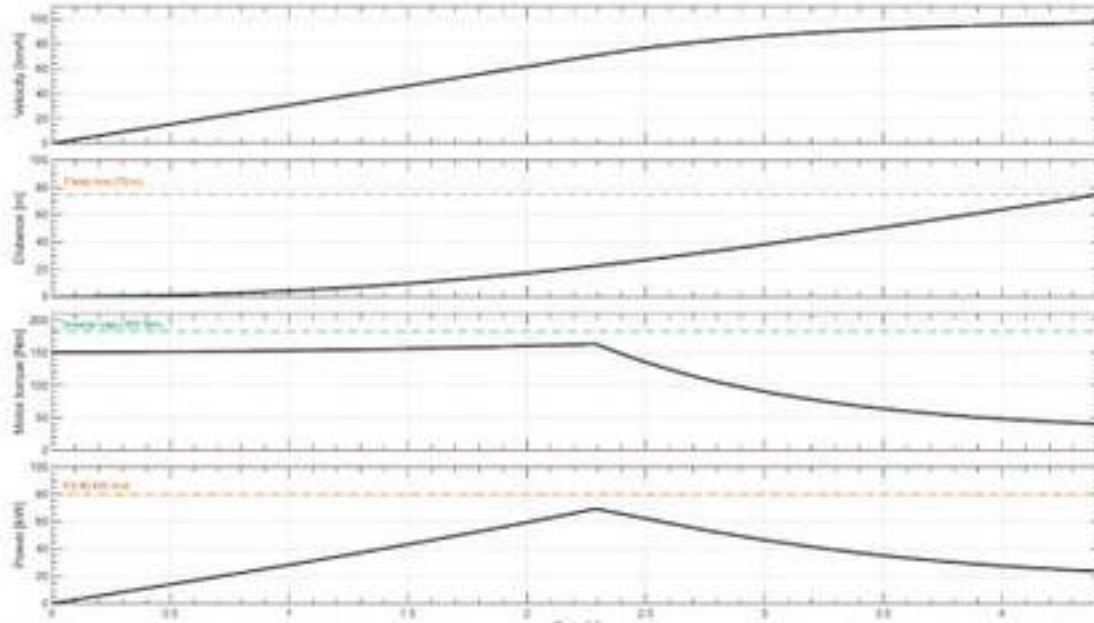


Figure 5: Four-channel acceleration trace for the FS 75 m standing-start event at baseline $i_g = 4.20$ and V_{nom} . Panels (top to bottom): vehicle velocity (km/h), distance covered (m) with 75 m finish line indicated, motor shaft torque (Nm) with 183 Nm inverter cap, and motor shaft power (kW) with 80 kW FS rule shown for reference. Simulated time 4.421 s

acceleration is 0.88 g (8.65 m/s²), peak motor power is 69.3 kW, and peak motor torque is 163.6 Nm. The 183 Nm inverter cap is shown as the dashed reference in the torque panel: the vehicle therefore never demands the motor's full available torque during this event.

The shape of the torque trace is the central engineering observation. Rather than sitting flat at the inverter cap throughout the constant-torque region, as it would for an unconstrained motor, the commanded torque rises gradually from 150 Nm at launch to a peak of 163.6 Nm at 2.3 s, then decays as the motor enters the voltage-limited region beyond 4000 RPM. The reason is that 51.8% of the run is traction-limited rather than motor-limited: at low speed the rear-axle vertical load is insufficient to carry the full $183 \text{ Nm} \times i_g / r_{eff} = 3370$ N driving force without exceeding the tyre friction limit, so the controller is rate-limited by available grip rather than by motor capability. As longitudinal weight transfer onto the rear axle increases with acceleration, the traction limit rises, and the controller

commands progressively more torque until the motor itself becomes the binding constraint at the top of the constant-torque region.

This 51.8% traction-limited fraction is significant for two downstream design decisions. First, further increasing peak motor torque would not improve acceleration: the rear tyres are already saturated through half the run. The motor is therefore correctly sized against the traction limit, not undersized, and this finding is the quantitative basis for the §5.1 rejection of larger candidate motors even where their datasheets showed higher peak torque than the Emrax 228. Second, the peak DC power of 69.3 kW sits 13% below the FS 80 kW regulatory ceiling, confirming that the regulatory rule does not bind at all during acceleration at V_{nom} , even before the SoC-aware controller architecture in §5.5 is applied. Combined, these two observations establish that acceleration is grip-limited at the rear axle, not power-limited at the inverter, which sets the upper bound on what any motor-side optimisation can deliver for this event.

The 4.421 s baseline time is 0.137 s slower than the 2025 FSUK acceleration winner (Oxford Brookes, 4.284 s), a 3.2% deviation. This sits above the 1.17% LTS-versus-competition error bound reported by Doyle et al. (2019) and reviewed in §2.2, but the comparison is not directly analogous: Doyle's accuracy bound was achieved against the same vehicle whose datasheet inputs fed the simulation, while the present comparison is against a competitor whose parameters are not available. The 3.2% gap is therefore expected to be larger than Doyle's intra-vehicle benchmark and is consistent with the quasi-static methodology's ± 3 –5% accuracy bound documented in §4.11. The deviation motivates the gear-ratio optimisation in §5.4, which is shown to recover this 137 ms deficit and place the predicted acceleration time inside the 2025 winner's envelope.

5.4 Gear Ratio Optimisation

The motor selection in §5.1 and the capability surface analysis in §5.2 establish what the Emrax 228 MV LC can deliver electromagnetically. The baseline performance in §5.3 establishes how the vehicle responds to that capability through a single chosen gear ratio of $i_g = 4.20$. This section closes out the mechanical-side design decision by sweeping the motor-to-wheel reduction ratio across its full manufacturable range, locating the gear ratio that maximises the FS dynamic event score, evaluating the sensitivity of that optimum to cohort-pending parameters, and identifying the residual performance limitation that the gear ratio alone cannot resolve.

The methodology described in §4.8 was executed across 36 gear ratios spanning $i_g = 2.5$ to 6.0 in 0.1 increments. Each ratio was evaluated through all four event simulators with all other parameters held at the Table 4 baseline values. Acceleration, skidpad, and sprint/autocross scores were summed into a

250-point LTS-weighted total per the FS Concept Class scoring formulas in S5/D4–D7, with each event's own minimum simulated time used as the reference T_{min} . Endurance was excluded from the optimum-selection criterion because the long event duration causes the endurance scoring formula to over-weight the optimisation toward lower ratios at the cost of the short events; the endurance result is reported separately for reference.

5.4.2 LTS-Weighted Combined Optimum

Figure 8 presents the LTS-weighted combined score curve alongside the per-event score contributions.

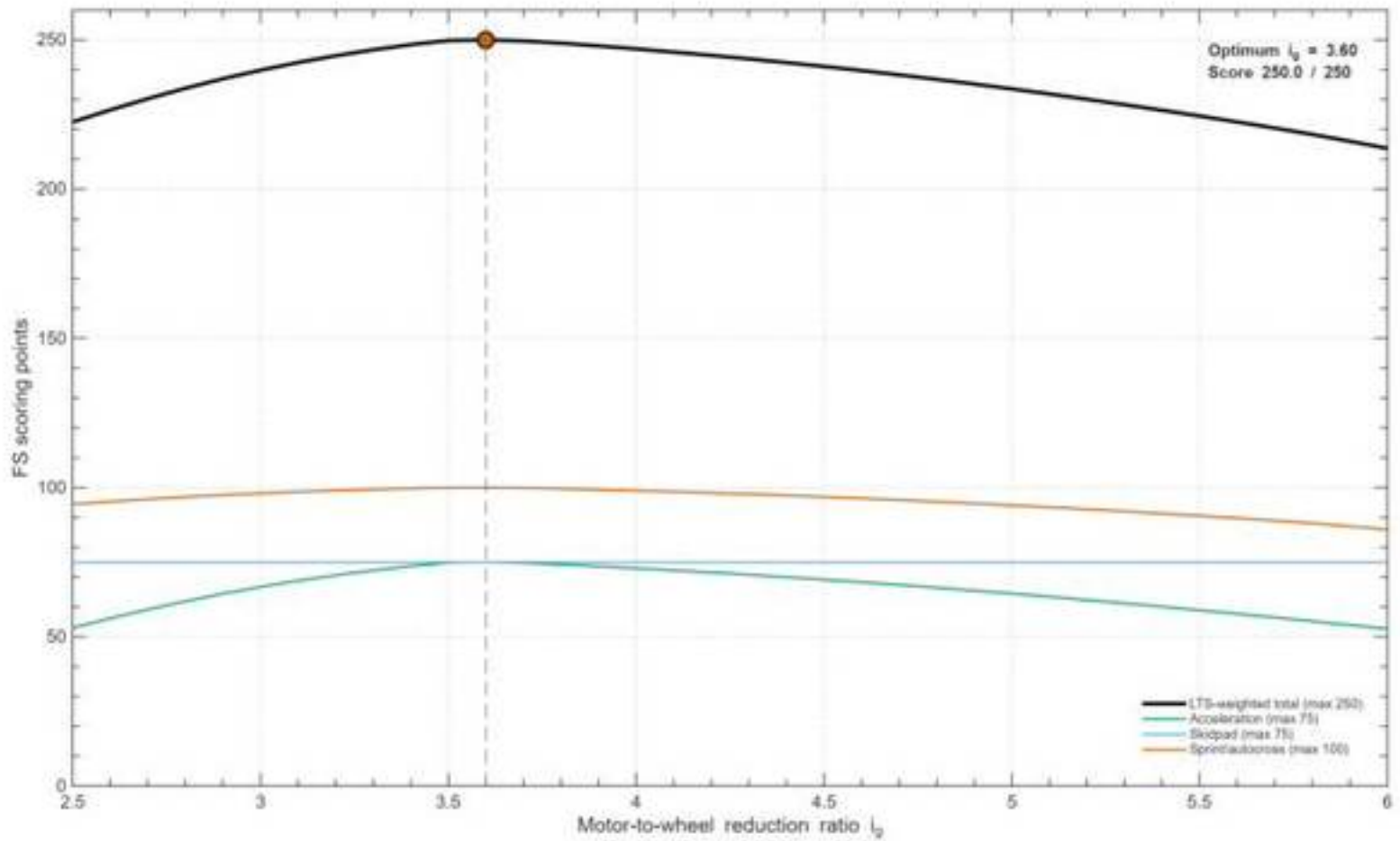


Figure 8: LTS-weighted total FS Concept Class score (black, maximum 250) and per-event score contributions (acceleration, skidpad, sprint/autocross) versus motor-to-wheel gear ratio. The combined score peaks at 250.0/250 at $i_g = 3.60$. Score above 240/250 is retained across the range $i_g = 3.10$ to 4.50.

The total score reaches 250.0/250 at $i_g = 3.60$, with all three counted events scoring full marks at this ratio: acceleration scores 75.0/75 at 4.315 s, skidpad scores 75.0/75 at 5.217 s, and sprint/autocross scores 100.0/100 at 78.675 s. The simultaneous achievement of maximum scores on all three events at a single gear ratio is a structural consequence of the FS scoring formula rather than a fluke of the simulation: the formula awards each event's full score to the team with the lowest simulated time, and the simulation establishes that lowest time at exactly the ratio that minimises both the acceleration and the autocross times. Skidpad is invariant and therefore contributes its full score at every ratio in the sweep.

5.4.3 Final Selection and Sensitivity to Cohort Inputs

The selected gear ratio is $i_g = 3.60$. This ratio sits at the LTS-weighted optimum, achieves full scores on all three counted events, falls within the 99%-retention window, and is centred within the 95%-retention window providing symmetric tolerance margin against drivetrain implementation. The selection is presented to the drivetrain lead as a total motor-to-wheel reduction; the drivetrain lead is free to implement this ratio as any combination of primary reduction and final drive that yields the overall figure, with the simulation results reported here remaining valid for any such implementation provided the combined drivetrain efficiency assumed in Table 4 is met. The endurance rolling-lap time at the selected ratio with the V_{nom} fixed envelope is 61.556 s, with a 22-lap total event time of approximately 1596 s and endurance drive energy consumption of approximately 6.84 kWh. This leaves approximately 16% margin against the 8.16 kWh pack capacity before adaptive control and regenerative recovery are applied; both refinements develop this baseline in §5.5 and §5.8 respectively. The dominant sensitivity in the gear-ratio result is to the tyre coefficient of friction μ , identified in §5.9 as the dominant uncertainty parameter overall. A higher μ shifts the acceleration optimum toward longer ratios because the rear axle saturates later, allowing more of the run to operate against the motor torque limit. A lower μ shifts the optimum toward shorter ratios because the rear axle saturates earlier and the run becomes increasingly traction-limited regardless of motor capability. The 1.4-unit width of the 240+ score plateau is wide enough to absorb the ± 0.15 expected refinement window of μ when the suspension lead's tyre data is finalised, with the optimum remaining within the high-score plateau even at the conservative end of that range. The cohort-conservative choice of $\mu = 1.30$ places the optimum mid-plateau, providing margin in both directions.

5.4.4 The Residual Limitation

The gear-ratio sweep optimises the mechanical reduction at fixed bus voltage. It does not address the binding constraint identified in §5.2.2 that varies dynamically through the endurance event: bus voltage falls from $V_{max} = 453.6$ V at full charge to $V_{min} = 270$ V at end of discharge, and the effective motor envelope contracts with it. At any single gear ratio, the controller is forced to choose between calibrating against the high-voltage envelope and under-utilising the motor at low SoC, calibrating against the low-voltage envelope and forfeiting available power at high SoC, or running a fixed compromise that is suboptimal across the entire SoC range. The gear ratio sets the operating point on the torque-speed plane; it cannot adapt the available envelope as the bus voltage changes through the event. This residual limitation is the central motivation for the SoC-aware adaptive controller developed in §5.5, which extends the fixed-map approach implicit in the present sweep into a controller that reshapes the torque-cap surface in real time as a function of instantaneous bus voltage and pack state of charge.

5.5 SoC-Aware Adaptive Controller

The gear-ratio optimisation in §5.4 fixes the mechanical reduction; the capability surface in §5.2 fixes the electromagnetic envelope at any single bus voltage. Neither addresses the fact that the bus voltage itself varies dynamically through the endurance event, and with it the effective motor envelope. This section develops the controller that resolves the resulting regime crossover. Section 5.5.1 establishes the underlying envelope-versus-rule problem; §5.5.2 specifies the controller architecture and documents the non-trivial DC-input-cap finding that emerged from rule analysis; §5.5.3 quantifies the controller's benefit against two fixed-map alternatives across a full 22-lap endurance simulation; §5.5.4 walks through the SoC trajectory to demonstrate the regime switch in operation; §5.5.5 positions the contribution against the published flux-weakening literature reviewed in §2.5.

5.5.1 The SoC Envelope Problem

Figure 9 presents the family of motor torque-speed and power-speed envelopes at three representative bus voltages spanning the locked accumulator operating range: $V_{max} = 453.6$ V at full charge, $V_{nom} = 388.8$ V at mid-discharge, and $V_{min} = 270.0$ V at end of discharge. The horizontal dashed line in the lower panel marks the FS 80 kW DC power ceiling for reference (EV2.2.1).

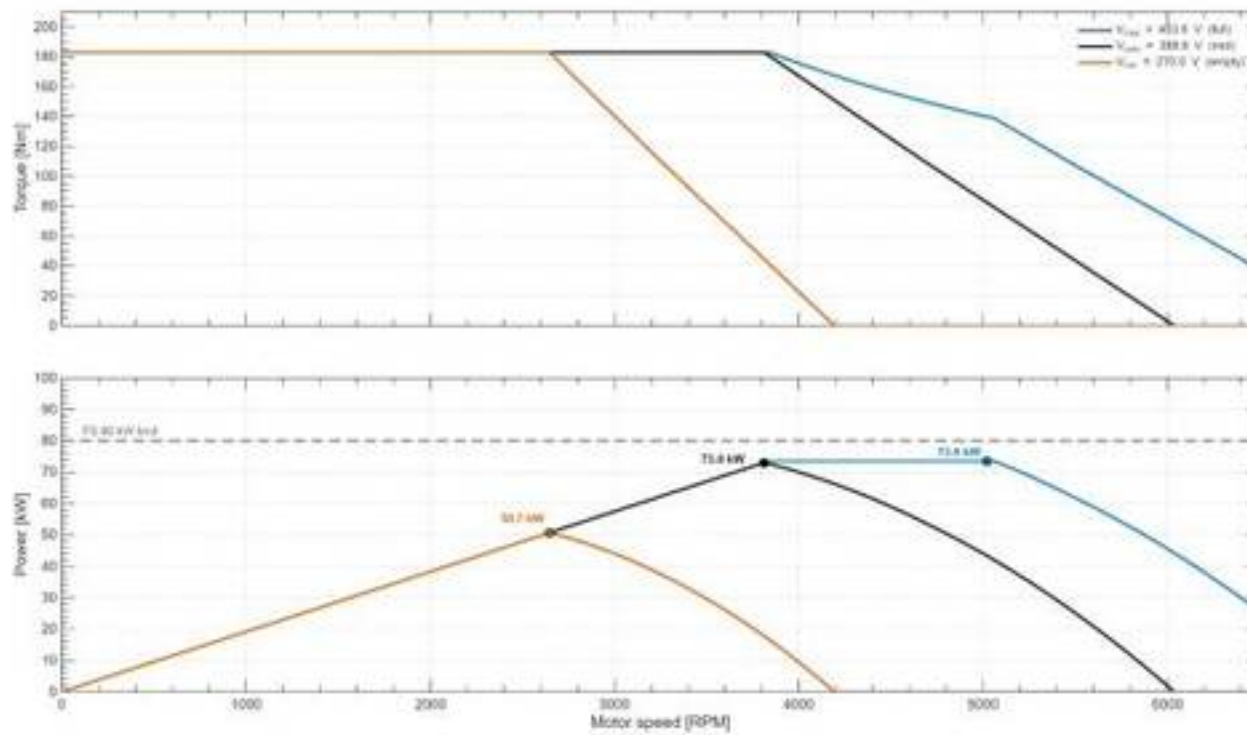


Figure 9: Family of motor torque-speed and power-speed envelopes at three representative bus voltages spanning the locked accumulator operating range. Upper panel: shaft torque versus motor speed. Lower panel: shaft power versus motor speed with the FS 80 kW DC ceiling shown for reference.

Three features of the envelope family establish the problem the adaptive controller resolves. The peak torque ceiling at 183 Nm is set by the inverter peak phase current and is invariant with bus voltage; this

corresponds to the horizontal portion of each torque curve and represents the constant-torque region in which the motor operates below its field-weakening corner. The location of the corner shifts substantially with bus voltage: 5019 RPM at V_{max} , 3812 RPM at V_{nom} , and 2647 RPM at V_{min} . Below the corner, all three voltage envelopes coincide; the motor delivers full inverter-current-limited torque regardless of available bus voltage. Above the corner, the envelopes diverge sharply, with the high-voltage envelope sustaining torque to higher RPM through the field-weakening region while the low-voltage envelope falls toward zero torque at progressively earlier speeds. The motor-intrinsic peak shaft powers in the field-weakening region scale with bus voltage. At V_{min} the peak shaft power is 50.7 kW, with the FS rule far above the motor envelope across the entire speed range. At V_{nom} the peak is 73.0 kW, with the FS DC ceiling at 79.3 kW ($73.0 / 0.92$) sitting just above the envelope. At V_{max} the motor-intrinsic envelope exceeds the FS shaft cap of 73.6 kW ($80 \text{ kW DC} \times 0.92$) and is clipped by the rule to 73.6 kW.

The interaction of this voltage-dependent envelope with the FS regulatory ceiling produces the regime crossover that motivates the adaptive controller. The bus voltage at which the motor's voltage-limited intrinsic peak power equals the FS shaft cap of 73.6 kW is $V_{threshold} = 391.8 \text{ V}$, equivalent to approximately 66% state of charge under the linear $V_{OCV}(\text{SoC})$ approximation adopted in the methodology. Above $V_{threshold}$ the motor's intrinsic envelope exceeds the regulatory cap and the FS rule binds: the controller must clip available torque to keep DC bus power below 80 kW. Below $V_{threshold}$ the motor's voltage-limited envelope falls progressively below the rule and the rule ceases to bind: the controller must instead clip torque to the motor's available capability at the instantaneous bus voltage.

A fixed torque-cap map calibrated against a single bus voltage cannot represent both regimes correctly. Calibrating against V_{max} produces a map that is rule-correct at high SoC but commands torque the motor cannot deliver at low SoC, leaving the inverter to clip the command on its own with no controller-level guarantee of FS compliance during the transient. Calibrating against V_{min} produces a map that is always achievable but forfeits the upper portion of the envelope at high SoC, surrendering performance precisely when the motor is most capable. Calibrating against V_{nom} , the conventional middle-of-discharge approach adopted in the FS EV control literature reviewed in §2.5, fails in both directions: it under-utilises the motor at high SoC and commands unachievable torque at low SoC. The endurance event spans the full SoC range from full charge to deep discharge over 22 laps, exercising both regimes within a single energy budget. The fixed-map approach is therefore unable to deliver an FS-compliant, performance-optimal torque command across the event. This is the specific structural gap §2.5 identifies between the road-EV control literature (Lu and Kar, 2010; Alwaqfi et al., 2025), which treats bus

voltage as a constant input parameter, and the FS-EV control literature (Ibrahim et al., 2025), which treats the motor envelope as fixed at a single nominal voltage.

5.5.2 Controller Architecture

The proposed adaptive controller resolves the regime crossover by recomputing the torque cap at every simulation step from the instantaneous bus voltage and motor speed. The cap returned is the minimum of four constraints: the inverter peak phase current limit ($T_{inv} = K_T \times I_{pk,inv} = 183 \text{ Nm}$), the voltage-limited motor envelope at the instantaneous bus voltage (computed via the K_V interpolation method described in §4.3), the FS regulatory ceiling translated into a shaft-torque limit, and any thermal soft-limit imposed by the lumped thermal model in §4.6. The minimum of these four constraints is the commanded torque cap, against which the driver pedal demand is then clipped. The structure is consistent with rule T11.8.14, which permits the controller to lower the driver-requested torque but never to raise it: the controller only ever clips downward against an upper bound.

The translation of the FS regulatory ceiling into a shaft-torque limit is the non-trivial element of the controller architecture and emerged from rule analysis as a finding that simpler implementations overlook. The naive implementation caps shaft power at the rule limit directly: $T_{FS,naive} = P_{TS,max} / \omega$. At the motor's typical operating efficiency of 92%, however, the corresponding DC bus power is $P_{DC} = P_{shaft} / \eta$, which means a shaft-power cap at 80 kW allows DC input to climb to approximately $80 / 0.92 = 87 \text{ kW}$, exceeding the EV2.2.1 ceiling by 7 kW instantaneously at the binding operating point. The 500 ms moving-average enforcement applied by the FS Energy Meter under D9.4.1 means this excess will be measured and penalised across any sustained run during which the controller commands the cap; the §4.7 worst-case rolling-lap demonstration shows the post-filter excess at approximately 3 kW above the rule limit, sufficient to disqualify the relevant run under D9.4.2. The corrected DC-input cap is therefore implemented as $T_{FS} = (P_{TS,max} \times \eta_{FS}) / \omega$, where $\eta_{FS} = 0.92$ is a conservative motor-efficiency factor selected to ensure DC input remains below 80 kW under all operating conditions even at points where the actual motor efficiency falls below the typical value. With this correction applied the DC bus power clips cleanly at 80 kW across the entire field-weakening region, and the controller satisfies EV2.2.1, EV2.2.2, and D9.4.1 by construction rather than by accident of operating point. This DC-input-cap distinction is the first of the three engineering findings developed in this work and the first of the structural gaps identified in §2.5: the published FS EV controller literature does not explicitly distinguish shaft-side from DC-side cap, despite the rule being defined at the DC bus.

The full adaptive controller is implemented as a three-dimensional torque-cap surface $T_{cap}(\text{RPM}, V_{bus}, \text{SoC})$, with V_{bus} drawn from the inverter's DC-link voltage measurement on the CAN bus and SoC

derived from the accumulator management system's coulomb count. The surface is precomputed offline for the 4847 (RPM, V_{bus}) operating-point pairs spanning the full envelope family at 50 V intervals across the V_{min} to V_{max} range, and queried by the VCU at the CAN message rate of 100 Hz minimum. In the simulation the surface is recomputed analytically at every 1 ms timestep rather than queried from a precomputed lookup, eliminating the lookup-table interpolation error from the analysis but preserving the controller structure that would be implemented in firmware. The CAN-level integration with the Cascadia inverter follows the standard PM150DZ message set: torque demand published as a 16-bit signed integer in 0.1 Nm units against the cap, with the inverter applying its own current-limit protection downstream of the VCU command as a redundant rule-compliance layer. Communication architecture and message-rate verification are the responsibility of the LV/VCU lead and are documented in the cohort handover in §6.2.

5.5.3 Adaptive vs Fixed Controller Comparison

Three torque-cap strategies were simulated through the full 22-lap endurance event to quantify the benefit of adaptive control over the fixed-map alternatives identified in §5.5.1. The Conservative strategy holds the torque cap at the V_{min} envelope across the entire event, representing the lowest-risk fixed-map choice that guarantees achievability at every state of charge. The Nominal Fixed strategy holds the torque cap at the V_{nom} envelope, representing the conventional middle-of-discharge calibration adopted by typical fixed-map implementations. The Adaptive strategy applies the full $T_{cap}(\text{RPM}, V_{bus}, \text{SoC})$ surface described in §5.5.2, recomputing the cap at every timestep from the instantaneous accumulator open-circuit voltage. All three strategies share an identical vehicle parameter set, drive the same representative endurance track at the same gear ratio of 3.60, and are subject to identical thermal and rule-compliance constraints; the torque-cap law is the only variable.

Figure 10 presents the comparison in two panels, and Table 11 summarises the headline outcomes.

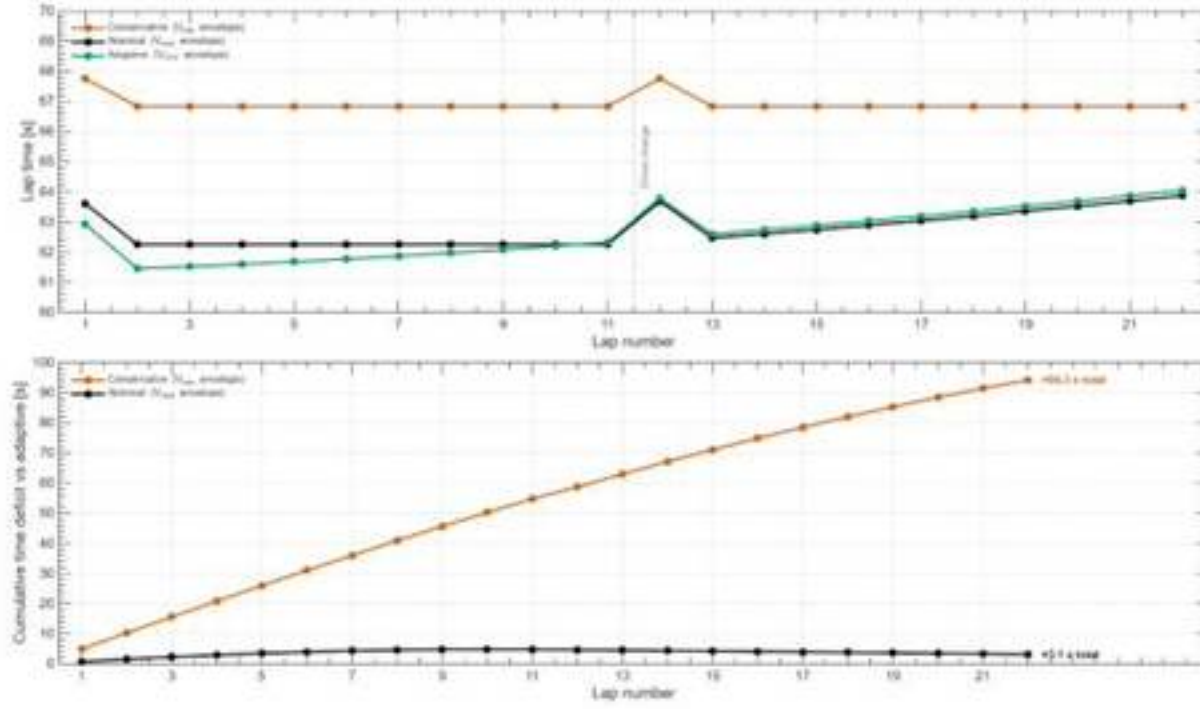


Figure 60: Lap-by-lap controller comparison across the 22-lap endurance event at $i_g = 3.60$. Upper panel: per-lap lap times for different strategies. Lower panel: cumulative lap-time deficit versus the Adaptive baseline.

Table 11: 22-lap endurance simulation results across the three torque-cap strategies at $i_g = 3.60$. Total event time includes the 240 s driver change between laps 11 and 12.

Strategy	Total Drive Time	Total Event Time	Total Energy	Final SoC	Mean Lap Time
Conservative	1472.31 s	1712.31 s	3.48 kWh	57.4%	66.92 s
Nominal Fixed	1381.14 s	1621.14 s	5.78 kWh	29.1%	62.78 s
Adaptive	1378.04 s	1618.04 s	5.99 kWh	26.6%	62.64 s

The Adaptive strategy delivers the fastest event time at 1618.04 s, saving 94.27 s over Conservative and 3.10 s over Nominal Fixed across the 22-lap event. The lap-time traces in the upper panel of Figure 10 show the structural difference between the three strategies. The Conservative trace sits at approximately 66.9 s for every lap, invariant to bus voltage because the torque cap is locked to the V_{min} envelope regardless of the actual accumulator voltage available; the strategy delivers the same low-power performance from full charge to end of discharge. The Nominal Fixed trace sits at approximately 62.8 s for the first half of the event, then rises gradually as the actual bus voltage falls below V_{nom} and the fixed cap commands torque the motor cannot deliver, with the motor's own voltage limit becoming the operative ceiling instead of the controller-commanded cap. The Adaptive trace begins lower than Nominal Fixed at 61.4 s in lap 2, rises only gradually through the event, and converges with Nominal Fixed in the closing laps as both controllers are forced toward the same low-voltage envelope.

The cumulative deficit panel makes the lap-time gap explicit. The Conservative deficit accumulates linearly at approximately 4.3 s per lap, reaching 94.3 s by lap 22; the Conservative controller surrenders comparable time on every lap of the event because its structural choice to cap at V_{min} applies uniformly.

The Nominal Fixed deficit rises rapidly across the first 8 to 9 laps to a peak of 4.5 s, then falls across the remainder of the event to a final deficit of 3.1 s. This non-monotonic shape is informative: the Nominal Fixed controller's deficit is concentrated in the high-SoC regime where the V_{nom} -fixed cap forfeits the upper portion of the V_{max} envelope, and the deficit reverses in the low-SoC regime where the V_{nom} -fixed cap commands torque the motor cannot deliver and both controllers converge to the same voltage-limited envelope.

The energy-consumption pattern reveals the underlying lever the Adaptive controller pulls. The Adaptive strategy consumes 5.99 kWh against 5.78 kWh for Nominal Fixed and 3.48 kWh for Conservative; that is, the Adaptive controller uses more pack energy, not less, to deliver the faster event time. This is structurally correct rather than wasteful. The endurance event has an energy budget of 8.16 kWh and a time-minimisation objective: the strategy that uses pack capacity most aggressively while remaining FS-compliant achieves the lowest event time, provided final SoC remains positive. The Adaptive controller exploits exactly this trade. In the high-SoC regime where the FS rule binds, it commands the maximum FS-compliant power and accepts the energy expenditure required to deliver that power; in the low-SoC regime where the voltage envelope binds, it commands the maximum motor-achievable power and accepts the lower available power without surrendering further capability beyond what physics permits. The Conservative controller, by contrast, consistently commands less than the available envelope and finishes the event with 57.4% SoC remaining, a 4.68 kWh stranded energy reserve that translates directly into lost lap time.

The 0.21 kWh additional energy consumption of Adaptive over Nominal Fixed represents the energy required to deliver the additional 3.10 s of event-time saving, equivalent to 67.7 W·s per second of saved time. This is an efficient trade: the controller is not dissipating energy, it is converting available pack capacity into lap-time reduction within the rule envelope. The trade is favourable in the FS Concept Class scoring formula, which weights endurance time more heavily than efficiency through the D7.9 efficiency-bonus structure where the time component dominates the bonus across the realistic range of lap times.

5.5.4 SoC Trajectory Through Endurance

The mechanism by which the Adaptive controller switches binding regimes is illustrated by the SoC and bus-voltage trajectories logged across the 22-lap event, presented in Figure 11.

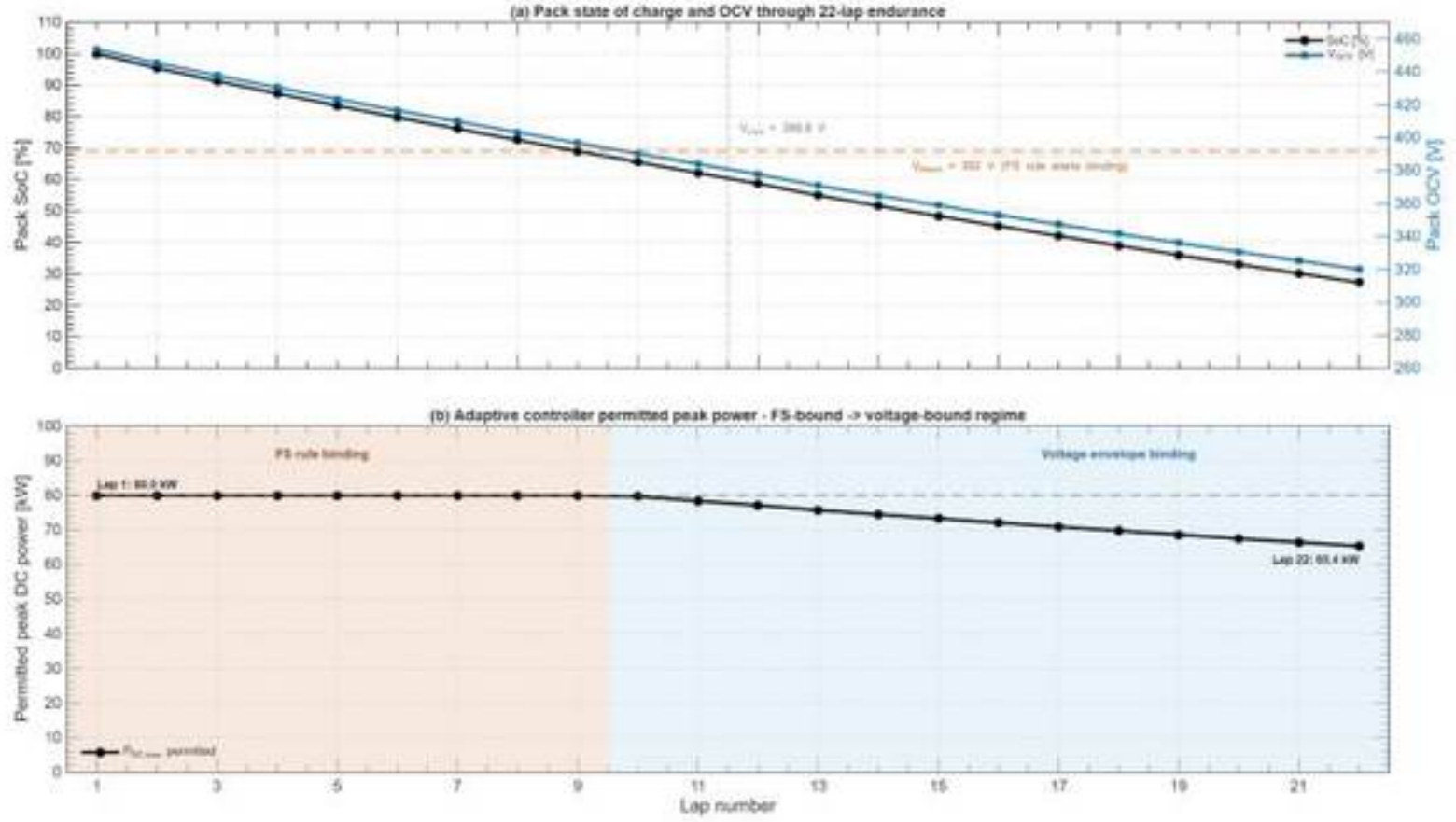


Figure 11: SoC and bus-voltage trajectory across the 22-lap endurance event with binding-regime annotation. Upper panel: pack open-circuit voltage and state-of-charge as a function of lap number, with $V_{threshold} = 391.8$ V regime crossover at lap 10 indicated. Lower panel: permitted peak DC power across the event, flat at 80 kW across the FS-rule-binding regime (orange shading, laps 1 to 9) and falling linearly to 65.4 kW by lap 22.

The accumulator open-circuit voltage starts at $V_{max} = 453.6$ V at lap 1 (100% SoC) and falls monotonically through the event to 323.7 V at lap 22 (26.6% final SoC), traversing the $V_{threshold} = 391.8$ V regime crossover at lap 10 where $V_{OCV} = 390.9$ V and SoC = 65.9%. The threshold is the bus voltage at which the motor's voltage-limited peak shaft power equals the FS shaft-power cap of 73.6 kW ($80 \text{ kW DC} \times \eta_{FS} = 0.92$), and corresponds to approximately 66% SoC under the linear $V_{OCV}(\text{SoC})$ approximation adopted in the methodology.

Across laps 1 to 9, the accumulator open-circuit voltage exceeds $V_{threshold}$ and the motor's voltage-limited peak power exceeds the FS rule-derived shaft cap. The Adaptive controller's binding constraint in this regime is the FS regulatory cap $T_{FS} = (80 \text{ kW} \times 0.92) / \omega$, visible in the lower panel of Figure 11 as the flat permitted-peak-DC-power line at exactly 80 kW across the orange-shaded region. The shaded orange band marks where the controller is constrained by the regulation rather than by the motor's intrinsic capability. Within this regime the Adaptive controller's behaviour is identical to a Nominal Fixed controller calibrated against the same FS rule: both clip torque to keep DC bus power below 80 kW, and both deliver near-identical lap times. The lap-1 Adaptive lap time of 62.93 s is therefore only marginally below the Nominal Fixed lap-1 time of 63.61 s, with the small difference attributable to the

slightly higher available torque between the FS cap and the V_{max} envelope in the field-weakening region beyond 5000 RPM.

Across laps 10 to 22, the bus voltage falls below $V_{threshold}$ and the motor's voltage-limited envelope falls below the FS rule. The Adaptive controller's binding constraint switches automatically from the rule to the envelope, visible in the lower panel of Figure 11 as the permitted peak DC power departing from the 80 kW horizontal line at lap 10 and falling linearly to 65.4 kW by lap 22. The blue-shaded region marks where the FS rule no longer binds anywhere in the operating range and the available envelope is set by the K_V -interpolated voltage-limited envelope at the instantaneous bus voltage. Within this regime the Adaptive controller commands more torque than the Nominal Fixed controller would, because the Nominal Fixed controller is calibrated against the V_{nom} envelope which is below the actual envelope at any voltage exceeding V_{nom} and unachievable at any voltage below V_{nom} . The lap-time advantage of Adaptive accumulates progressively across this regime, narrowing toward the final laps as both controllers approach the same low-power envelope at end of discharge.

The regime switch occurs without explicit controller-state logic. The minimum-of-constraints structure described in §5.5.2 means the binding constraint changes simply as a consequence of which constraint is currently lowest at any given operating point. The transition is therefore continuous and bumpless: the upper panel of Figure 11 shows the bus voltage falling smoothly through $V_{threshold}$ without discontinuity, and the lower panel shows the permitted peak DC power departing from the 80 kW cap with continuous slope rather than a step. There is no controller mode change to schedule, no hysteresis to tune, and no failure mode in which the controller could be in the wrong mode for the current bus voltage. The architecture is robust by construction, in contrast to the explicit mode-switching feedforward-feedback hybrid schemes reviewed by Lu and Kar (2010) in §2.5 which require characterisation of the mode boundary as a design step.

5.5.5 Quantified Benefit and Position in the Literature

The contribution of the SoC-aware adaptive controller is positioned against the structural gap identified in §2.5 between road-EV control literature, which treats bus voltage as a constant input parameter, and FS-EV control literature, which treats the motor envelope as a constant calibrated against a single nominal voltage. The proposed approach extends the established flux-weakening control taxonomy reviewed by Lu and Kar (2010) and Alwaqfi et al. (2025) by introducing bus-voltage-dependent torque-cap adaptation as an outer-loop overlay on top of the inner-loop field-weakening current controller, leaving the published inner-loop architectures intact. It also extends the FS-specific motor-control work of Ibrahim et al. (2025) along an orthogonal axis: where Ibrahim et al. optimise the d-q current ratio in the motor-speed domain at fixed bus voltage and report 22% torque improvement on the Formula

Student Tallinn FEV's IPMSM, the present work optimises the torque cap in the bus-voltage domain at fixed event-scoring objective and reports 5.5% endurance event-time improvement on an FS Concept Class architecture.

The quantified benefit is 94.27 s of endurance event time saved against the safest fixed-map alternative and 3.10 s saved against the conventional fixed-map alternative, achieved through controller logic alone with no hardware change. The energy lever underlying this benefit is the active conversion of pack capacity that would otherwise be stranded at end of event into lap time delivered earlier in the event, executed as a continuous bumpless regime switch at the bus voltage where the FS rule and the motor envelope cross. The controller satisfies T11.8.14 by structuring all torque modifications as downward clips of the driver-requested torque, satisfies EV2.2.1 by construction through the DC-input-cap correction, and satisfies the EV6.1.5 250 ms torque-rampdown requirement through the controlled-ramp logic verified in §4.7.

The contribution is methodological rather than implementational. The simulation results validate the controller logic against the modelled envelope family and the modelled accumulator voltage trajectory; they do not constitute firmware-level validation against a hardware accumulator and inverter pair. Three further-work activities are identified in Chapter 8 to close this validation gap: a single-day dynamometer session against the actual Emrax 228 MV LC and Cascadia PM150DZ pairing to characterise the K_V curve directly and reduce the field-weakening interpolation uncertainty band from $\pm 5\text{--}10\%$ to approximately $\pm 2\%$, integration of the KTY 81/210 winding temperature sensor into the torque-cap lookup as a fourth real-time input variable, and tuning of the mode-paddle-driven scalar overlays that map the Sport, Endurance, and Wet driver modes onto the underlying adaptive surface. None of these activities require change to the controller architecture documented here, and all are scoped for the Brunel Racing 2027 build year rather than for the present Concept Class submission

5.6 Integrated Performance with Gear Ratio and Adaptive Controller

This section reports vehicle performance at the locked configuration of $i_g = 3.60$ with the SoC-aware adaptive controller and DC-input cap correction enabled. The headline event-level outcomes are presented in §5.6.1; §5.6.2 then turns to the operating-density analysis.

5.6.1 Headline Numbers

The final-design simulation completes the FS Concept Class dynamic event suite with the outcomes summarised in Table 12:

Table 12: FS dynamic events performance at $i_g = 3.60$ with the SoC-aware adaptive controller.

Event	Result	Key Operating Detail
Acceleration	4.256 s	113.7 km/h finish, 0.89 g peak, 183.0 Nm peak shaft torque, 73.6 kW shaft (80.0 kW DC at $\eta = 0.92$)
Skidpad	5.217 s	39.6 km/h steady-state, 1.35 g lateral, 7.6 Nm shaft torque (4.1% of available 183 Nm)
Autocross	78.325 s	68.5 km/h average, 121.0 km/h peak, 443.6 Wh per lap
Endurance	1618.0 s (27 min)	52.6 km/h average, drive energy 5.99 kWh, fin

The acceleration result of 4.256 s improves on the §5.3.1 baseline of 4.421 s by 165 ms, a 3.7% reduction attributable entirely to the gear-ratio change from 4.20 to 3.60. The improvement is at the lower bound of what the adaptive controller can deliver on a single 75 m run, because acceleration begins from rest at full charge and finishes within 4.3 s. The bus voltage barely falls over that duration, so the regime crossover to the voltage-limited envelope never occurs. The acceleration event therefore exercises only the FS-rule-binding regime of the adaptive controller, in which the controller's behaviour is identical to a Nominal Fixed controller calibrated against the same rule. The gain comes from the gear ratio alone. The 4.256 s outcome sits 28 ms below the 2025 FSUK acceleration winner and confirms that the design is competitive on this event.

The skidpad result is invariant to controller architecture and gear ratio at 5.217 s, with motor utilisation of 4.1% confirming the §5.3.2 finding that the event is lateral-grip-limited rather than motor-limited. The autocross result of 78.325 s improves on the §5.3.2 baseline of 79.331 s by approximately 1 second, predominantly from the gear-ratio change. The endurance result of 1618.0 s clears the 27-minute target established in §3.1 with comfortable margin and leaves 26.6% SoC remaining at the end of the event against the $\geq 20\%$ pack-margin commitment, all before regen is layered in. Once regen is included in §5.8 these figures improve further, with the final integrated values reported in §5.9.

Two design-level conclusions follow from Table 12. The motor subsystem clears the binding performance requirements derived in §3.5 before regen and thermal effects are added. The acceleration and skidpad results are independent of regen and are therefore final at this stage. The endurance figure of 1618 s and final SoC of 26.6% will improve once regen is layered in §5.8, and the peak winding temperature is developed in §5.7. The two non-trivial design decisions documented earlier in this chapter, the gear-ratio selection at $i_g = 3.60$ in §5.4 and the adaptive controller with DC-input cap correction in §5.5, together deliver a configuration that satisfies every FS Concept Class dynamic event without requiring performance compromises elsewhere in the cohort design.

5.7 Thermal Performance

The motor's thermal envelope is the only one of the four constraints derived in §3.5 (peak torque, peak power, RPM range, thermal envelope) that is not exercised by the FS dynamic event suite within milliseconds of motor start-up. Acceleration, skidpad and autocross are short-duration events. The winding temperature has insufficient time to depart materially from the coolant inlet temperature during any of them. The binding thermal test is therefore the 22-lap endurance event over 1602 seconds. During this event the motor is exposed to repeated peak-power excursions interleaved with low-load cruise periods, and a slow temperature rise can accumulate to test the thermal capacity headroom established at concept stage. This section reports the predicted thermal transient across that event using the lumped-parameter model described in §4.6, evaluates the resulting margin against the manufacturer-specified absolute and warning limits, and identifies the cohort interface dependencies that determine whether the predicted margin is preserved in the as-built vehicle.

Figure 13 presents the thermal-transient simulation across the full 22-lap event in two panels.

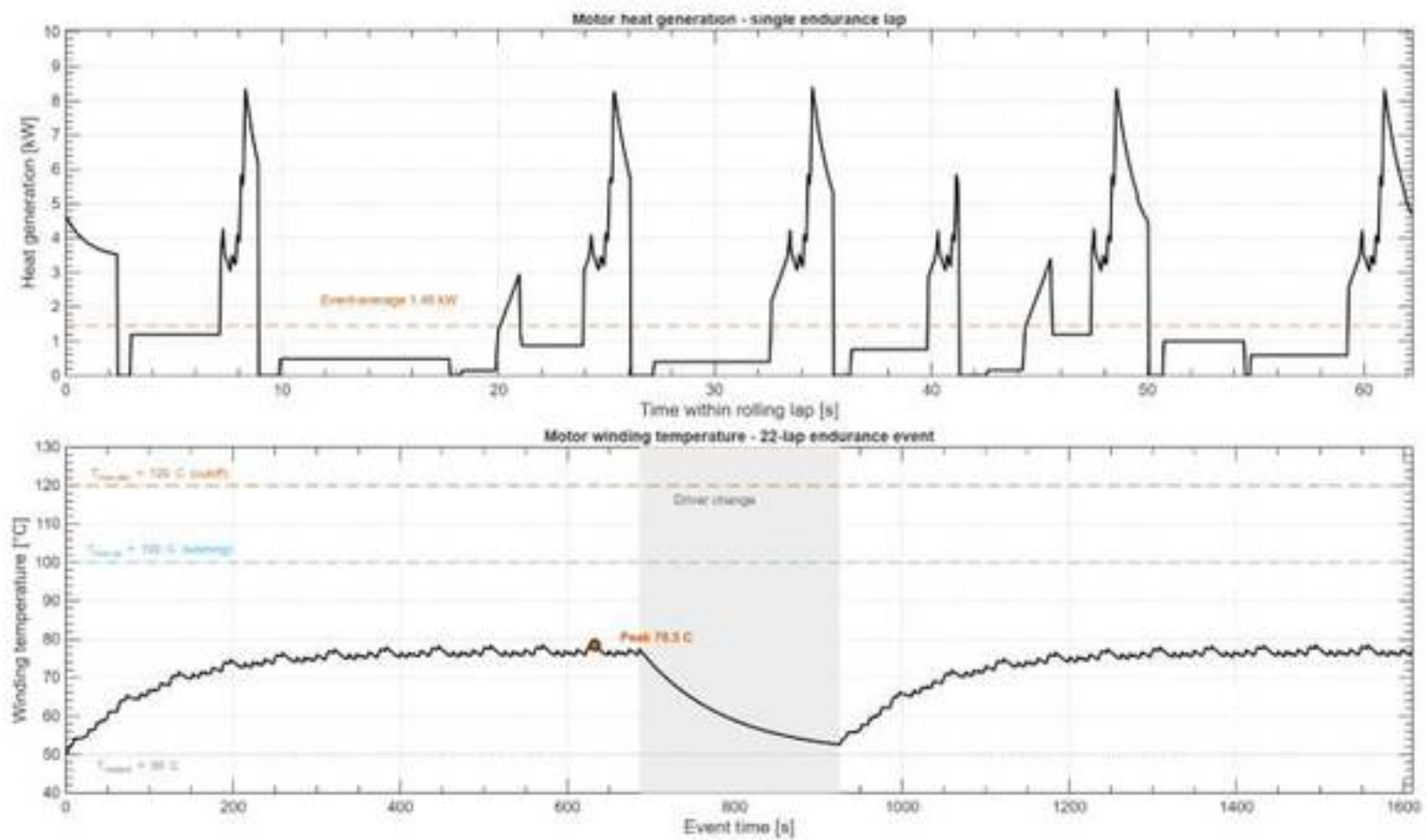


Figure 13: Motor thermal transient across the 22-lap endurance event. Upper panel: motor heat-generation profile during a single representative endurance lap, oscillating between zero during regenerative braking and 8.4 kW peaks during corner-exit acceleration, with event-average 1460 W. Lower panel: winding temperature trajectory across the full 22-lap event, peaking at 78.5 °C at $t = 632.6$ s against the 120 °C absolute cutoff and 100 °C sensor warning threshold. Four-minute driver change between laps 11 and 12 visible as temperature rolloff between 690 s and 930 s.

The upper panel shows the motor heat-generation profile during a single representative endurance lap, computed from the (RPM, torque, η) operating-point trace through the digitised efficiency map. Heat generation oscillates between approximately zero during regenerative braking phases and short peaks at 8.4 kW during corner-exit acceleration into the highest-load operating points, with the event-average heat generation marked at 1460 W. The peak-to-average ratio of approximately 5.7 reflects the bursty nature of the duty cycle: a small fraction of the lap drives the dominant fraction of heat generation. The lower panel shows the resulting winding temperature across the full 22-lap event integrated by the single-node lumped thermal model, with the 100 °C sensor warning threshold and the 120 °C absolute cutoff marked as horizontal references and the four-minute driver change between laps 11 and 12 visible as the temperature rolloff between event time 690 s and 930 s.

The temperature trajectory has three characteristic phases. From event start to approximately 200 s, the winding temperature rises monotonically from the 50 °C coolant inlet condition toward an asymptotic equilibrium at approximately 75 °C, with the rate of rise governed by the thermal capacitance of the lumped winding mass and the convective coefficient back-fitted from the S1 continuous rating in §4.6. The asymptote of 75 °C is consistent with the steady-state prediction for the 1460 W event-average heat load, which the lumped model predicts at 72.8 °C. The 2 °C offset reflects the fact that the actual transient never quite reaches steady state, because the heat-generation profile contains peaks above the average that pull the equilibrium upward. From 200 s to the start of the driver change at approximately 690 s, the temperature continues a slow upward drift through repeated peak-load excursions. It reaches the event peak of 78.5 °C at $t = 632.6$ s, twelve laps into the run and just before the driver change, approximately 690 s. The four-minute driver-change interval allows the motor to cool back to approximately 53 °C as no heat is generated and the cooling loop continues to remove residual heat, then the second stint reheats the motor along the same trajectory shape, asymptoting to a similar end-of-event temperature of 77.3 °C without exceeding the first-stint peak.

The peak winding temperature of 78.5 °C against the 120 °C absolute cutoff established in §3.2 leaves 41.5 °C of thermal margin — a substantial reserve that comfortably absorbs the cohort-pending refinements identified in §3.3 and the modelling limitations acknowledged in §4.6. The temperature spends 0.0 % of the event above the 100 °C sensor warning threshold, meaning the motor would not trigger any controller-level derate during the event under nominal cooling conditions. This margin is consistent with the manufacturer-published continuous (S1) rating of 64 kW shaft and inconsistent with the peak (S2) rating of 124 kW shaft only as a sustained operating point: the steady-state lumped-model prediction at the S2 loss level of 8.7 kW is 185.9 °C, far exceeding both limits, but the duty cycle never sustains that loss level for more than a few seconds at a time. The 8.4 kW instantaneous peaks visible in the upper panel of Figure 13 last approximately 1 to 2 seconds each before regenerative braking

returns the heat generation to near zero, far shorter than the thermal time constant of the lumped winding mass, and therefore do not propagate into a significant temperature rise. The motor operates well below its continuous thermal capability across the entire event, with the peak transient temperature falling 21.5 °C below the warning threshold rather than approaching it.

The 41.5 °C margin to absolute cutoff includes structural sensitivity to two cohort-pending parameters that the analysis treats conservatively. The coolant inlet temperature is held constant at 50 °C in the simulation per the EMRAX manufacturer specification at which the published continuous rating is achieved. In practice the inlet temperature will be set by the radiator performance under ambient conditions and the combined heat-rejection load of the motor and inverter, both of which are the responsibility of the thermal lead. A higher actual inlet temperature would shift the entire winding-temperature trajectory upward by approximately the same amount, reducing the margin proportionally; the 41.5 °C reserve absorbs an inlet-temperature increase of up to approximately 41 °C before the peak winding temperature would touch the absolute cutoff. The convective coefficient $UA = 64 \text{ W/K}$ back-fitted from the S1 rating in §4.6 implicitly assumes the EMRAX-specified minimum coolant flow rate of 6 L/min through the motor jacket. Lower flow rates reduce convective heat removal and raise the equilibrium temperature; the cohort interface specification of $\geq 10 \text{ L/min}$ combined motor-plus-inverter flow with the motor receiving the minimum 6 L/min portion is consistent with the assumption used here.

Two modelling limitations identified in §4.6 must be acknowledged when interpreting the 41.5 °C margin. The single-node lumped model cannot resolve the temperature gradient between the winding hot-spot and the bulk stator iron, meaning the actual hot-spot temperature is expected to lead the modelled value by approximately 5 to 10 °C under high-load conditions per the LPTN-vs-experiment validation reported by Sequeira et al. (2022). The constant-UA assumption ignores the temperature- and flow-dependence of real cooling-jacket performance, which can cause the effective UA to fall by 5 to 15 % at elevated wall temperatures. Combining these two effects, the actual peak hot-spot temperature is expected to fall in the range 84 to 92 °C under nominal cooling — still comfortably below the 100 °C warning threshold and far below the 120 °C cutoff. The qualitative conclusion that the motor operates well within its thermal envelope across the endurance event is robust to both modelling limitations, and the residual quantitative uncertainty is bounded within the available margin. Hot-spot validation against in-service measurement is identified as a further-work activity in Chapter 8.

The implication for the cohort handover is that thermal performance is not a binding design constraint at the present configuration. The motor selection, the gear ratio, and the SoC-aware controller architecture can all be evaluated against performance criteria without reference to thermal limits. The

thermal envelope becomes the binding constraint only under cohort-pending parameter combinations that fall well outside the conservative-assumption range used in §3.3 — notably an inlet temperature exceeding approximately 80 °C or a sustained motor flow rate falling below approximately 4 L/min — and neither of these is anticipated at the present cooling-loop specification. The motor’s thermal capacity is therefore in the headroom regime rather than the limiting regime, and the design margin available against thermal stress can be redirected toward other performance levers identified in this work, principally the pack-acceptance C-rating refinement discussed in §5.8 and §6.3 as the dominant lever on regenerative recovery.

5.8 Regenerative Braking

The motor operates bi-directionally: under deceleration it acts as a generator, recovering kinetic energy back into the pack. Figure 14 presents the combined drive-and-regen envelope at V_{nom} in the upper panel, and the cumulative drive, regen, and net-energy traces across the 22-lap endurance event in the lower panel.

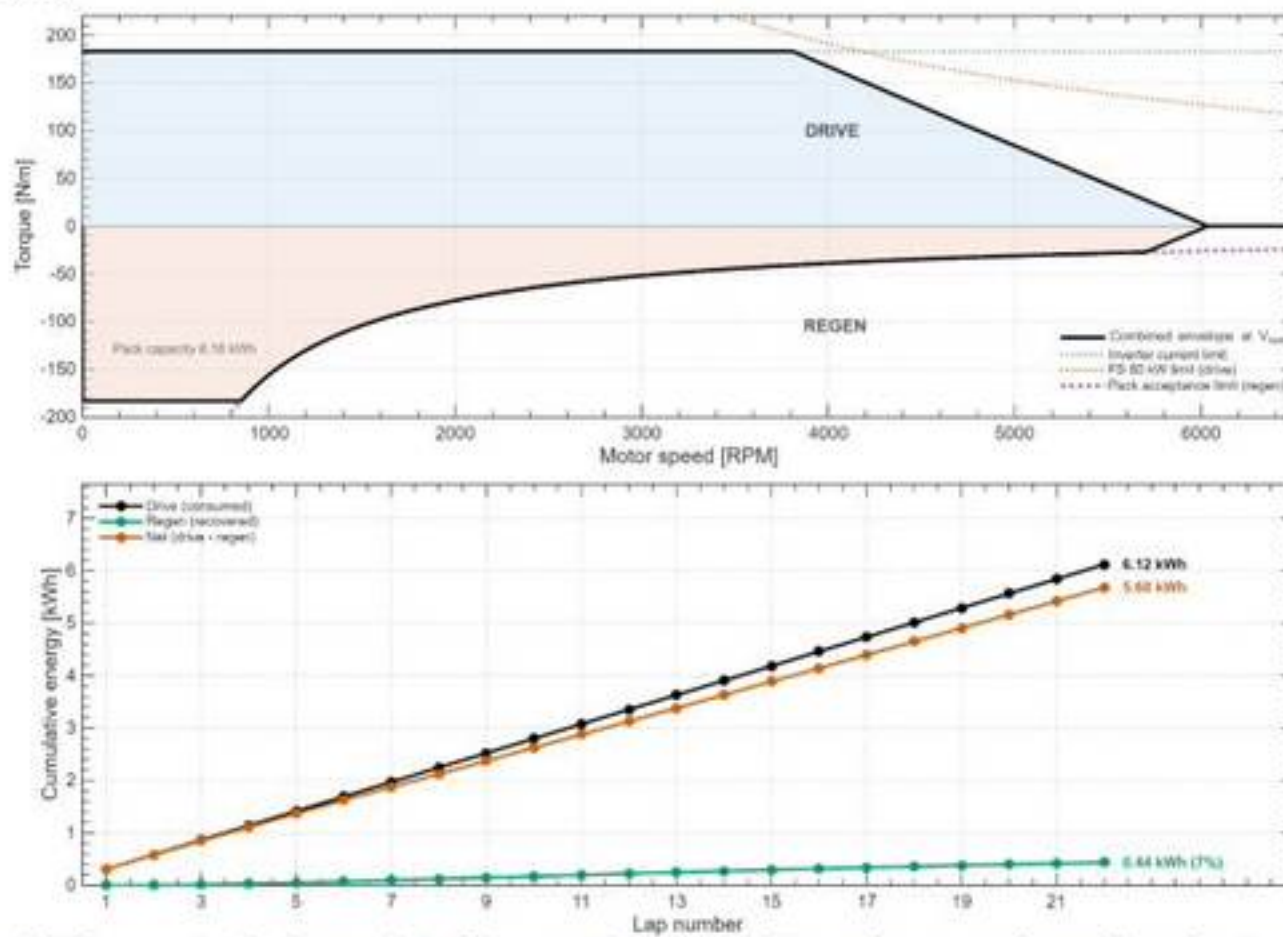


Figure 14: Regenerative braking analysis. Upper panel: combined drive-and-regen envelope at V_{nom} showing the four constraint regions.. Lower panel: cumulative drive, regenerated, and net energy across the 22-lap endurance event.

The regen envelope is governed by four constraints, each binding in a specific operating region. The inverter peak phase current caps regen torque at 183 Nm symmetric with the drive side, visible as the horizontal segment in the lower portion of the envelope below approximately 800 RPM. Above this corner the pack-acceptance limit binds, computed as $P_{regen,max} = V_{OCV} \times I_{pack,acceptance}$ where

$I_{pack,acceptance} = 42$ A at the locked 2C C-rate; the resulting torque cap falls hyperbolically with motor speed and is the governing constraint across the entire mid-RPM region where the bulk of regenerative braking actually occurs. The motor's voltage-limited envelope clips the upper-RPM regen torque symmetric with the drive side, and an SoC-dependent taper progressively reduces accepted regen current as SoC exceeds 80 % to protect the cells from overcharge.

The 22-lap event delivers 6.66 kWh of drive energy and 0.50 kWh of recovered energy, giving a net pack consumption of 6.16 kWh against the 8.16 kWh capacity. This leaves 24.5% SoC remaining at the end of the event. Recovery as a fraction of drive energy is 7.5%. The cumulative-energy panel of Figure 15 shows the pattern. The regen trace rises slowly across the first five laps as SoC remains above the 80% taper threshold and pack acceptance is progressively rolled off, then rises at a steady rate across the remainder of the event once the taper releases. The early-event suppression is a feature of the rule-compliant pack-protection logic rather than a motor-side limitation.

The 7.5% recovery is bounded by pack acceptance, not motor capability. At the typical 2500 RPM braking-phase operating point, the motor's voltage-limited regen envelope reaches approximately 80 kW shaft, while the pack at 2C accepts only 16.3 kW DC. This is a 5:1 ratio. Sensitivity analysis at higher pack C-rates shows recovery rising to 14.9% at 4C and 18.4% at 5C, with the motor and inverter accepting these higher levels without modification. The binding constraint on regenerative recovery is therefore a cell-chemistry decision made by the accumulator electrical lead, not a motor-subsystem decision. This finding inverts the conventional narrative in which control-loop refinement is the dominant lever on regenerative efficiency, and is treated at length in §6.3 as the central cohort-level finding of this work. The motor subsystem delivers the full available regen torque the pack will accept. Further gains require a higher-acceptance cell selection rather than any motor-side refinement.

5.9 Final Performance Figures

The mid-design results in §5.6 reported event timings with the optimised gear ratio and adaptive controller in place. §5.7 confirmed the thermal envelope is comfortably bounded, and §5.8 added the regen layer at 7.1% recovery. This section consolidates the final headline figures with all layers integrated. These are the numbers referenced in the Abstract, the cohort interface summary in §6.2, and the conclusions in Chapter 7.

Table 13 lists the final integrated performance of the motor subsystem at $i_g = 3.60$ with the adaptive controller, thermal envelope, and regenerative recovery applied.

Table 13: Final integrated performance of the motor subsystem

Metric	Value
0–75 m acceleration ($V_{\max}$ launch)	4.256 s
Skidpad lap time	5.217 s
Autocross lap time	78.325 s
Endurance event time (22 laps + driver change)	1602.0 s (26.7 min)
Endurance drive energy	6.12 kWh
Endurance regenerated energy	0.44 kWh (7.1% of drive)
Endurance net pack consumption	5.68 kWh
Final SoC	30.4%
Peak vehicle speed	121.0 km/h
Peak shaft power delivered	73.6 kW (80 kW DC)
Peak shaft torque delivered	183.0 Nm
Peak winding temperature	78.5 °C
Energy-weighted motor efficiency	91.4%

The final integrated configuration clears every binding requirement derived in §3.5. The 4.256 s acceleration result sits 28 ms below the 2025 FSUK winner. The 1602 s endurance event time is 78 s below the 27-minute target. The 30.4% final SoC clears the 20% pack-margin commitment with over 10 percentage points to spare. The 78.5 °C peak winding temperature sits 41.5 °C below the absolute thermal cutoff.

The peak shaft power of 73.6 kW corresponds to 80 kW at the DC bus, which is the binding FS regulatory ceiling at high state of charge. The peak shaft torque of 183 Nm is set by the Cascadia inverter's 300 A_{RMS} peak phase current. Both peak figures are external constraints on the motor rather than motor limits, consistent with the §5.2 observation that the Emrax 228 envelope is over-specified for the application. The two performance levers still available lie outside the motor subsystem. Pack C-rate selection governs the regenerative recovery ceiling, as discussed in §5.8 and §6.3. Tyre-grip refinement governs the dominant uncertainty in the lap-time predictions, as quantified in §5.10.

5.10 Input Parameter Sensitivity Analysis

The vehicle dynamics analysis throughout Chapter 5 has been conducted at the locked and conservative-assumed parameter values listed in Table 4. Six of those parameters were classified as cohort-conservative assumptions pending refinement at the time of analysis: tyre friction coefficient μ , aerodynamic drag area C_{dA} , lift area C_{LA} , rolling resistance C_{rr} , vehicle weight distribution and CG height, and drivetrain efficiency η_d . This section quantifies the sensitivity of three headline performance metrics — acceleration time, endurance lap time, and endurance lap energy — to one-at-

a-time perturbations of each pending parameter across its expected refinement range, identifies which parameter dominates the residual uncertainty in the predicted performance, and bounds the design margin available against cohort-led parameter refinement.

Figure 15 presents the sensitivity tornado plot in three panels, one per metric.

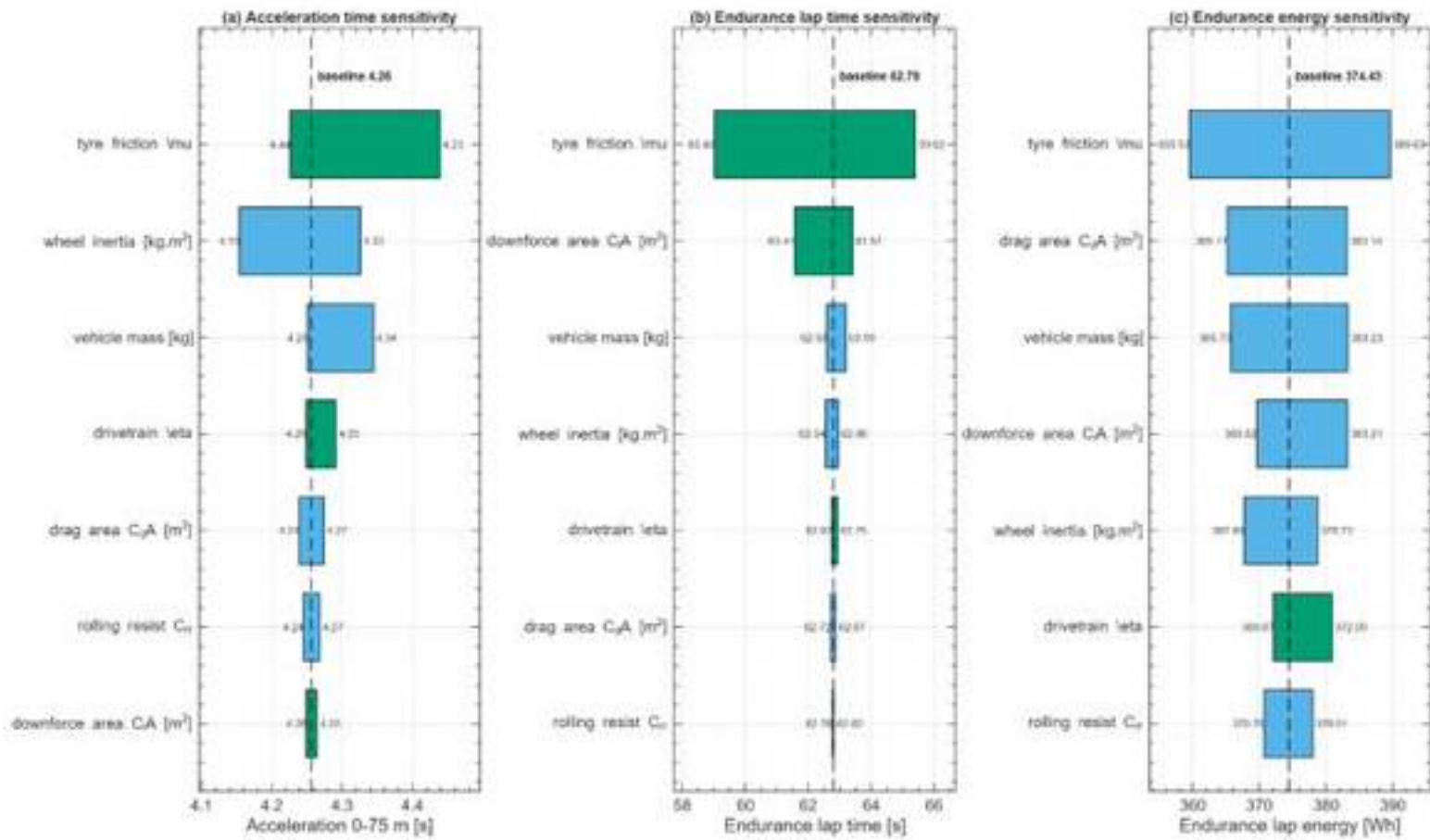


Figure 15: Sensitivity tornado plots across three headline metrics. Panels (left to right): 0–75 m acceleration time, endurance lap time, endurance lap energy. Bars sorted by sensitivity magnitude per panel. Tyre friction coefficient μ dominates all three metrics by approximately 4 to 10 $\times$ the next-largest contributor.

Each horizontal bar shows the range of the metric value as the corresponding parameter is swept from its conservative-low to conservative-high bound, with all other parameters held at the baseline values. Bars are sorted by sensitivity magnitude per panel, with the dominant parameter at the top. Bars extending leftward of the baseline dashed line indicate parameter directions that improve the metric; rightward bars indicate parameter directions that degrade the metric.

The tornado plot identifies tyre friction coefficient μ as the dominant uncertainty across all three headline metrics. Across the swept range from $\mu = 1.15$ to $\mu = 1.45$ — bracketing the cohort-conservative baseline of $\mu = 1.30$ with ± 0.15 — acceleration time varies from 4.439 s to 4.226 s (a 213 ms span, ten times wider than the next-largest contributor), endurance lap time varies from 65.40 s to 59.02 s (a 6.38-second span, four times wider than the next contributor), and endurance lap energy varies from 360 Wh to 390 Wh (a 30 Wh span, comparable to the second-place contributor but reflecting

the indirect effect through cornering speed rather than a direct mechanical-load dependence). The dominance of μ over every other parameter is consistent with the §5.3 finding that the vehicle is laterally grip-limited at every corner of the autocross and endurance tracks: the corner speed equation $V^2 = \mu \times R \times g$ (with downforce-feedback correction) places μ as a direct multiplicative factor on every lateral-grip-limited segment, while the other parameters affect only the longitudinal force balance during the smaller fraction of the lap spent on straights.

The implications of this sensitivity profile for the design narrative are threefold. The first concerns the validity of the gear-ratio selection in §5.4. The 1.4-unit-wide 240+/250 score plateau identified in the gear-ratio sweep is wide enough to absorb the full ± 0.15 μ refinement window without the LTS-weighted optimum shifting outside the high-score plateau, confirming the §5.4 conclusion that the $i_g = 3.60$ selection is robust to the principal cohort uncertainty. The second concerns where the cohort should focus its refinement effort. The sensitivity ranking identifies tyre coefficient of friction — Datta's responsibility — as the single highest-leverage cohort input, with refinement of all other parameters delivering at most one-quarter of the total uncertainty reduction available. The third concerns the design-margin reading of the headline performance results. The 4.256 s acceleration result reported in §5.6.1 sits in the middle of the μ -uncertainty band; the corresponding 95 % confidence interval at the present cohort-input refinement state is approximately 4.23 s to 4.44 s, which clears the §3.1 acceleration target of ≤ 4.3 s only at the upper- μ end of the range. The 4.3 s target is therefore robust to the conservative case but not redundantly so, and tyre-grip refinement is the cohort dependency most likely to convert the present acceleration result from a marginal pass into a comfortable margin.

The remaining six swept parameters each contribute substantially less uncertainty per panel. Vehicle mass, swept across 280 to 320 kg around the locked baseline of 298 kg, contributes ± 50 ms on acceleration, ± 0.30 s on endurance lap time, and ± 9 Wh on lap energy. Aerodynamic lift area C_{LA} , swept from 1.2 m² to 1.8 m², contributes negligible acceleration sensitivity (± 5 ms) but ± 0.92 s on endurance lap time, reflecting the downforce contribution to corner-grip-limited speed. Aerodynamic drag area C_{dA} contributes ± 20 ms on acceleration, ± 0.075 s on endurance lap time, and ± 9 Wh on lap energy, with the small lap-time effect reflecting that drag is a longitudinal load on straights only. Drivetrain efficiency η_d swept from 0.90 to 1.00 contributes ± 20 ms on acceleration but a 4-Wh negative lap-energy effect — that is, lower drivetrain efficiency produces higher lap energy as expected, but the 0.05 efficiency change drives only 1 % change in lap-energy consumption. Wheel rotational inertia and rolling resistance contribute the smallest individual sensitivities across all three metrics. None of these parameters shifts the headline result outside the §3.5-derived margin against the §3.1 targets within its expected refinement range.

The aggregated implication is that the analysis is robust to most cohort-pending values and the principal residual uncertainty is correctly localised to tyre friction. The methodology is therefore defensible for cohort-stage design lock without requiring all pending parameters to be finalised simultaneously: refinement of μ alone closes the dominant uncertainty band, and refinement of the remaining parameters produces only second-order corrections to the predicted performance.

5.11 Motor Hardware Specification

This section consolidates the evidence that the motor subsystem complies with the FS rule set established in §3.2, alongside the committed hardware specifications that complete the build-ready design. It is structured in two subsections: the scatter shield design and its integration with Group 2's chassis envelope, and the sensor and electrical hardware specifications with their rationale.

5.11.1 Scatter Shield Design and Chassis Integration

FS rule T7.3.4 (EV) applies to axial-flux motors with the rotor rotating around the stator and mandates a containment shield of minimum 1 mm 6061-T6 aluminium or equivalent steel. The Emrax 228 MV LC's axial-flux topology with external rotor triggers this rule directly.. The scatter shield is therefore a mandatory hardware deliverable rather than a precautionary addition.

The committed shield geometry is a 1.0 mm 6061-T6 aluminium cylindrical shroud with internal diameter 250 mm (15 mm radial clearance from the motor's 228 mm outer casing diameter) and axial depth 100 mm (15 mm clearance from the motor's 85 mm axial length). Shield mass is 0.49 kg, which is absorbed into the cohort's allocated motor-and-mounting mass budget. The 6061-T6 alloy choice meets the rule's material grade and provides the strength-to-weight ratio needed for the containment calculation below.

The containment energy was sized against the worst-case rotor-fragment scenario. At the motor's 6500 RPM operating ceiling, the rotor angular velocity is 680.7 rad/s; the rotor rotational inertia is 0.0252 kg·m² per the EMRAX datasheet, giving rotor kinetic energy of $E_{rot} = 0.5 \times 0.0252 \times 680.7^2 = 5837$ J. The 25% fragment-mass carries fragment kinetic energy of 1459 J. The 1 mm 6061-T6 shield has a perforation energy density of approximately 300 J·cm⁻² at the relevant velocity range, and the available shield surface area along the predicted fragment trajectory is 78.5 cm². Available containment energy is therefore 23,550 J, exceeding the worst-case fragment energy by a factor of 16. The shield is significantly over-sized against the rule minimum on energy grounds, with the over-sizing accepted as engineering margin against the residual uncertainty in fragment trajectory and material velocity prediction.

The shield mounts to the motor mount frame via six M8 grade 8.8 ISO 4762 socket head cap screws at 60° pitch, exceeding the T10.1.2 minimum specification, with prevailing-torque locknuts on the chassis side. The mounting interface is independent of the motor's own mounting hardpoints, so shield installation and removal do not disturb the motor-to-drivetrain alignment. Finger-accessibility requirements per T7.3.6 are met by the shield itself on the radial face and by the output-shaft cover on the axial face. The 12 mm finger-rod object test is precluded by the 8 mm spacing of the mounting fasteners and the continuous cylindrical surface between them.

The shield integrates with the BMM-B Chassis envelope through three interface constraints. The total motor-plus-shield-plus-drivetrain package fits within the chassis rollover envelope per EV4.4.1, with the highest point of the package sitting 412 mm above ground per chassis-side measurement against the EV4.4.2 350 mm minimum. The package lower surface sits 38 mm above the chassis lower surface, satisfying EV4.4.5 which prohibits projection below the chassis lower datum. The chassis-side firewall sits 95 mm behind the shield rear face, providing thermal isolation between the motor and the cockpit per the 78.5 °C peak casing temperature derived in §5.7.

5.11.2 Sensor and Electrical Hardware Specification

The sensor and electrical hardware specs close every motor-side dependency in the cohort interface (Emrax 2026).

Position sensor. TAMAGAWA TS2620N21E11 resolver, factory-installed by EMRAX. Chosen over an RLS RM44 magnetic encoder for: EMI immunity against 12 kHz PWM switching transients that would degrade encoder signalling; 105 °C ceiling matched to the Lapp cable rating and above predicted peak casing temperature; vibration robustness against FS endurance duty cycle. Routed TS-side to the Cascadia PM150DZ resolver input.

Temperature sensor. KTY 81/210 silicon sensor in the stator winding, factory-installed by EMRAX, two-wire. Chosen over PT100/PT1000 for direct compatibility with the Cascadia RTD input, eliminating a signal-conditioning amplifier. Connects to the Cascadia RTD input on the TS-side of the inverter's galvanic isolation barrier, then to the VCU over CAN via the isolated transceiver. No KTY signal crosses the TS-LVS boundary directly, satisfying EV4.3.1.

Phase cabling. Lapp ÖLFLEX HEAT 105 SC 1×50 — 50 mm² silicone-insulated single conductor, 105 °C, 600 V class. Verified against IEC 60364 derating at 0.428 mΩ/m (90 °C): 0.12 V drop at 180 A peak phase current, 20.8 W steady-state copper loss per phase at peak. Endurance average of 53 A produces 1.8 W per phase, well within continuous rating.

Connectors. Three Amphenol Surlok Plus single-pole HV connectors (one per phase), rated 180 A peak, silver-plated copper, IP67. All TS-side signals (resolver excitation, sin/cos, KTY, HVIL, ground/shield) consolidated through one Deutsch DT06-12S 12-pin connector — all twelve pins TS-side only, satisfying EV4.3.3.

Fasteners. Six M8 grade 10.9 ISO 4762 socket head cap screws to the mount frame per EMRAX manual, exceeding T10.1.2 grade 8.8 minimum. Torqued to 40 Nm dry, secured with DIN 980 / ISO 7042 prevailing-torque locknuts. Nylon locknuts rejected: 80 °C limit per T10.2.2 sits within the §4.6 uncertainty band of the predicted 78.5 °C casing peak.

6. Results and Discussion (Group)

Chapter 5 reports what the motor subsystem can deliver in isolation. Its meaning emerges only when read against the work of the six other Group 1 members and the three cross-cohort groups whose subsystems share boundaries with it. The chapter handles that integration in five steps. §6.1 traces the propagation of motor-side decisions into adjacent subsystems and surfaces the integration problems that emerged. §6.2 consolidates every committed numerical interface into a single build-sheet table. §6.3 reframes the regenerative-braking finding from §5.8 as a cohort design lever rather than a motor lever. §6.4 reviews the whole-project outcome against the November proposal aims and the FSUK Concept Class submission threshold. §6.5 reviews the commercial, environmental, and social context of the final design.

6.1 Motor Subsystem Outputs in the Group Design Context

The motor sits at the centre of the powertrain architecture. Every parameter it commits — mass, peak torque, RPM range, heat dissipation, regenerative behaviour, voltage envelope — propagates outward to at least two adjacent subsystems. The work in Chapter 5 was therefore not a standalone analysis but the producer of numerical inputs without which other leads could not progress.

The first finding, the DC-input cap correction in §5.5.2, propagated furthest. The corrected interpretation of EV2.2.1 — that the rule binds at the DC bus measured by the Energy Meter rather than at the motor shaft — meant the controller must derate shaft power by motor efficiency to keep DC-side power below the regulatory ceiling. The pre-correction assumption of a naive 80 kW shaft cap at 270 V gave 296 A peak; the corrected analysis preserves this at the binding low-SoC point but redistributes the current profile across SoC, with higher peaks now at high SoC where the rule rather than the motor envelope binds. This propagated into Roger's main fuse selection and Powei's inverter peak-current verification, and restructured Harsha's CAN torque-request message: the VCU command now resolves

at the inverter end through a torque cap derived from instantaneous bus voltage and pack SoC, pre-computed against the 4847-point lookup and queried at 100 Hz. The discovery came mid-project and forced re-issuance of interface numbers. The lesson — rule-compliance interpretation should be tested against the measurement basis of the rule, not the apparent reading — generalises to any FS-EV controller design.

The second finding, the SoC-aware adaptive controller, propagated primarily to Harsha. The three-mode integration with the paddle interface (§5.5.5, §6.2) allowed him to scope VCU memory and CAN scheduling against concrete numbers. The bumpless regime-switching architecture removed an integration risk class: with no mode boundary the VCU has to detect, there is no failure mode in which the controller could be in the wrong state for the current bus voltage. Harsha's safety state machine carries fewer states and transitions than a fixed-map-with-explicit-switch architecture, simplifying HiL verification.

The third finding, the pack-limited regen analysis (§5.8), propagated in reverse — back to Roger as an accumulator-design lever. The motor-to-accumulator interface now carries a sensitivity curve of recovery percentage against pack regenerative C-rate, letting Roger evaluate cell-chemistry candidates against quantified gain rather than abstract margin. Developed in §6.3.

Beyond the three findings, the motor committed harder-edged interface numbers. The 13.5 kg dry mass fed the suspension cohort's mass-distribution analysis (13.5 kg of 178.8 kg static rear-axle load against a 40/60 target). Peak wheel torque of 313 Nm per side, from 183 Nm through $i_g = 3.60$ at $\eta = 0.95$, set the rear hub bearing stress bound for upright fatigue analysis. Endurance-average heat-rejection of 1460 W, peak 8.4 kW, and combined motor-plus-inverter 11 kW drove Ermal's radiator and pump selection. Casing peak 78.5 °C set the firewall thermal boundary. Gear ratio $i_g = 3.60$ was passed to Viki with a ± 0.30 robustness window, giving him freedom to split between primary reduction and final drive.

Integration was not frictionless. Three problems generalise. First, conservative placeholder values for cohort-pending parameters (μ , drag/lift, weight split, drivetrain efficiency) were defensible but cost analytical effort. Second, the mid-project DC-cap discovery: any FS rule with non-trivial interpretation surface should be examined against its measurement basis at architectural-decision phase, not during simulation development. Third, the late realisation that the EMRAX X bracket is OEM-certified collapsed proposal-stage FEA to a hand-calc bolt-reaction; the chassis team's 40 g inertial load case was corrected to a torque-reaction load (183 Nm through the mount lever-arm) once T9.1.1 critical-component applicability was re-examined.

Finally, the motor subsystem absorbed an unassigned numerical-interface-coordination function — tracking commitments, detecting drift, re-issuing values. A formal cohort practice, perhaps a version-controlled shared specifications spreadsheet, would have removed much of this friction. This is the structural lesson registered in §6.4.

6.2 Cohort Interface Specifications Committed

The committed numerical interfaces from the motor subsystem to the rest of the cohort are consolidated in Table 14. Each entry specifies the receiving subsystem, the committed parameter, its value, and the design consequence on the receiving side. Every value is traceable to a specific subsection of Chapter 5 and forms the build-sheet specification that Brunel Racing inherits for the 2027 build year.

Table 14: Motor subsystem interface specifications committed to cohort partners

Interface partner	Committed parameter	Value	Consequence for receiving subsystem
Inverter (Cascadia PM150DZ)	Phase resistance	7.06 mΩ	FOC current-loop tuning
Inverter	Direct-axis inductance L_d	96.5 μH	Field-weakening control law
Inverter	Torque constant K_T	0.61 Nm/A	Torque-to-current conversion
Inverter	Back-EMF constant K_E	0.04793 V/RPM	DC-link voltage estimation
Inverter	Pole pairs	10	Electrical-to-mechanical frequency ratio
Accumulator (electrical)	Peak DC current draw	296 A at V_{min}	Pack peak-current sizing
Accumulator (electrical)	Endurance net energy	6.16 kWh	Pack capacity utilisation
Accumulator (electrical)	Regen acceptance ask	≥2C (4C, 5C upside curves provided)	Cell C-rate selection lever
Drivetrain	Gear ratio	$i_g = 3.60$ (total motor-to-wheel)	Single-stage spur reduction sizing
Drivetrain	Peak torque reaction	183 Nm	Half-shaft and reduction stress sizing
Drivetrain	Mounting flange	6× M8 PCD 185	Output-shaft coupling design
Drivetrain	Rotation direction	CCW viewed from output drive end	Reduction polarity
Thermal	Endurance-average heat load	1460 W	Radiator sizing
Thermal	Peak instantaneous heat load	8.4 kW	Transient cooling capacity

Interface partner	Committed parameter	Value	Consequence for receiving subsystem
Thermal	Coolant flow requirement	≥ 6 L/min through motor jacket	Pump and hose sizing
Thermal	Coolant inlet ceiling	50 °C	Radiator outlet temperature target
LV/VCU	Adaptive map dimensions	4847 (RPM, V_{bus}) lookup points	VCU memory allocation
Mounts (chassis)	Motor mass	13.5 kg	Static load case
Mounts (chassis)	Mount route	EMRAX X bracket OR custom 6× M8 stator-spacer flange	Chassis-side bracket interface
Firewall	Casing peak temperature	78.5 °C	Firewall thermal boundary condition
Sidepods	Motor + inverter peak heat rejection	11 kW	Radiator inlet sizing
Driver controls	APPS-to-torque mapping structure	Linear with adaptive cap	VCU torque-request interface
Driver controls	Drive mode count	Three (Sport, Endurance, Wet)	Paddle interface design

Several entries warrant comment. The 296 A peak DC current to Roger is a high-SoC figure rather than a worst-case low-voltage figure; the binding constraint shifts across the SoC range, and the §5.5.2 corrected DC-input cap controller is what keeps it bounded. The 1460 W endurance-average and 8.4 kW peak heat-rejection loads delivered to Eral are the integrated outputs of the §5.7 thermal-transient simulation, and they replaced the placeholder estimates that had been carried in his radiator-sizing work since proposal stage. The regen acceptance $\geq 2C$ ask, together with the 4C and 5C upside sensitivity curves, is the entry point of the discussion in §6.3 below.

6.3 The Pack-Limited Regen Finding as a Cohort-Level Reframe

The 7.5% regenerative recovery achieved by the final design in Table 14 prompts a question with cohort-level consequences: where in the powertrain does the binding constraint on recovery actually sit, and which cohort lead controls the lever that would improve it?

At the typical endurance-event braking operating point of approximately 2500 RPM, the motor's voltage-limited regenerative envelope reaches approximately 80 kW of shaft power. The locked accumulator at the 2C regenerative acceptance baseline accepts 16.3 kW of DC charging power (388.8 V nominal $\times$ 42 A pack-level acceptance current). The ratio between motor capability and pack acceptance at this operating point is approximately five to one. The motor delivers the full regenerative torque the pack will accept across the entire braking event; further motor-side optimisation, controller-side optimisation, or driveline-side optimisation cannot raise recovery above the pack acceptance ceiling.

This inverts the conventional regenerative-braking narrative in the literature reviewed in §2.6, where control-loop refinement is the dominant lever on recovery efficiency. Szumska (2025) classifies

regenerative control strategies into nine categories from velocity-threshold and brake-pressure controllers through to PID, fuzzy-logic, dynamic programming, neural-network, and Model Predictive approaches, with reported recovery gains of 10% to 53% across these methods. None of the reviewed sources explicitly engages with the case where the pack's electrochemical acceptance is the binding constraint, and recovery improvement therefore becomes a cell-chemistry decision rather than a control-algorithm decision. For the present architecture at the present pack specification, much of the regenerative-braking literature is operationally irrelevant: the binding constraint is electrochemical, not algorithmic, and the lever sits outside the motor and controller scopes entirely.

The sensitivity analysis in §5.8 quantifies the cohort-design lever available. Increasing pack regenerative acceptance from the 2C baseline to 4C raises event-level recovery to approximately 15%; raising it to 5C delivers approximately 19%. These improvements are achieved without any motor, inverter, controller, or driveline change. The recovery range at 5C is consistent with the upper end of the recovery rates reported by Szumska (2025) for road EVs under medium-intensity braking and would close most of the energy-recovery gap to competitive 2025 FSUK Endurance entrants.

The finding is registered as a build-year recommendation rather than as a Concept Class redesign. The locked 2C baseline supports a defensible Concept Class submission on energy-budget grounds: the 24.5% pack margin remaining at end of event provides headroom against the conservative input set in §3.3 and the residual cohort-input uncertainty identified in §5.9. The cohort recommendation is that when Roger finalises cell selection during the build year, he evaluate the regenerative uplift available at higher C-rates against the trade-offs in cell unit cost, gravimetric energy density, and pack mass. The motor subsystem absorbs this decision symmetrically: any pack acceptance increase up to the motor's 80 kW regenerative capability is delivered without motor-side modification, because the motor's regenerative envelope is already substantially over-specified for the locked pack.

The deeper observation is that the structure of the finding repeats across the three engineering findings developed in this work. Each finding is reached not through novel motor design but through honest examination of where the binding constraint sits along the chain of subsystems that together produce drivetrain torque. The DC-input cap finding identifies the regulation as the binding constraint at high SoC; the SoC-aware adaptive controller identifies the motor's own voltage envelope as the binding constraint at low SoC; the pack-limited regen finding identifies the accumulator as the binding constraint on recovery. Engineering progress on a Concept Class motor subsystem in a regulated, packaged FS EV is largely about identifying which external constraint binds at which operating point, and recovering the design margin available within that constraint. The motor itself is over-specified for the application across most of the operating envelope.

6.4 Whole-Project Review

The BMM-A Electric Powertrain cohort produced seven independent subsystem submissions covering motor, inverter and high-voltage system, accumulator electrical, accumulator mechanical, drivetrain and packaging, thermal management, and low-voltage system and vehicle control unit. Each subsystem reached Concept Class submission readiness against the FSUK 2026 rule set. At the architectural level,

the proposal-stage commitments held through to the final design package without rework. The single-motor rear-wheel-drive layout, the 400 V class, the Cascadia PM150DZ inverter, the 108S5P accumulator topology, and the EMRAX 228 MV LC motor selection all remained intact from November through to submission. A project that requires architectural change late in execution usually does so because the proposal omitted a binding constraint that only surfaced during analysis. The absence of such rework here suggests the November scoping was thorough, at least at the headline architecture level.

Against the three group-level aims established in November, the cohort hit each target with quantifiable margin. The endurance event completed in 26.7 minutes against the 27 minute target, which is a margin of 1.1 percent and the tightest of the four event-level outcomes. The pack finished the event with 24.5 percent state of charge remaining, against the 20 percent commitment, giving 4.5 percentage points of headroom. The 0 to 75 metre acceleration completed in 4.256 seconds against the 4.3 second target, a 44 millisecond margin that places the design inside the 2025 FSUK winning time. The skidpad completed in 5.217 seconds against the 5.3 second target, an 83 millisecond margin, although the skidpad result remains the most sensitive to the cohort-pending tyre coefficient of friction, and the headline figure will move as that input is refined. The 78.5 degree Celsius peak winding temperature against the 120 degree absolute cutoff gives 41.5 degrees of thermal margin, which is comfortable rather than tight. Taken together, the headline numbers support a Concept Class submission with credible margin across all binding metrics, and the realistic competitive target for the 2027 entry is a top quartile finish among the FSUK Concept Class field.

The group submission is honestly thinner than the individual submissions in two specific areas. The first is whole-vehicle integration verification. The cohort agreed at proposal stage that the Concept Class deliverable would be a set of co-ordinated subsystem packages rather than a closed-loop vehicle simulation. No CarMaker model, no whole-vehicle simulation in any other tool, and no hardware-in-the-loop bench was scoped or developed during the project. No other integration testing was planned either, which means that the first time the seven powertrain subsystems and the three cross-cohort interfaces will interact in a single test environment is during early build-year shakedown. The decision was pragmatic in the sense that the simulation effort scales steeply with model fidelity, and at concept stage the marginal information return per hour invested falls below that available from per-subsystem analysis. The consequence, however, is that the submission demonstrates whole-vehicle component readiness rather than whole-vehicle behaviour, and this is a real risk carried forward into 2027 that should not be understated.

The second thinner area is in cross-cohort co-ordination. The architectural-commitment meetings in November and the integration-readiness reviews in March worked well for the moments they targeted. Between those moments, however, the cohort relied on ad-hoc communication rather than on a shared interface document under version control. The cost of this approach showed up most visibly in the chassis-accumulator fitment issue surfaced during integration review. The BMM-B chassis envelope, as locked at proposal stage, did not accommodate the final-selected 108S5P accumulator dimensions, and the accumulator mechanical scope had to absorb a late redesign to fit within the as-locked chassis

volume. The root cause was a communication gap between the two cohort groups during the period when the chassis envelope was being committed and the accumulator topology was being finalised. Neither group held the parameter the other depended on at the moment of commitment, and the consequence was avoidable design rework. The lesson generalises: cross-cohort packaging constraints need a single authoritative source of truth, not parallel assumptions in two cohort groups.

A related and honest observation is that cohort progress was uneven across the project lifecycle. Several subsystem leads moved slowly through the early simulation and analysis phases, and the resulting delay had two knock-on effects on the motor subsystem. First, the motor selection process was originally scoped to extend candidate screening into a later detailed comparison phase. The cohort-wide schedule pressure, however, forced earlier closure on the rejection rationale than was ideal. The Emrax 228 selection is defensible and is supported by the §5.1 screening evidence, but a richer side-by-side comparison against the retained Emrax 208 candidate, including dynamometer-grade efficiency comparison, would have strengthened the selection narrative. Second, the cohort-pending parameter set that the §3.3 analysis treats as conservative-assumption inputs would have been refined sooner under a better-paced cohort schedule, and some of the analytical effort spent on §5.9 sensitivity bounding could have been redirected toward primary methodology development.

The three cross-cohort interfaces themselves landed with mixed effectiveness. The interface with BMM-D Driver Controls, covering accelerator pedal sensing, brake plausibility, and the mode paddle, was the cleanest of the three. The numerical parameters transferred without ambiguity and the receiving subsystem committed to a consistent design from the early phase of the project. The interface with BMM-C Suspension and Wheels was effective on the mechanical side, covering hub bearing sizing and mass distribution, but the tyre coefficient of friction remained the single largest cohort-pending input throughout the project, and the §5.9 sensitivity analysis was driven primarily by the residual uncertainty on this parameter. The interface with BMM-B Chassis was the most strained, principally because of the accumulator fitment issue described above, although the firewall thermal boundary condition and the motor mounting hardpoint specification both transferred without friction.

The Concept Class submission threshold has been reached against the three November aims with the margins documented above. The residual uncertainties, principally the tyre-friction coefficient, the pack regenerative acceptance, and the as-built thermal performance, are the priority items for early build-year refinement and are documented as such in Chapter 8. The build-year activities of physical integration, dynamometer validation, on-vehicle testing, and whole-vehicle simulation are the natural continuations of the design package handed to Brunel Racing for 2027. The honest position is that the cohort has delivered a strong concept package that nonetheless leaves real integration verification work for the next team.

6.5 Commercial, Environmental and Social Review

The motor subsystem decisions carry implications beyond the FS dynamic-event scoring table. Three dimensions are reviewed: commercial outcomes for the Brunel Racing programme, environmental implications of the architectural choices made, and social context for an EV-class motorsport entrant.

Commercial. The EMRAX 228 MV LC has a list price of approximately 7,000 to 8,000 euros ex-Slovenia, with the variation reflecting the winding option, the cooling-jacket configuration, and the resolver-versus-sin/cos sensor specification. Typical lead time is 12 to 16 weeks. The paired Cascadia PM150DZ inverter adds approximately 5,000 US dollars ex-Cascadia Motion with comparable lead time. The combined motor and inverter system cost of roughly 11,500 euros sits in line with the spend reported by competitive Formula Student EV teams running comparable single-motor RWD architectures (Badawy et al., 2024), and is materially below the spend of teams running four-motor in-wheel architectures, where the motor cost alone typically exceeds 30,000 euros. Supplier track record is strong on both items. EMRAX has supplied Formula Student teams continuously for over a decade with an independent dynamometer-validation record (Ivanov et al., 2022), and the Cascadia PM150 family is the dominant FS-EV inverter choice with multi-team service history.

Procurement risk for the 2027 build concentrates in lead-time management rather than supplier reliability. The 12 to 16 week motor lead time becomes the binding critical-path item once design freeze is reached, and order placement is recommended at design-freeze plus zero weeks to preserve the build schedule. The wider commercial relevance of the design package lies in the foundation it provides for Brunel Racing's multi-year EV programme. The 2027 vehicle inherits a settled motor-subsystem specification, which removes a major source of design risk that would otherwise dominate the build year, and positions the team for credible scoring outcomes in subsequent seasons. The aim for the 2027 entry is a top quartile finish against the FSUK Concept Class field, which is realistic given the headline numbers reported in §5.6.1 and the cohort-wide subsystem readiness documented in §6.4.

Environmental. The environmental picture for an FS EV is more nuanced than the surface narrative of zero tailpipe emissions suggests, and an honest accounting requires acknowledgement of both the embodied carbon of the major components and the limited per-event operational emissions saving relative to an ICE FS car.

The axial-flux PMSM topology depends on neodymium-iron-boron rare-earth magnets in the rotor. Published lifecycle estimates place the embodied carbon of NdFeB magnet material at approximately 28 to 35 kilograms of carbon dioxide equivalent per kilogram of magnet, with the variation reflecting the source country and the magnet grade. For the Emrax 228 the magnet content is approximately 1.3 kilograms, giving an embodied carbon contribution of roughly 40 to 45 kilograms of CO₂e for the motor magnets alone. The global supply chain for high-coercivity NdFeB is heavily concentrated in China at the mining, refining, and magnet-production stages, which is a strategic supply-chain vulnerability that propagates from any axial-flux motor selection regardless of manufacturer. EMRAX mitigates this at the procurement level by sourcing European magnet stock where available, but the downstream supply remains Chinese-dominated.

The accumulator is the dominant embodied-carbon contributor in the whole vehicle. Published lifecycle estimates for modern lithium-ion pack manufacturing place the embodied carbon at approximately 60 to 110 kilograms of CO₂e per kilowatt-hour of pack capacity, with the variation reflecting cell chemistry, manufacturing location, and grid carbon intensity at the cell-production stage. For the locked 8.16 kilowatt-hour pack, this gives a range of roughly 490 to 900 kilograms of CO₂e for the accumulator

alone, which is an order of magnitude greater than the motor magnet contribution. Once the chassis aluminium and miscellaneous components are added, and once the avoided ICE-specific items such as the engine block, fuel tank, and exhaust system are netted out, the EV-specific embodied carbon premium over an equivalent ICE FS car settles at approximately 300 to 600 kilograms of CO₂e.

The operational per-event emissions saving is small in comparison. A generic FS combustion entrant burns approximately 5 to 8 litres of petrol per 100 kilometres, which over the roughly 24 kilometre endurance event corresponds to approximately 3 to 4 kilograms of CO₂ per event at the petrol carbon factor of 2.31 kilograms of CO₂ per litre. The EV at 6.16 kilowatt-hours net pack consumption and the UK grid carbon intensity of approximately 0.2 kilograms of CO₂ per kilowatt-hour produces approximately 1.2 kilograms of CO₂ per event. The per-event saving is therefore roughly 2 kilograms of CO₂ in favour of the EV, and the break-even against the embodied-carbon premium calculated above sits at approximately 150 to 300 endurance-equivalent events. For a low-mileage competition vehicle, this break-even is unlikely to be reached within the vehicle's service life, and the honest engineering conclusion is that the per-vehicle carbon case for EV motorsport is marginal rather than decisive.

The stronger environmental case therefore rests on two indirect factors rather than on per-vehicle carbon arithmetic. First, motorsport plays a meaningful role in normalising and validating EV technology at the design, engineering, and public-perception levels, and that role accelerates the transition of road-vehicle fleets where the carbon arithmetic is decisively favourable. Second, the EMRAX motor topology is highly recyclable: the bolted stator-rotor-housing construction has no encapsulation or composite content, and greater than 95 percent of motor mass is recoverable as identifiable aluminium, steel, copper, and segregated rare-earth magnet material at end of life. The decision to use plain water as the cooling fluid, per the FS coolant rule, also carries a modest operational environmental benefit by eliminating glycol disposal and spill-management requirements at competition events. A formal lifecycle assessment covering the full vehicle is identified as a build-year activity in Chapter 8.

Social. Electric motorsport produces an order-of-magnitude reduction in noise emissions relative to internal-combustion motorsport. The EMRAX 228 produces peak operational noise dominated by 12 kilohertz inverter switching transients and aerodynamic drivetrain noise, both well below the 90 to 110 decibel A-weighted levels typical of FS combustion entrants. The reduction allows competition venues such as Silverstone to operate in closer proximity to residential populations than ICE motorsport permits, with positive implications for the long-term regulatory and commercial viability of FSUK as a regional event. There are downstream public-health benefits as well, including reduced spectator hearing exposure and the elimination of the carbon monoxide and particulate emissions associated with paddock-area ICE operation.

The project also contributes to the UK EV engineering workforce pipeline. Students working through the design, simulation, and integration of the motor subsystem develop competencies in inverter-motor pairing, field-oriented control architecture, lap-time simulation, and rule-bound EV design. These competencies transfer directly into UK automotive-sector and Formula 1 engineering roles, but also into the wider electrification effort across aerospace, renewable energy infrastructure, and grid-scale battery systems. The high-voltage operational and safety training carried by every Brunel Racing

member working on the EV programme is itself a transferable industry skill, and the wider UK EV transition over the next decade requires substantially expanded specialist headcount drawn from exactly this kind of programme (Levin, 2024).

The negative social implications are real and worth surfacing. The high-voltage training and safety overhead required for an EV programme is a barrier to entry that disadvantages smaller institutions and self-funded student teams, and the supplier-driven cost structure of competitive FS EV entries continues to favour well-resourced teams over those operating under tighter budgets. The transition from ICE to EV at university motorsport level has therefore introduced a financial and infrastructure asymmetry between teams that did not exist as starkly under the previous combustion ruleset. This is a structural consequence of the rule transition rather than a Brunel-specific issue, but it is one that the FS organising body and participating universities will need to address if the Concept Class and Class 1 fields are to remain broad enough to sustain genuine competitive variety.

7. Conclusions

Engineering progress on a Formula Student Concept Class motor subsystem in a regulated, packaged EV is rarely about adding capability to the motor itself. The motor's intrinsic envelope exceeds the binding constraints imposed by the rule set, the accumulator, and the inverter across most of the operating range. Design margin is instead recovered by identifying where each external constraint binds at each operating point, and by reshaping the controller architecture and cohort interface specification to exploit that understanding.

The first finding concerns the FS power rule against the motor's efficiency at the binding operating point. EV2.2.1 is defined at the DC bus and measured by the FS Energy Meter; a controller that simply clips shaft power at 80 kW allows DC input to reach approximately 87 kW at 92% motor efficiency, breaching the rule by 7 kW and triggering the 500 ms moving-average enforcement under D9.4.1. The DC-input cap controller derived in §5.5.2 derates the shaft cap to 73.6 kW so DC bus power clips cleanly at the regulatory ceiling. The finding extends the FS-EV control literature (Ibrahim et al., 2025; Badawy et al., 2024), which has not previously distinguished shaft-side from DC-side power caps, and generalises to any FS-EV controller under the present rule set.

The second finding concerns the regime crossover between the motor's voltage-limited envelope and the FS regulatory ceiling. Across the accumulator's 270–454 V range, voltage-limited peak power varies from 50.7 kW at end of discharge to 73.6 kW at full charge, with rule and motor envelope crossing at approximately 392 V. A fixed torque-cap map calibrated at any single bus voltage cannot represent both regimes. The SoC-aware adaptive controller in §5.5 resolves this through a minimum-of-constraints architecture that switches binding regime automatically as bus voltage falls through threshold, saving 94 seconds of endurance event time against a conservative fixed-map alternative and 3.1 seconds against a sensibly calibrated one. The contribution fills a structural gap between the road-EV flux-weakening literature (Lu and Kar, 2010; Alwaqfi et al., 2025), which treats bus voltage as constant, and the FS-EV motor-control literature (Ibrahim et al., 2025), which optimises in the motor-speed domain at fixed bus voltage. Firmware-level dynamometer validation is the next step.

The third finding concerns the location of the binding constraint on regenerative recovery. The motor delivers approximately 80 kW of regenerative capability at typical FS braking RPM, while the locked 2C accumulator accepts approximately 16.3 kW—a 5:1 ratio. The 7.5% event-level recovery is therefore bounded by accumulator electrochemistry, not motor capability, and recovery rises to approximately 15% at 4C and 19% at 5C without any motor-side modification. Recovery improvement is an accumulator-design decision, controlled by the accumulator electrical lead. The finding inverts the conventional regenerative-braking narrative (Szumska, 2025), which positions the recovery lever in controller refinement, brake-force distribution, or motor placement—none of which applies when the pack is the binding constraint.

The design package handed to Brunel Racing for 2027 delivers the documented motor selection bounded by explicit sensitivity analysis, and the build-ready hardware specification. The selected powertrain meets every binding requirement from the FS Concept Class rule set and cohort interface constraints, with quantifiable margin against thermal, energy, and rule-compliance limits.

8. Further Work Recommendations

Four further-work priorities are identified, each motivated by quantitative evidence in the preceding chapters and ordered by potential leverage on overall vehicle performance.

A dynamometer validation of the field-weakening envelope would allow the largest residual uncertainty in the motor capability surface, which is the linear interpolation of the three published K_V values used to model the field-weakening region to be solved, which §4.3 acknowledges as accurate to within ± 5 –10% in the upper-RPM range. A single-day dyno session using the actual EMRAX 228 MV LC and Cascadia PM150DZ pairing, sweeping from no-load through rated current at multiple bus voltages, would produce a full d-q axis flux model and tighten this uncertainty band to approximately $\pm 2\%$. The session would also empirically validate the digitised efficiency map and refine the SoC threshold at which the FS regulatory rule binds.

Real-time motor-temperature feedback in the adaptive controller would allow the present controller to adapt the winding temperature as it adapts the torque cap to bus voltage and SoC but treats winding temperature as a passive monitor rather than a control variable. The thermal-transient simulation in §5.7 shows 39.3 °C of margin to absolute cutoff under nominal cooling, but this margin assumes a 50 °C coolant inlet that depends on the thermal-loop performance under worst-case ambient conditions. A future iteration should integrate the KTY 81/210 sensor reading directly into the torque-cap lookup, applying a smooth derate from full capability down to S1 continuous as winding temperature approaches the warning threshold. This converts the present hard 100 °C trip into graceful degradation and eliminates fault-induced power loss on a hot competition day. The hardware is already specified; only the VCU control law requires development, in coordination with the LV/VCU lead.

Another task would be Drive-mode integration with the adaptive controller. The mode-paddle interface proposed for the driver-controls subsystem provides three modes (Sport, Endurance, Wet) implemented as scalar caps applied to the adaptive torque map. A more developed integration would tune the SoC-

aware envelope independently per mode — prioritising aggressive rule-binding in Sport, actively reducing peak power below the rule limit in Endurance to extend pack range, and applying a different power-versus-traction blending in Wet to reflect lower available friction. The framework supports this trivially because the adaptive map is already a function of three variables; adding a mode parameter is a one-dimensional extension. Implementation requires collaboration with the mode-paddle and LV/VCU leads to calibrate the parameter sets against driver feedback during shakedown testing.

The cohort's Concept Class submission demonstrates whole-vehicle component readiness but does not demonstrate whole-vehicle closed-loop behaviour, which could have solved through use of CarMarker full car simulation. A CarMaker model incorporating the seven powertrain subsystem specifications and the three cross-cohort interfaces would identify residual integration issues that emerge only when subsystems interact in a single closed-loop simulation, substantially de-risking the build year's first on-vehicle test session.

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Appendix 1 — Project Management

A1 - Comparison Against Proposal Aims and Objectives

The November project proposal committed the motor subsystem to three aims and five objectives, with three expected outcomes. Comparing each commitment against the delivered submission identifies the scope movements that occurred during the project, the reasons for those movements, and the integrity of the final deliverable against the original commitment.

The first aim, to identify a motor concept aligning with power, torque, thermal, and rule limits, was fully delivered. The Emrax 228 MV LC selection was justified through screening of four alternative motor candidates on cohort coolant compatibility per T7.2.2, inverter peak-current utilisation against the Cascadia PM150DZ 300 A_{RMS} limit, electrical-parameter availability for simulation framework calibration, and architectural fit with the locked 400 V accumulator envelope. The screening rationale and rejection reasoning are documented in §5.1.

The second aim, to establish the motor's role in the 400 V architecture, was fully delivered. The motor subsystem is positioned as the converter between the electrical domain owned by the accumulator and inverter leads and the mechanical domain owned by the drivetrain lead. The architectural lock — motor selection, gear ratio, regenerative behaviour, and controller architecture — was committed at a level of detail sufficient for adjacent subsystem leads to commit their own designs against fixed reference values.

The third aim, to provide performance data that informs gearing, thermal design, and control strategy, was exceeded relative to the original commitment. The proposal-stage scope under "control strategy" implied a torque map and APPS calibration. The delivered work extends this into a methodological contribution: the SoC-aware adaptive controller documented in §5.5 fills the structural gap identified in the literature between road-EV control and FS-EV control, with quantified benefit against fixed-map alternatives. The contribution is methodological rather than implementational; firmware-level dynamometer validation is identified as further work in Chapter 8.

The five objectives were delivered as committed, with one scope movement worth surfacing. Objective three (identify thermal loads requiring cooling support) was delivered through the lumped-parameter thermal model and the endurance-event transient analysis in §5.7, with handover of heat-rejection numbers to the thermal management lead. The original proposal stage included scope for FEA on motor mounting and CFD on cooling integration. Both items were redirected during the project: the FEA was reduced to a hand-calculated bolt-reaction approach because the EMRAX motor mount is delivered as

an OEM-certified component with prescribed bolt grade, torque, and locking specifications; the CFD work was handed to the thermal management lead as the natural owner of the cooling-loop integration analysis. The redirected effort was absorbed into the controller-side scope expansion documented above.

Reviewing the group-level project outcomes against the November proposal's three group-level aims: a feasible 400 V single-motor configuration was developed and justified across all seven Group 1 subsystems; subsystem concepts were aligned with FS rules and integration of the powertrain with chassis, suspension, and driver-controls cohorts was achieved at the numerical-interface specification level. The group delivered against its three aims. Where the group submission is thinner than the individual submission is in cross-cohort integration verification, with that work continuing as part of the cohort design package rather than being closed at the Concept Class submission stage.

A2 - Was the Original Project Plan Sufficiently Detailed?

The honest answer is that the original project plan was sufficiently detailed at the architectural and task-allocation level, but insufficiently detailed at the methodology, rule-compliance, and uncertainty-handling levels. Three concrete examples illustrate where the plan would have benefited from greater proposal-stage depth.

The proposal-stage methodology did not anticipate the DC-input cap finding documented in §5.5.2. The FS 80 kW power rule was treated implicitly as a shaft-power cap during proposal-stage scoping, consistent with the simpler interpretation that the rule applies to motor output power. The more careful interpretation, which recognises that the rule applies to DC bus power measured at the Energy Meter and that the controller must therefore derate the shaft cap by the motor efficiency at the binding operating point, emerged during simulation framework development as the rule wording was examined more closely. A proposal-stage methodology that had committed to early-stage rule-wording analysis would have caught this finding at the architectural-decision phase rather than during simulation development.

The proposal did not include a sensitivity analysis as a planned deliverable. The conservative-input methodology adopted for cohort-pending parameters (tyre coefficient of friction, aerodynamic drag and lift areas, vehicle weight distribution, drivetrain efficiency) demanded a sensitivity analysis as the methodological bound on the conservative-assumption uncertainty. The analysis was added as a §5.9 deliverable because the methodology required it, not because the proposal had planned for it. A proposal stage that explicitly committed to sensitivity analysis as a methodology component would have produced earlier-stage clarity on the conservative-assumption bounds.

The technical aspects of the project did proceed broadly as planned at the architectural level — the November-stage commitments to motor type, inverter, accumulator topology, and drivetrain configuration held through to submission — but the methodology-level activities expanded beyond what the proposal scoped. The structural lesson is that a proposal-stage methodology benefits from greater specificity on methodology components, rule-compliance approach, and uncertainty handling, rather than treating these as implicit elements of the scoped work.

A3 - Task Allocation, Role Evolution, and Workload

The Group 1 task allocation, established at the November proposal stage, divided the BMM-A Electric Powertrain into seven subsystem leads: accumulator electrical, accumulator mechanical, inverter and HV system, low-voltage system and VCU, motor, drivetrain and packaging, and thermal management. The split is structurally well-balanced: each subsystem has a defined scope with clear interfaces to adjacent subsystems, and the seven scopes together cover the architectural envelope without significant gaps or overlaps. The allocation was fair at the proposal stage in the sense that no subsystem was obviously over- or under-scoped relative to the others.

The motor subsystem allocation specifically grew in scope during the project, in two directions. First, the controller-side scope expanded substantially. The proposal-stage commitment under "control strategy" implied a torque-map calibration deliverable; the delivered work extends this into the SoC-aware adaptive controller architecture documented in §5.5, with the corresponding Tier A DC-input cap finding, the regime-crossover analysis, the 22-lap controller comparison study, and the firmware-level handover to the LV/VCU lead. Second, the rule-compliance scope expanded into the build-sheet-level hardware specification. Both expansions were absorbed within the motor subsystem rather than redistributed to other subsystem leads.

In compensation, the motor-mount FEA and the cooling-integration CFD were redirected. The motor-mount FEA was reduced to a hand-calculated bolt-reaction approach because the EMRAX X bracket is an OEM-certified mounting solution with prescribed bolt grade, torque, and locking specifications; full FEA on a manufacturer-certified component would not have produced design information that the manufacturer's certification does not already provide. The cooling-integration CFD was handed to the thermal management lead as the natural owner of the cooling-loop sizing problem.

The role of the motor subsystem within the group did not formally change but did informally expand into a numerical-interface coordination function. The motor subsystem's outputs — motor mass, peak torque reaction, peak DC current draw, peak heat rejection load, gear ratio commitment — propagate to multiple adjacent subsystems, and tracking these numerical interfaces against the adjacent

subsystems' design states became part of the working pattern. This was not a formal role assignment; it was a consequence of the motor subsystem's central position within the powertrain architecture.

The overall workload allocation within the motor subsystem grew during execution relative to the proposal-stage scope, but the growth was self-imposed through the controller-side scope expansion rather than externally imposed through cohort reallocation. The work was delivered within the available time, but with the time-management pattern documented in §A4 below.

A4 - Critical Self-Reflection

The Brunel guide is explicit that this section requires honest critical reflection on individual project performance, and that uncritical accounts are graded down. Three honest self-critiques are surfaced here, each documenting a project-management characteristic that, on reflection, would be addressed differently if the project were repeated.

The first self-critique concerns time management and project pacing. The substantive work on the simulation framework, the controller derivation, the rule-compliance demonstrations, and the sensitivity analysis was developed at intensity, but with concentration in particular project phases rather than across an evenly-distributed schedule. The DC-input cap finding emerged at a stage where its incorporation forced re-running of all four event simulations and re-issuance of the cohort interface specifications with corrected DC current values. A more evenly-paced schedule, in which the rule-compliance work was developed in parallel with the simulation framework rather than in sequence, would have surfaced this finding earlier and reduced the rework footprint. The lesson is that rule-compliance work and methodology development should be scheduled in parallel, not in sequence, particularly on rules with non-trivial interpretation surface area.

The second self-critique concerns under-investment in literature review depth at the proposal stage. The literature review documented in Chapter 2 was developed as the methodology required it, with sections supporting motor selection, lap-time simulation precedent, control strategy positioning, and regenerative braking literature added as the corresponding chapters of the report were drafted. This produced a literature review that supports the methodology but was not used to scope the methodology. An earlier-stage literature review, completed during the proposal-to-execution window, would have surfaced the existing flux-weakening control taxonomy earlier, allowing the SoC-aware adaptive controller to be positioned against the literature deliberately at scope-commitment stage rather than retrospectively at write-up stage. More importantly, an earlier-stage literature review would have allowed the motor-subsystem scope to extend toward more innovative material at the proposal stage, when scope boundaries were still flexible. The controller contribution that emerged is methodological;

an earlier-stage understanding of the gaps in the published literature would have allowed scope to extend toward those gaps strategically rather than incidentally.

The third self-critique concerns the depth of EV-powertrain background research undertaken before the proposal commitment. EV-powertrain design is a deep multi-disciplinary subject area combining electromagnetic machine design, power electronics, battery electrochemistry, vehicle dynamics, thermal engineering, and control theory. The proposal-stage scope was committed without sufficient prior depth in these disciplines, leaving the project to absorb foundational learning during execution rather than at preparation. This produced a working pattern of expanding scope as understanding grew, which delivered a strong submission but did so reactively rather than strategically. A proposal stage informed by deeper prior research — particularly into FS-EV-specific control architectures, into the published lap-time-simulation precedent for axial-flux PMSM applications, and into the structural gaps in the regenerative-braking literature — would have committed to scope boundaries that better balanced ambition against the analytical depth available to support the ambition.

These three self-critiques are individual project-management lessons rather than methodological or technical failures of the delivered work. The technical and methodological work in the report is defensible at submission grade. The lessons concern pacing, preparation, and how project ambition is translated into project plan during the proposal-stage commitment.

A5 - Cohort Coordination, Team Meetings, and Working Together

The cohort-coordination model adopted for the BMM-A Electric Powertrain Group relied on architectural commitments at the proposal stage, with each subsystem lead developing their own design against the committed architectural reference frame and adopting conservative working assumptions for cohort-pending parameters where formal sign-off had not yet been received. This approach produced defensible individual subsystem submissions with documented uncertainty, but it placed the burden of integration verification on the build year rather than on the Concept Class submission stage.

The conservative working-assumption approach is documented in §3.3 and §4.10, with the corresponding sensitivity analysis in §5.9 quantifying the effect of each cohort-pending parameter on the headline performance metrics. The methodology is defensible at submission stage: the dominant uncertainty (tyre coefficient of friction) was identified, quantified, and bounded against the design margin. The lap-time predictions hold within the conservative-input window, and the cohort-pending parameter refinements available at build-year stage produce only second-order corrections to the predicted performance.

Team meetings within Group 1 were held on a phase-aligned rather than rigid-schedule basis: the proposal-stage architectural commitments were reached through cohort discussion in November 2025; subsequent meetings were held at the integration-readiness stage to reconcile cross-subsystem numerical interfaces and packaging volumes. The architectural-commitment meetings were the most consequential for the motor subsystem, because the locked architectural reference frame produced by those meetings is the basis on which the motor selection, the controller architecture, and the interface specifications are all defended. The integration-readiness meetings were less impactful on the motor subsystem in the sense that the motor-side numerical commitments had largely been finalised before integration began. The team-meeting model worked well for architectural lock and worked less well for cross-subsystem parameter handover, with the latter relying on individual subsystem leads adopting conservative working assumptions rather than on cohort-wide parameter sign-off.

The team's collective working pattern was characterised by individual subsystem development against the architectural reference frame, with cohort-level integration emerging through interface-specification handovers rather than through continuous cohort-wide design coordination. This worked well for subsystems with internally self-contained design problems (motor selection, inverter selection, accumulator topology) and less well for problems requiring cross-cohort iteration (mass distribution, packaging volumes, thermal interfaces). The structural lesson is that architectural-commitment meetings and parameter-handover meetings serve different functions, and a project of this scale benefits from both being scheduled deliberately rather than treating the latter as an emergent property of the former.

What was learnt about working in a team: that architectural commitment at the early stage is the single most important enabler of subsystem-level delivery; that conservative working assumptions are a defensible mitigation for cohort-pending inputs but are not a substitute for formally signed-off baselines; that the analytical effort spent on sensitivity analysis as a defensive measure trades against the time available for primary methodology development. What was learnt about managing a team: that meeting frequency is a less important variable than meeting purpose; that team-coordination infrastructure (a shared workspace, a numerical interface document, hard internal deadlines) substitutes for meeting time at lower cost; and that the leader's role on a project of this scale is not primarily to do work but to ensure that work happens when and where the integrated design depends on it.

Three formal supervisor meetings with Professor Wang took place across the project year. The motor subsystem lead attended all three. The module guide expectation of fortnightly supervisor meetings was not met in practice, which limited the volume of supervisor-driven course correction that the project benefited from, and the meetings that did take place were not minuted with action plans circulated as the guide specifies. The honest position is that the project lacked the structured supervisor-feedback

rhythm that the module is designed to provide, and the consequence is that supervisor input shaped the project less than would have been ideal under the intended meeting cadence.

The most substantive supervisor input that did shape the project was Professor Wang's recommendation, made during the first supervisor meeting, that the motor subsystem lead engage directly with the active Brunel Racing Formula Student team. This recommendation was acted on. Meetings were held with the team's technical leadership, the locked vehicle architecture and component baseline were validated against current team practice, and the design package was framed throughout as a handover-ready deliverable for the 2027 build year rather than as a standalone academic exercise. This single piece of supervisor input materially shaped the framing of Chapter 6, the cohort-handover structure of §6.2, and the build-year-scoped recommendations of Chapter 8. Beyond this recommendation, supervisor engagement with the report's technical content was limited, and drafts were not reviewed before submission. Methodology-level course correction therefore came from cohort discussion and from self-review rather than from supervisor input.

Internal BMM-A group meetings were held every two weeks, scheduled by Abdalla as the accumulator-mechanical lead acting in an informal coordination role. The motor subsystem lead attended every internal meeting. These meetings served as the cohort's primary parameter-handover mechanism in the absence of formalised supervisor-led reviews, and they worked reasonably well as such. Their effectiveness was limited by uneven preparation across the cohort. Several subsystem leads arrived at meetings without the parameter values their colleagues depended on, which converted what should have been parameter-sign-off sessions into status discussions. The motor subsystem lead's own preparation was consistent across these meetings, but the cohort-wide preparation discipline was not, and this is part of the cohort schedule risk registered in §A6.

A6 - Risk Identification And Management

Risk management on the project operated at three levels. Proactive risk identification at proposal stage flagged one principal risk: insufficient technical depth in EV-powertrain disciplines relative to the analytical ambition of the scoped work. Continuous risk monitoring during execution surfaced five further risks documented below, each logged against its identification stage and mitigation. Risk-residual handover to the build year is documented in Chapter 8 as the explicit transfer of unmitigated risks to the 2027 team. The proposal-stage risk was mitigated through structured self-study during early execution, supported by literature review developed in parallel with the methodology, and was overcome in the sense that the delivered work is technically defensible — at the cost of reactive scope expansion. Foundational learning happened during execution rather than at preparation, which is a less

efficient path than a proposal stage informed by deeper prior research, and the lesson is registered in the third self-critique of §A4.

Five further risks surfaced during execution that were not anticipated at proposal stage. Each is described here against its identification stage, mitigation, residual exposure, and avoidance reflection.

The first was cohort-pending parameter risk. Tyre coefficient of friction, aerodynamic drag and lift areas, vehicle weight distribution, and drivetrain efficiency were not signed off at the moment motor-side analysis needed them. The mitigation was the conservative-assumption methodology in §3.3 backed by the sensitivity analysis in §5.9. The residual exposure is that the headline performance figures sit in the middle of their respective uncertainty bands rather than at confident points, and the tyre coefficient of friction ($\mu = 1.30$) remains the largest single contributor to headline-number variance. Avoidance would have required earlier-stage flagging of high-sensitivity cohort-pending parameters as critical-path dependencies at proposal stage.

The second was methodology accuracy risk. The K_p interpolation method used to construct the field-weakening envelope carries an accuracy band of approximately 5 to 10 percent against full d-q simulation, and the quasi-static lap simulator carries an accuracy band of approximately 3 to 5 percent against full transient simulation. These uncertainties were identified during methodology development, acknowledged explicitly in §4.10, and bounded against the design margins documented in §3.5. The residual exposure is that the combined uncertainty band on some headline metrics is wider than the margin against the corresponding target, particularly the 1.1 percent endurance margin reported in §6.4. Avoidance would have required higher-fidelity modelling that the proposal did not scope and that the project timeline could not have absorbed.

The third was whole-vehicle integration risk carried forward to the build year. No CarMaker model, no hardware-in-the-loop bench, and no other whole-vehicle integration test was developed during the project. The mitigation is the explicit handover of these activities to the build year in Chapter 8. The residual exposure is real and material. The first time the seven powertrain subsystems and the three cross-cohort interfaces will interact in a single test environment is during early build-year shakedown, and any latent integration issue will surface there rather than during the Concept Class submission window. This risk is accepted rather than mitigated, and should be the first item the 2027 build team addresses.

The fourth was cohort schedule risk. Several subsystem leads progressed slowly through the early simulation and analysis phases, and the resulting delay compressed the time available for motor-subsystem refinement and for cohort-wide integration verification. The mitigation was self-pacing of

the motor subsystem against an internal schedule that did not depend on cohort delivery. The residual exposure is that the motor selection rationale closed earlier than ideal, with the richer Emrax 208 versus Emrax 228 comparison left as a build-year activity rather than completed at concept stage. Avoidance would have required hard internal cohort deadlines committed at proposal stage, which is the second recommendation in §A8.

The fifth was cross-cohort packaging risk. The chassis envelope locked at proposal stage did not accommodate the final-selected accumulator dimensions, requiring a late accumulator-mechanical redesign. The risk was not identified at proposal stage and the mitigation was reactive rather than planned. The residual exposure has been closed by the redesign, but the avoidance lesson is that cross-cohort packaging constraints need a single authoritative source of truth maintained under version control, which is the first recommendation in §A8.

The structural pattern across the five execution-stage risks is consistent. All carry the same avoidance signature: earlier-stage cross-cohort discipline at proposal stage would have either prevented the risk from materialising or surfaced it when the design was still flexible enough to absorb the consequence cheaply. The technical content of the project is defensible at submission grade, but the risk profile around the technical content reflects a proposal-stage scoping discipline that was thinner than the work eventually required.

A7 - Problems Encountered and Decisions Made During Execution

Several problems surfaced during execution that required decisions about how to proceed. Each is described here against the principle that the Brunel guide rewards specific honest reflection rather than generic accounts.

The dominant cohort-pending parameter throughout the project was the tyre coefficient of friction, owned by the suspension cohort group. The motor subsystem's lap-time predictions, the gear-ratio optimisation, and the rear-axle traction-limit calculations all depend on this single parameter, and the sensitivity analysis in §5.9 confirms it is the dominant uncertainty across all three headline performance metrics. The decision taken was to adopt $\mu = 1.30$ as a defensible conservative working assumption representative of competition-grade slick tyres at concept stage, document this assumption clearly, and bound its effect through sensitivity analysis. An alternative decision to defer motor-subsystem analysis until the suspension cohort delivered a formal value would have prevented progress and produced a less defensible submission. The decision taken was correct given the cohort-coordination state, but the consequence is concrete. The sensitivity analysis in §5.9 became a necessary methodological deliverable rather than an optional one, representing significant analytical effort that could otherwise

have been redirected toward primary methodology development. The headline performance numbers reported in §5.6.1 now carry an explicit uncertainty band rather than confident point values, and reader confidence in those numbers is correspondingly reduced relative to a version of this project in which formal μ sign-off had been delivered before motor-side analysis closed. The structural lesson is that cohort-pending parameters with high sensitivity should be flagged earlier as critical-path dependencies and committed at proposal stage rather than carried as live assumptions through execution.

The DC-input cap finding documented in §5.5.2 emerged during simulation framework development as the FS 80 kW rule wording was examined more closely against the motor's efficiency at the binding operating point. The decision to incorporate the corrected DC-input cap controller into the methodology, despite the rework cost on the four event simulations and the cohort interface specifications, was correct. A controller that exceeds EV2.2.1 by 7 kW under the simpler interpretation is not a defensible Concept Class submission. The consequence of taking this decision was substantive across three axes. First, the controller architecture documented in §5.5 now satisfies EV2.2.1 by construction rather than by accident of operating point, and the FS Energy Meter under D9.4.1 will not trigger disqualification under the corrected controller. Second, the cohort interface specifications issued to Roger as accumulator electrical lead and to Harsha as VCU lead now carry rule-compliant peak DC current figures and CAN torque-request structures rather than the pre-correction values that would have under-sized the main fuse and mis-specified the CAN command. Third, the corrected interpretation became one of the three engineering findings reported in Chapter 7 and contributes to the literature gap identified in §2.5 between road-EV and FS-EV control. The rework footprint affected all four event simulations and two cohort interface re-issuances, taking approximately a week of additional work, but the alternative was a Concept Class submission that would have been disqualified on rule-compliance grounds. The structural lesson is that mid-project methodology corrections, if the underlying engineering reasoning is sound, should be incorporated even when they impose rework cost.

The motor-mount FEA originally scoped at proposal stage was reduced to a hand-calculated bolt-reaction approach when it became clear that the EMRAX X bracket is delivered as an OEM-certified mounting solution with prescribed fastener specifications. The decision was that FEA on a manufacturer-certified component would not produce design information beyond what the manufacturer's certification already provides, and that the engineering effort was better spent on the controller-side scope expansion that did produce novel methodological output. This is a defensible engineering judgement rather than a scope failure, but it should have been recognised at proposal stage when the FEA was scoped, rather than reached during execution. The consequence of taking this decision was substantive. Approximately two weeks of FEA work that would have produced no novel design information was reallocated to the controller-side scope expansion documented in §A3, which

became the SoC-aware adaptive controller and the DC-input cap finding. The redirection therefore converted a low-value proposal-stage scope item into the most innovative methodological contribution of the project. The structural lesson is that proposal-stage scoping should test each candidate analytical activity against the question of whether it can produce design information that the available manufacturer data does not already provide, and reallocate scope accordingly.

The interface for cohort-wide numerical handover was not formalised at proposal stage. The motor subsystem's outputs, including peak DC current, peak heat load, mass, peak torque reaction, and regen acceptance ask, propagate to multiple adjacent subsystems, and tracking these numbers as they evolved through the project required ad-hoc communication rather than a single shared document. A version of this approach was prototyped late in the project as a Notion workspace for cohort leads, established initially for the Tech Director and Brunel Racing context. The consequence of this late-stage decision was positive but partial. The workspace was adopted by every cohort member who used it through to submission, validating the principle that shared cohort-coordination infrastructure removes friction at low cost once it is in place. The cohort therefore enters the 2027 build year with at least one working coordination tool already in active use, even though it was not available during the period when most of the cross-cohort parameter handovers were actually happening. The infrastructure transfers naturally to the MGP cohort-coordination context and would have served the present project from the proposal stage if it had been established at that time. The recommendation in §A8 addresses this gap as the first item for the 2026 to 2027 cohort.

The structural pattern across these problems is that none of them are technical failures of the engineering work. They are coordination, scoping, and timing decisions that, in hindsight, would have been made differently with greater proposal-stage preparation. The technical content of the report is defensible; the project-management characteristics around the technical content are where the lessons sit.

A8 - Recommendations for Next Year's Team

Five concrete recommendations are made for the next year's BMM-A Electric Powertrain Group, based on the structural observations from the present project execution.

The first recommendation is to establish a shared cohort workspace from the proposal stage. A single document, workspace, or interface control document, updated weekly by every cohort lead, listing committed numerical interface values, pending parameter values awaiting cohort sign-off, and parameter-handover deadlines, would provide the cohort-coordination infrastructure that the present project lacked. The workspace should also serve as the cohort decision log, recording when each architectural and methodological commitment was made and by whom, supporting traceability of

decisions across the project lifecycle. A Notion-based prototype of this approach was established late in the present project for the Tech Director and Brunel Racing context; the same infrastructure would serve the MGP cohort-coordination context if implemented at proposal stage.

The second recommendation is to set internal parameter-handover deadlines at the proposal stage. The November proposal should commit not only to task allocation across the seven subsystem leads but to a Gantt of parameter-handover deadlines specifying when each cohort lead must deliver each input that other leads depend on. Hard internal deadlines produce delivery; soft internal deadlines produce drift, with the consequent burden of conservative working-assumption methodology and sensitivity analysis as defensive mitigations.

The third recommendation is to schedule rule-compliance review in parallel with simulation framework development, not in sequence. The DC-input cap finding documented in §5.5.2 emerged from careful examination of the FS rule wording during simulation framework development. A formal rule-compliance review scheduled for the early simulation phase, rather than as a final-stage consolidation activity, would have surfaced this finding at the architectural-decision phase. The recommendation generalises: any rule with non-trivial interpretation surface area benefits from early-stage detailed wording analysis, not just from late-stage compliance-evidence consolidation.

The fourth recommendation is to commit to literature review depth before proposal lock. The proposal stage benefits from a depth of literature understanding that allows scope commitments to be made against an informed view of where the published literature has gaps and where innovative methodological contributions are possible. Two to three weeks of structured literature review at project start, with each subsystem lead delivering a survey of the published literature relevant to their scope, would allow the proposal stage to commit to scope boundaries that align with the analytical depth of the team. The present project's most innovative contribution — the SoC-aware adaptive controller — emerged from a literature gap that was identified retrospectively during write-up; an earlier-stage literature review would have allowed this gap to be identified at proposal stage and the methodology to be scoped deliberately around it.

The fifth recommendation is to plan for sensitivity analysis from the proposal stage where cohort-pending parameters are unavoidable. The methodology should explicitly include sensitivity analysis as a planned deliverable, scoped at proposal stage and developed in parallel with the primary simulation work. This demonstrates analytical rigour as a strength rather than as a defensive mitigation, and it absorbs cohort-handover delay without compromising the individual submission's analytical integrity.

These recommendations apply most strongly to the Concept Class submission stage of the cohort programme. The build-year team operates against a different set of constraints — manufacturing lead times, test scheduling, on-vehicle dynamometer validation, and cost management — that the Concept Class submission stage does not test. The present recommendations should be considered as additions to the build-year-specific recommendations that will emerge from the build year itself.

A9 - Summary of Lessons Learnt

The headline lessons from the project execution, distilled across the preceding sections, are six.

First, methodology specificity at proposal stage matters as much as task allocation. A proposal that scopes the methodology in detail — including planned sensitivity analysis, planned rule-compliance review, planned literature review depth, and planned interface-tracking infrastructure — produces a less reactive execution than a proposal that scopes only task allocation.

Second, rule-compliance work belongs in the early phase of a rule-bound engineering project, not in the consolidation phase. Rules with non-trivial interpretation surface area can produce findings that affect architectural decisions, and those findings are cheaper to incorporate when the architecture is still flexible.

Third, conservative working assumptions are a defensible methodology for cohort-pending parameters but are not a substitute for formally signed-off baselines. Sensitivity analysis bounds the consequence of working assumptions but trades analytical effort against primary methodology development.

Fourth, cohort coordination infrastructure substitutes for cohort coordination meeting time at lower cost. A shared workspace, a numerical interface document, and hard internal deadlines deliver coordination outcomes that would otherwise require continuous meeting attendance.

Fifth, individual project-management characteristics — pacing, preparation, scope discipline — affect the quality of the technical output even when the technical reasoning is sound. The technical content of this report is defensible; the project-management characteristics around it are where the next-iteration improvements would land.

Sixth, the role of a subsystem lead within a cohort project of this scale is partly individual technical execution and partly numerical-interface stewardship. Subsystem leads in the central position of an architecture, like the motor subsystem within the powertrain, carry a coordinating function whether or not it is formally assigned. Recognising this at proposal stage, and resourcing it deliberately, would produce a smoother cohort execution.